

Spectrum Power 5
Information Model
Management (IMM)

5.21

Reference Guide

History

Preface

Table of Contents

Information Model Management

1

Operational Steps

2

User Interface Description

3



Note

Please observe safety notes and warnings for your own safety.

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History

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04-2-01	11.2006	Update for SPC 4.2

Preface

Safety Notes

This manual does not constitute a complete catalog of all safety measures required for operating the product in question because special operating conditions might require additional measures. For application specific safety measures refer to the respective project documentation. However, it does contain notes that must be adhered to for your own personal safety and to avoid damage to property. These notes are highlighted with a warning triangle and different keywords indicating different degrees of danger as stated in the Warning Conventions.



WARNING

WARNING indicates a potentially hazardous situation which, if not avoided, could result in death or serious injury.

- ✧ Follow all advice instructions to prevent death or serious injury.



CAUTION

CAUTION indicates a potentially hazardous situation which, if not avoided, may result in minor or moderate injury.

- ✧ Follow all advice instructions to prevent minor or moderate injury.



Note

NOTE indicates important information about the product, the handling of the product, or the part of the documentation in question, to which special attention must be paid.

Qualified Personnel

Only competent and authorized personnel should work with this product after becoming thoroughly familiar with all warnings, safety notices, operating instructions and maintenance procedures.

Use As Described

The product must not be used for any other purposes than that described in the technical documentation. If it is used together with third-party devices and components, these must be recommended or approved by

Siemens AG. The successful and safe operation of this product is dependent on adequate transportation and proper handling, storage, installation, operation and maintenance.

Scope

This Reference Guide provides specific and detailed information on how to use a particular product or product component:

- Typical workflows and operations
- Description of all user interfaces and displays



Note

Please note that the screenshots used in this document contain sample data which may not be available in some systems.

Typical Users

This Reference Guide is designed for users that are already familiar with operational and technical aspects of power generation and power transmission and distribution as well as Spectrum Power 5 product concepts:

- **System Engineers**
The System Engineer is able to install and to customize the system. He needs deep knowledge about the internal structures and processes of the network control system.
- **Operators**
The Operator is controlling the network or the plants with the system therefore he needs to know how the network control system has to be operated.
- **Data Engineers**
The Data Engineer translates the real network of a utility or the generation plants to the data model of the network control system. He also maintains the data model.
- **Technical Specialists**
The Technical Specialist is responsible for a special technical area which is related to the network control system. This can be the planning department, which plans changes in the network and uses network analyses applications of the network control system or the RTU department, which has to connect the RTU to the network control system. The Technical Specialist has to know how specific functions of the network control system work in detail.

Structure of this Manual

- **Introduction:**
Basic information about the Spectrum Power 5 component
- **Workflow / Operations:**
Main use cases, workflows and operational steps for optimal use of the component
- **References:**
Reference and description of user interfaces and dialogs

Documentation Conventions

Example	Meaning
File > Open	Selecting the menu item Open on the File menu
F1	Pressing the F1 function key on the keyboard
Control+o	Holding down the Ctrl key and pressing the letter o on the keyboard
Control+Shift+o	Holding down the Ctrl key and the Shift key and pressing the letter o on the keyboard
Left-click	Clicking the left mouse button
Right-click	Clicking the right mouse button
Shift-click	Holding down the Shift key on the keyboard and clicking the left mouse button

Questions?

This document is part of a set of manuals that describes the complete Spectrum Power 5 product.

Various other product modules may be mentioned or discussed in this document. For more detailed information or if you have any questions about these Spectrum Power 5 products, please contact your Siemens representative.

Table of Contents

	History	3
	Preface	5
1	Information Model Management	15
1.1	Introduction	16
1.2	Workflows	17
1.2.1	Introduction	17
1.2.2	Modify an Instance and Activate the Change	17
1.2.3	Create a New Type	17
1.2.4	Modify an Instance in a Time-Based Job Model	17
1.2.5	Modify an Instance Using a Test System	17
1.2.6	Modify an Instance in an HMM Configuration	18
1.3	Keyboard Shortcuts	19
2	Operational Steps	21
2.1	Starting to Work with IMM	22
2.2	Managing Time-Based Jobs	25
2.3	Configuring Archiving Using the IMM Admin Tool	27
2.3.1	Introduction	27
2.3.2	Configuring Job Archiving	27
2.3.3	Enabling and Disabling Job Archiving	27
2.3.4	Job Activation	30
2.3.5	Managing Archives and Long Term Archives	30
2.3.6	Troubleshooting Job Archiving Problems	35
2.3.7	Removing the Archiving IMM Jobs Function	35
2.4	Adding New Instances of an Existing Type	37
2.5	Adding a New Curve	40
2.6	Adding 2 New Attributes to the Existing Type Switch	41
2.7	Adding the New Type Recloser and an Instance of This Type	45
2.8	Modifying Instances Using the Table Editor	53
2.9	Modifying Instances Using the Bulk-Instance Editor	54
2.10	Linking Instances to CFE Data Points	57
2.11	Configuring ICCP Data Objects	58
2.12	Activating the Data Changes	60
2.13	Exporting and Importing Engineering Data	62
2.14	Modifying the Presentation Logic for the Circuit Breaker with Automatic Reclosing Function	65
2.15	Testing Jobs Using the Test System	70

2.16	Handling Change Requests Within a Hierarchically Organized Control Center System	72
3	User Interface Description	77
3.1	Information Model Management	78
3.1.1	Structure of the Information Model Management	78
3.1.2	Menu Bar	79
3.1.2.1	Overview	79
3.1.2.2	File	80
3.1.2.3	Edit	80
3.1.2.4	View	82
3.1.2.5	Insert	84
3.1.2.6	Tools	84
3.1.2.7	Help	86
3.1.3	Toolbar	86
3.1.4	Tree View	89
3.1.4.1	Introduction	89
3.1.4.2	Tree	89
3.1.4.3	Dataset Tree	89
3.1.4.4	Type Tree	90
3.1.4.5	Instance Overview Tree	92
3.1.5	Working Area	94
3.1.5.1	Introduction	94
3.1.5.2	Editors	94
3.1.5.3	Tabs	95
3.1.5.4	Table Editor	95
3.1.5.5	Handling of Views	99
3.1.5.6	Dialogs	100
3.1.5.7	Wizards	101
3.1.5.8	Information Tabs	101
3.1.6	Log Section	102
3.1.7	Status Bar	105
3.1.8	Progress Indication Window	105

3.2	General Engineering Applications	106
3.2.1	Job Management User Interface	106
3.2.1.1	Overview	106
3.2.1.2	Dataset Tree	106
3.2.1.3	Job Management UI Tab - Job List	108
3.2.1.4	Job Management UI Tab - Job Context Menu	110
3.2.1.5	Test System Tab / Production System Tab	113
3.2.1.6	Change Request List	115
3.2.1.7	Emergency-Backup Control Center Tab	117
3.2.2	Session Properties Tab	118
3.2.3	Change Log	119
3.2.4	Report Dialog	121
3.2.4.1	Overview	121
3.2.4.2	Report Dialog for Types	121
3.2.4.3	Report Dialog for Instances	123
3.2.4.4	Report Viewing Window	124
3.2.5	Report Generator Dialog	125
3.2.5.1	Overview	125
3.2.5.2	Predefined Reports	126
3.2.5.3	User-Defined Reports	127
3.2.6	Import Dialog	128
3.2.7	Export Dialog	131
3.2.8	Instance Search Dialog	132
3.2.9	Import to Test System	134
3.2.10	Import to Production System	135
3.2.11	Logs	135
3.2.11.1	IMM Status	135
3.2.11.2	Task Log	136
3.2.11.3	Validation Log	136
3.2.11.4	Activation Log	136
3.2.11.5	Import Log	137
3.2.11.6	Job History	138
3.2.11.7	Finalized Jobs	138
3.2.11.8	Conflict Log	139
3.2.11.9	Notification Log	139

3.3	Instance Editors	141
3.3.1	Single-Instance Editor	141
3.3.1.1	Introduction	141
3.3.1.2	General Tab	142
3.3.1.3	Attributes Tab	143
3.3.1.4	Links Tab	144
3.3.1.5	Protocol Tabs	145
3.3.1.6	Options Dialog	146
3.3.1.7	Instance Creation Dialog	146
3.3.2	Table Editor	148
3.3.2.1	Introduction	148
3.3.2.2	Table View	148
3.3.2.3	Protocol Tab	150
3.3.2.4	Communication View	151
3.3.2.5	Options Dialog	151
3.3.3	Bulk-Instance Editor	153
3.3.3.1	Introduction	153
3.3.3.2	Query Designer	154
3.3.3.3	Result Table	157
3.3.4	Curve Editor	158
3.3.4.1	Introduction	158
3.3.4.2	Curve Definition Editor	159
3.3.5	Color Picker	160
3.3.5.1	Introduction	160
3.3.5.2	Color Palette	161
3.3.6	Common Calculation Editor	162
3.3.6.1	Introduction	162
3.3.6.2	Script Tab	162
3.3.6.3	Trigger Tab	163
3.3.6.4	Buttons	165
3.4	Special Editors	166
3.4.1	Communication Editor for CFE	166
3.4.1.1	Introduction	166
3.4.1.2	Measurement Table	166
3.4.1.3	RTU Table	167
3.4.2	Communication Editor for ICCP	168
3.4.2.1	Introduction	168
3.4.2.2	Overview	169
3.4.2.3	Basic Editor Components	169
3.4.2.4	ICCP Client View	170
3.4.2.5	ICCP Server View	172

3.5	General Editors	174
3.5.1	Type Editor	174
3.5.1.1	Introduction	174
3.5.1.2	Editor Context Menu	175
3.5.1.3	General Tab	175
3.5.1.4	Attributes Tab	177
3.5.1.5	Enumerations Tab	181
3.5.1.6	Hierarchy Tab	183
3.5.1.7	References Tab	186
3.5.2	Decision Table Editor	188
3.5.2.1	Introduction	188
3.5.2.2	Editor Context Menu	189
3.5.2.3	General Tab	194
3.5.2.4	Preview Tab	196
3.5.2.5	Main Area Tab	197
3.5.2.6	Left Top Tab	199
3.5.2.7	Left Bottom Tab	199
3.5.2.8	Right Top Tab	200
3.5.2.9	Right Bottom Tab	200
3.5.2.10	Graphic File Browser Dialog	200
3.6	Wizards	202
3.6.1	Introduction	202
3.6.2	Energy Accounting Wizard	202
3.6.2.1	Introduction	202
3.6.2.2	Definition of the Number of Import and Export Accounting Points	203
3.6.2.3	Configuration of Accounting Points	204
3.6.2.4	Definition of Measuring Links to Accounting Points	205
3.6.2.5	Definition of Tariff Rule Links, Control, and Forecast Area Links to Accounting Points	206
3.6.2.6	Saving Accounting Points	207
3.6.3	Script Definition Wizard	208
3.6.3.1	Introduction	208
3.6.3.2	Definition of the Script and Its Attributes	208
3.6.3.3	Definition of Formal Script Operands	209
3.6.3.4	Definition of the Script Body	211
3.6.4	Calculation Wizard	212
3.6.4.1	Introduction	212
3.6.4.2	Definition of the Measuring Result and the Calculation Method	212
3.6.4.3	Definition of Operands	216
3.6.5	Script Wizard	217
3.6.5.1	Introduction	217
3.6.5.2	Definition of the Script Call/Execution	217

1 Information Model Management

1.1	Introduction	16
1.2	Workflows	17
1.3	Keyboard Shortcuts	19

1.1 Introduction

Spectrum Power Information Model Management (IMM) gives you the ability to enter and maintain all power system related engineering data for use within Spectrum Power control center systems within a CIM-based central repository. Engineering data consists of domain data like equipment, measurements and topology, as well as single line displays and configuration data of the Spectrum Power control center system like hardware and software deployment and application configuration. A set of editors allow data access and data definition optimized for the different engineering workflows a data engineer needs to perform. Complete, partial, and incremental import and export of engineering data is provided through CIM-RDF and XDF formats based on the W3C standard XML. IMM also controls the activation of data changes to the Spectrum Power control center runtime databases. The Information Model Management is based on an object-oriented data model using a Relational Database Management System (RDBMS).

Based on the Microsoft Windows standard, IMM provides one common user interface for data entries covering all aspects of data engineering. The IMM user interface can be viewed and accessed via Microsoft Internet Explorer. It provides different views of the data according to your intention or workflows.

The Graphics Editor user interface for creating and maintaining graphical displays and worldmaps is based on the Siemens WinCC Graphics Designer. It has been adapted with features optimized for the IMM workflows.

1.2 Workflows

1.2.1 Introduction

This section contains examples for workflows in IMM.

For detailed information on the individual operating steps, please refer to chapter Operations.

1.2.2 Modify an Instance and Activate the Change

- Start IMM.
- Create a job.
After the job has been created, the job is loaded automatically.
- Start the single instance editor.
- Navigate to an instance in the tree view and modify an attribute.
- Click the save button in the toolbar.
The data modification is stored in the job.
- Quit the job.
- Validate the job, prepare it for activation and activate it.
Now the data modification is available in the data model.

1.2.3 Create a New Type

- Start IMM.
- Open the root job.
The job is loaded.
- Navigate to a type in the tree view, create a type, and define the attributes.
- Click the save button in the toolbar.
The data modification is stored.
- Quit the job.
The new type is available.

1.2.4 Modify an Instance in a Time-Based Job Model

- Create a new private job.
- Modify an instance.
- Quit the job.
- Publish and activate the job.

1.2.5 Modify an Instance Using a Test System

- Create a job on the production system and modify an instance.
- Quit the job.
- Transfer the data model to the test system.
- On the test system, import the job from the production system.
- Test the job on the test system.

- If necessary, make modifications in the job.
- On the production system, import the job from the test system.
- Activate the job on the production system.

1.2.6 Modify an Instance in an HMM Configuration

- Create and validate a job on a second-level control center.
- Create a change request.
- Send the change request.
- On another second-level control center, download and accept the change request.
- Import the change request into a new job.

1.3 Keyboard Shortcuts



Note

Keys may be all lowercase, all uppercase or mixed lowercase and uppercase unless explicitly described.

To	Press
Close Information Model Manager	Alt-fx or Alt-F4

Accessing Menus

To display	Press
File menu	Alt-f
Edit menu	Alt-e
View menu	Alt-v
Search an instance	Alt-es
Insert menu	Alt-i (lowercase I)
Tools menu	Alt-t
Help menu	Alt-h

Viewing Status Information

To view	Press
Session Properties Page	Alt-fp

Selecting a Data View

To Select	Press
Type view	Alt-vt or Ctrl-t
Instance view	Alt-vi or Ctrl-o
Instance Editors cascading menu	Alt-ve
Instance Editors Single View	Alt-ves
Instance Editors Table View	Alt-vet
Instance Editors Automatic Generation View (this functionality is currently not supported)	Alt-vea
Communication Editor CFE	Alt-vecc

Navigating in the Overview Tree

To	Press
Select top entry of Overview Tree	Home
Select bottom entry of Overview Tree	End
Display next page of entries in Overview Tree	Page Down
Display previous page of entries in Overview Tree	Page Up
Expand subtree	Right arrow key
Collapse subtree	Left arrow key
Select first entry of subtree	Right arrow key
Select Parent entry of subtree	Left arrow key

Select next entry in sequence regardless of hierarchy	Down arrow key
Select previous entry in sequence regardless of hierarchy	Up arrow key
Update Overview Tree	Alt-vr

Editing Engineering Data

To	Press
Insert new type	Alt-it
Insert new instance	Alt-ii or Ctrl-i
Mark selected instance for copy	Alt-ec
Mark selected instance for cut	Alt-et
Paste marked instance with internal links	Alt-ep
Save currently edited type or instance	Alt-fs or Ctrl-s
Delete currently selected type or instance	Alt-ed
Undo modifications performed in the currently selected editor	Ctrl-z

Accessing Engineering Applications and Tools

To open	Press
Report Dialog	Alt-tr
Import Dialog	Alt-fi or F2
Export Dialog	Alt-fe or F3
Graphics Editor	Alt-tg

Accessing Help Information

To view	Press
Help topics	Alt-hh or F1
Help information about Work Pane content	Alt-hw
Version information about the Information Model Manager	Alt-ha



2 Operational Steps

2.1	Starting to Work with IMM	22
2.2	Managing Time-Based Jobs	25
2.3	Configuring Archiving Using the IMM Admin Tool	27
2.4	Adding New Instances of an Existing Type	37
2.5	Adding a New Curve	40
2.6	Adding 2 New Attributes to the Existing Type Switch	41
2.7	Adding the New Type Recloser and an Instance of This Type	45
2.8	Modifying Instances Using the Table Editor	53
2.9	Modifying Instances Using the Bulk-Instance Editor	54
2.10	Linking Instances to CFE Data Points	57
2.11	Configuring ICCP Data Objects	58
2.12	Activating the Data Changes	60
2.13	Exporting and Importing Engineering Data	62
2.14	Modifying the Presentation Logic for the Circuit Breaker with Automatic Reclosing Function	65
2.15	Testing Jobs Using the Test System	70
2.16	Handling Change Requests Within a Hierarchically Organized Control Center System	72

2.1 Starting to Work with IMM

The goal of this task is to start IMM and to create a job for editing data.

Starting the Information Model Management

- Select **Start > Spectrum > Information Model Management** from the Windows Start menu. Alternatively you can start the IMM from the Spectrum Power 5 PAT entry IMM.

An instance of the Microsoft Internet Explorer opens. By default, the Internet Explorer is connected to the Power System Object Server (PSOS).

The **Job Management** user interface of the Information Model Management opens. The Job Management user interface allows you to control all data entry and engineering activities of the system.



Note

You can open multiple instances of the Information Model Management on multiple clients - which are all connected to the same PSOS server.

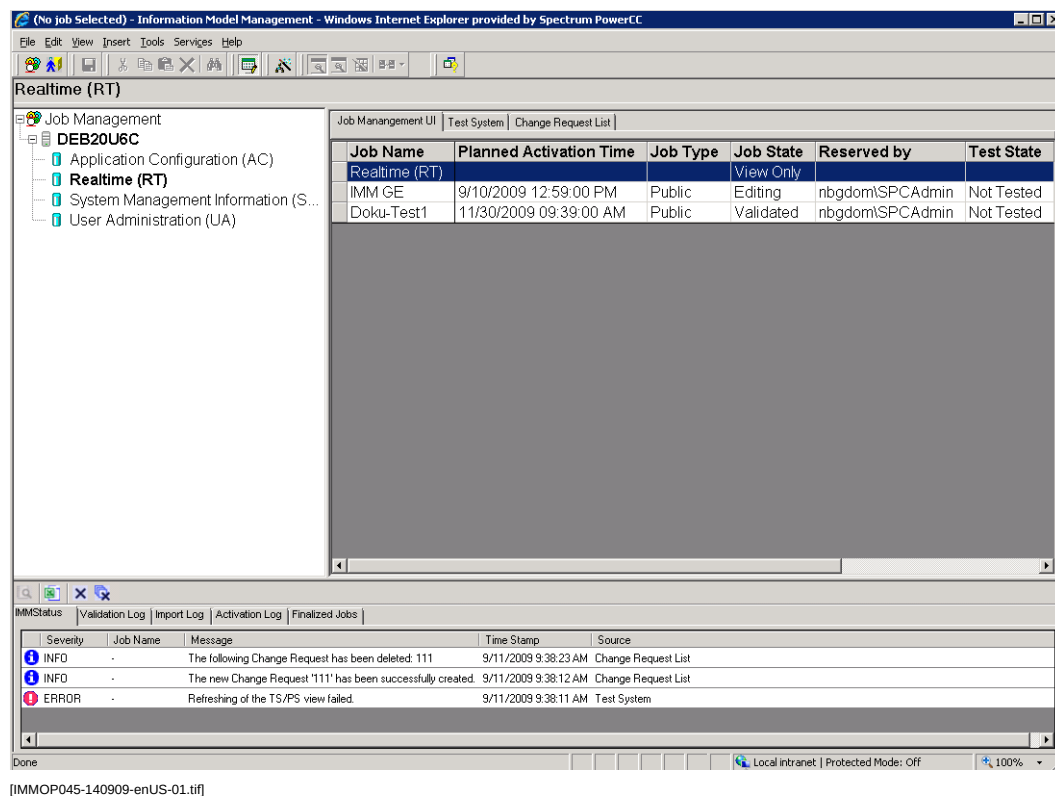


Figure 2-1 Job Management User Interface



Note

For information on the structure and content of the Job Management user interface, refer to [3.1.1 Structure of the Information Model Management](#).

Logging On

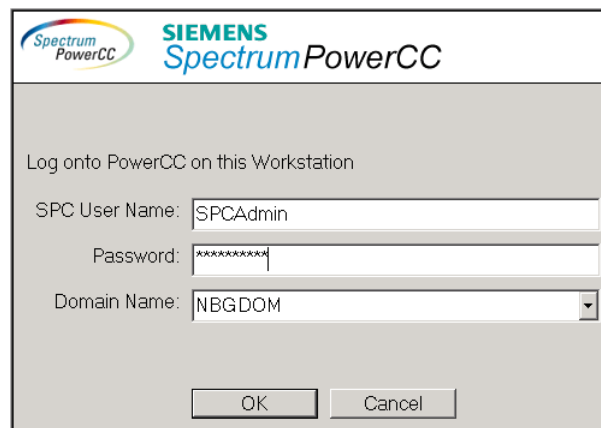
To edit IMM instance data you must have special user access rights. Log on as Spectrum Power Control Center user with IMM modification rights.



Note

For detailed information on user access rights, refer to the [Reference Guide System Software \(SYSW\)](#).

- Click the toolbar button **Logon** in the Information Model Management. (This step is not necessary if you are already logged on to Spectrum Power 5.)
The logon dialog opens.



[IMMOP002-291008-enUS-01.tif]

Figure 2-2 Logon Dialog

- Enter the user name and the password. If necessary, select the appropriate domain name.
- Click the button **Logon**.
You are logged on.

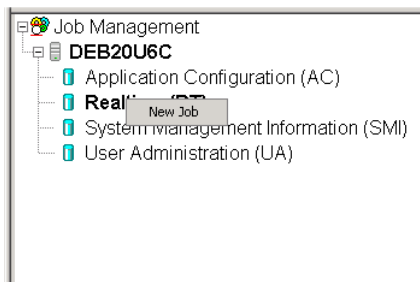
Creating a New Job



Note

This section describes the creation of a job in the dataset Realtime (RT) for the interlock-based job model configuration of IMM. For creation of a job in a time-based job model configuration, refer to section [2.2 Managing Time-Based Jobs](#).

- Right-click the dataset **Realtime (RT)** on the tree view of the Job Management user interface.
The **Create Job** context menu opens:



[IMMOP047-150909-enUS-01.tif]

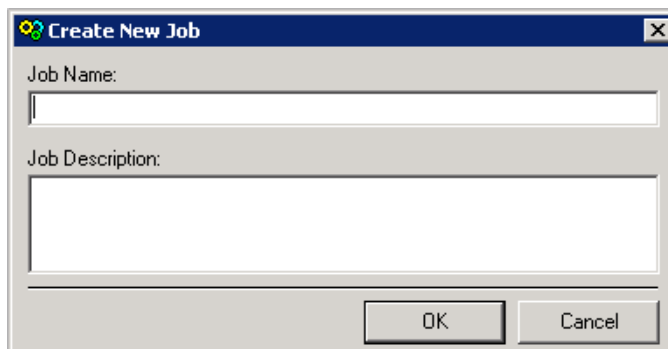
Figure 2-3 Create Job Context Menu

- Left-click the context menu item **New Job**.
The **Create New Job** window opens.

**Note**

Job management is only supported for the real-time dataset and requires appropriate access rights. The Create Job context menu can be disabled depending on the selected dataset and your access rights.

- Enter a new job name into the text box **Job Name** and a comment into the text box **Job Description** (optional).



[IMMOP005-101212-enUS-01.tif]

Figure 2-4 Create New Job Window (Interlock-Based Model)

- Click the **OK** button.
The top level of available instances is displayed on the tree view of the Information Model Management framework.

2.2 Managing Time-Based Jobs

The goal of this task is to create a private time-based job, to publish it, and to reschedule the job.

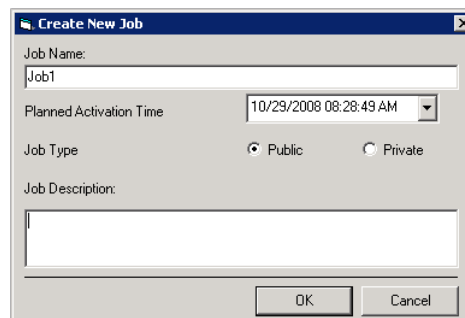


Note

This operating sequence can only be performed if the time-based job model option has been selected during system installation. For additional information on the time-based job model for data entry, refer to the [Technical Specification Information Model Management \(IMM\)](#).

Creating a New Time-Based Job (Private)

- Right-click the dataset **Realtime (RT)** on the tree view of the Job Management user interface.
The Create Job context menu opens.
- Left-click the context menu item **New Job**.
The **Create New Job** window opens.
- Enter a new job name into the text box **Job Name** and a comment into the text box **Job Description** (optional).
- Select the desired activation time and select the option button **Private** to create a private job. The desired activation time is the scheduled date for the activation of the job. It defines the sequence, in which different jobs are applied to the data model. You can modify this time later.



[IMMOP004-291008-enUS-01.tif]

Figure 2-5 Create New Job in a Time-Based Job Model

- Click the **OK** button.
The top level of available instances is displayed on the tree view of the Information Model Management framework.

Publishing a Private Job

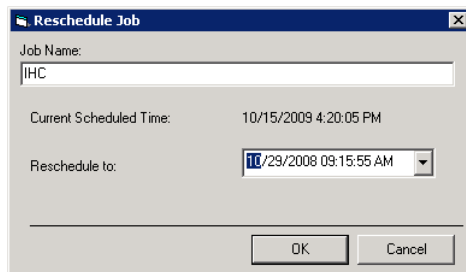
Only public jobs can be activated. Therefore private jobs must be published, before they can be activated.

- Right-click the desired private job in the job list.
- In the context menu select the item **Publish**.
Now the job list shows the job type **Public** for the selected job.
(Only possible if there are no dependencies with other jobs in the system.)

Rescheduling a Job

To modify the planned activation date of a job, perform the following steps:

- Right-click the desired job in the job list. In the context menu select the item **Reschedule**. The **Reschedule Job** display opens.



[IMMOP005-291008-enUS-01.tif]

Figure 2-6 Reschedule Job Window

- Select a new activation date in the text box **Reschedule To**.
 - Click the **OK** button.
- The new planned activation date is displayed in the job list.
(Only possible if there are no dependencies with other jobs in the system.)

2.3 Configuring Archiving Using the IMM Admin Tool

2.3.1 Introduction

The **IMM Job Archiving** function allows you to view the data model in the past.

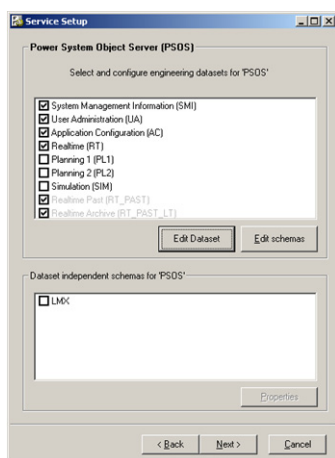
For additional information on the time-based job model and the interlock-based job model for data entry, refer to [Technical Specification Information Model Management \(IMM\)](#).

2.3.2 Configuring Job Archiving

By default the Realtime (RT) dataset is configured so that **Job Archiving** can be enabled once import of the bulk data into the root dataset is complete, you can edit data through jobs. During ServiceSetup the structures are put in place for **Job Archiving** and space is reserved for the Past dataset (RT_PAST) and long-term archive (RT_PAST_LT).

ServiceSetup is run during Spectrum Power 5 installation and creates the IMM datasets needed for **Job Archiving**.

In the ServiceSetup **Select and configure engineering datasets for PSOS** section, you can see that the **Realtime Past (RT_PAST)** and the **Realtime Archive (RT_PAST_LT)** datasets are checked.



[RGIMM001-270411-enUS-01.tif]

Figure 2-7 ServiceSetup: Selecting and Configuring Datasets for PSOS

2.3.3 Enabling and Disabling Job Archiving

After the Realtime (RT) dataset is first created, you can enter data into the root dataset. Once the system is brought online, jobs are used to incrementally activate data updates to the online system. The status of the root dataset is changed from **READY** to **VIEW ONLY** and it is no longer allowed to modify data directly in the root dataset.

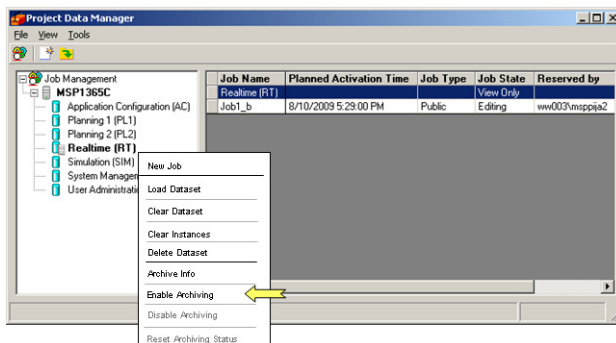
Job Archiving can begin anytime after import of the bulk data into the root dataset is complete and you can edit data through jobs.

Enabling Job Archiving

You need to have administrative rights to enable **Job Archiving** from the **IMM Admin Tool**. It is done by right clicking on the RT dataset and selecting **Enable Archiving** from the drop down menu. You will be prompted

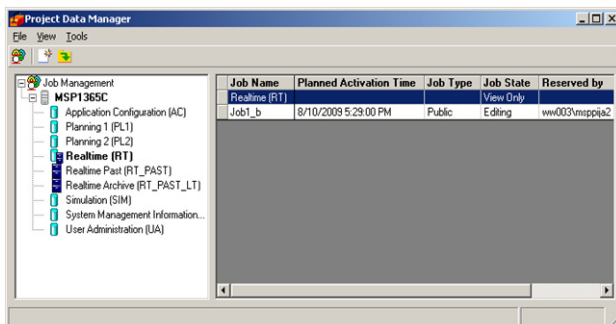
for an optional comment. Enable **Job Archiving** will take some time as all activated data is copied to the RT_PAST dataset. After **Job Archiving** is enabled, the following read only datasets will appear:

- Realtime Past (RT_PAST)
This is the active archive where activated jobs are stacked up and saved.
- Realtime Archive (RT_PAST_LT)
This is where long term archives can be loaded and examined.



[RGIMM002-270411-enUS-01.tif]

Figure 2-8 Enabling Job Archiving

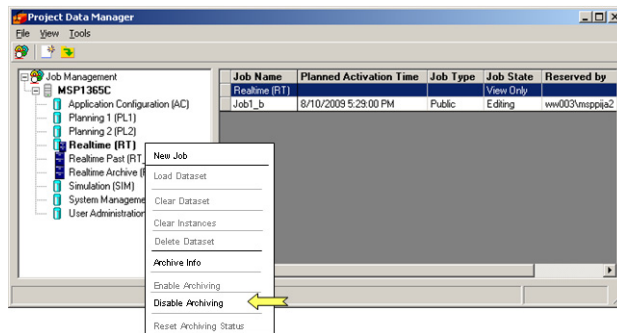


[RGIMM003-280411-enUS-01.tif]

Figure 2-9 Job Archiving Enabled

Disabling Job Archiving

While **Job Archiving** is enabled, **Clear Dataset**, **Clear Instances**, **Delete Dataset**, and **Reset [to Ready]** are grayed out for the RT dataset. These options are not allowed while archiving is enabled. If you have administrative rights, you can disable archiving. **Disable Archiving** is available in the drop down menu while archiving is enabled.



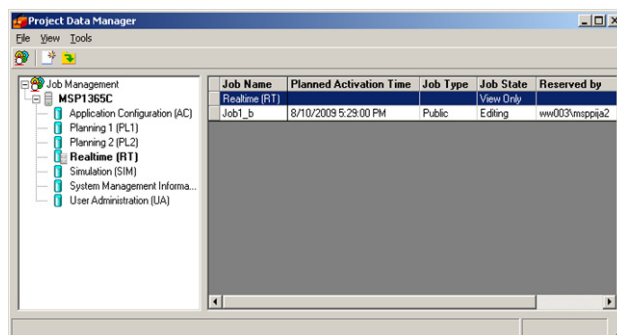
[RGIMM004-280411-enUS-01.tif]

Figure 2-10 Disabling Job Archiving

After **Job Archiving** is disabled, the Realtime Past (RT_PAST) and the Realtime Archive (RT_PAST_LT) datasets are removed.

**Note**

All past information stored in the Realtime Past (RT_PAST) dataset is lost unless a long term archive is created.

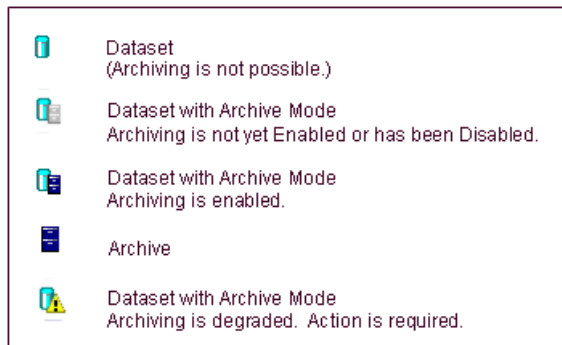


[RGIMM005-280411-enUS-01.tif]

Figure 2-11 Job Archiving Disabled

Dataset Icons

It is important to know whether a dataset configured with **Archive Mode** is actually archiving or not. This is done with the dataset icons in the job management tree pane. The icons are a visible indication of a reminder to enable archiving once you are ready.



[RGIMM006-101212-enUS-01.tif]

Figure 2-12 Dataset Icons

When the **Archiving is degraded** symbol is displayed, view the **Archive Log** to determine the problem and fix it.

2.3.4 Job Activation

When a job is archived to a Past time based dataset, its data changes along with its job logs, activation logs, validation logs, and import logs are preserved. Moving the job to the Past dataset is an important part of processing the job. IMM ensures that the job is archived before it is finalized.

Whenever a job activation successfully sends data changes to the online data repository, the data changes are also recorded in the Past dataset. The job changes are archived together with time parameters equal to the activation time. That is, the Past dataset reflects the online data over time. Archiving a job does not interfere with the actual activation and hence performance of activation of a job. Activation to the system occurs before the job is archived.

The same processing takes place for a job that is undone. When an undo activation successfully sends data changes to the online data repository, the undo changes are also recorded in the Past dataset. The undo changes are archived together with time parameters equal to the undo activation time. If a job is activated and then undone, 2 entries for the same job will exist in the Past dataset. While the job name will be the same, the job time will be different.

When **Job Archiving** is enabled, a job that is being activated has 2 additional states:

- Archiving
- Archived

Likewise, a job whose activation is being undone has these 2 additional states:

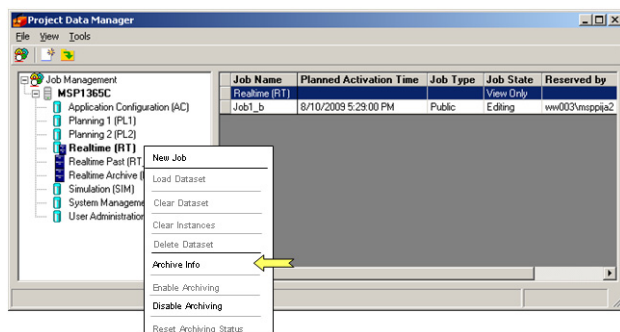
- Undo Archiving
- Undo Archived

2.3.5 Managing Archives and Long Term Archives

There are 3 functions for managing archives:

- Creating a long term archive
- Loading a long term archive
- Managing the size of the active archive RT_PAST (Rebase)

You can access these functions from the **IMM Admin Tool** by right clicking the RT dataset and selecting **Archive Info** from the menu.



[RGIMM007-280411-enUS-01.tif]

Figure 2-13 Archive Info

Archive Info in the IMM Admin Tool

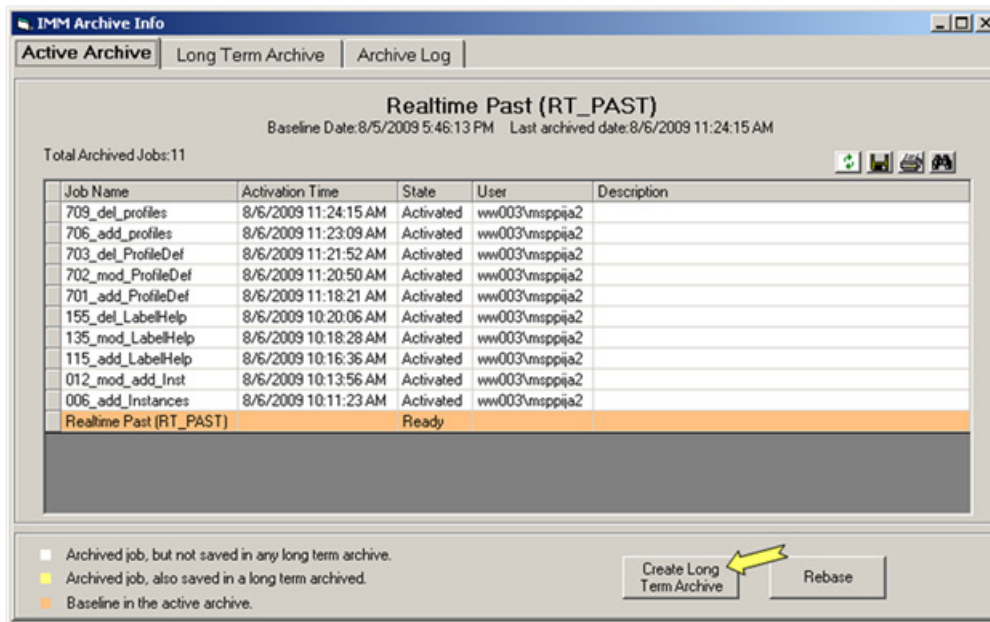
Archive Info opens a display where you can execute any of the 3 archive management functions. It displays information about the active archive (RT_PAST), long-term archives, and the archive log. Based on this information, you can make proper choices about creating long term archives and managing the size of the current active archive.

Selecting Archive info from the menu displays 3 tabbed pages:

- Active Archive (main)
 - Displays information about the current active archive
 - Allows you to create a long term archive
 - Allows you to rebase, or create a new day 0 baseline date, in order to reduce the size of the active archive
- Long Term Archives
 - Displays information about the long term archives that were created
 - Allows you to load a long term archive
- Archive Log
 - Displays log information such as when archiving was enabled and disabled

Creating a Long Term Archive

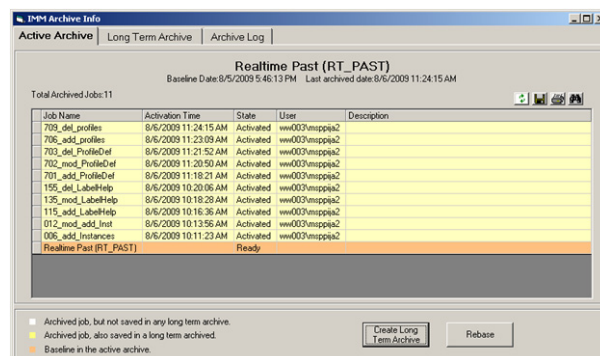
Creating a long term archive from Realtime Past (RT_PAST) is done from the main tab, **Active Archive**, where the archived jobs displayed in chronological order. To create a long term archive, press **Create Long Term Archive**. You will be prompted for an optional comment.



[RGIMM008-101212-enUS-01.tif]

Figure 2-14 Creating a Long-Term Archive

After the long term archive is completed, jobs are highlighted in yellow. Jobs highlighted in yellow are saved to a long-term archive. The baseline is highlighted in orange.



[RGIMM009-280411-enUS-01.tif]

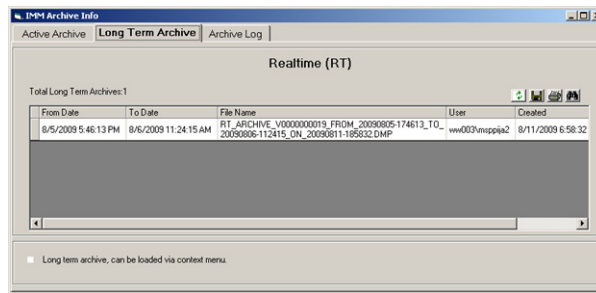
Figure 2-15 Long Term Archive is Created

The physical location of the long term archive is stored in the `longterm_archive_dir`. You can move the file, but you are responsible for the location and security of the file. The filename is generated based on the dataset version, time range of the data and current time:

RT_ARCHIVE_<Version>_FROM_<Begin timestamp>_TO_<End timestamp>_ON_<Current timestamp>.dmp

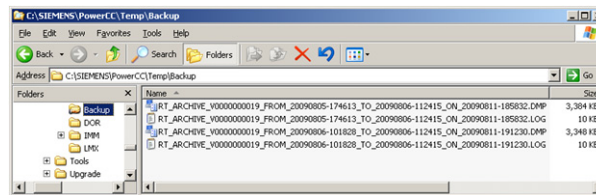
Where date and time format is: `yyyymmdd-hhmmss`

For example, *RT_ARCHIVE_V0000000019_FROM_20090805-174613_TO_20090806-112415_ON_20090811-185832.dmp*



[RGIMM010-280411-enUS-01.tif]

Figure 2-16 Viewing the Created Long Term Archive



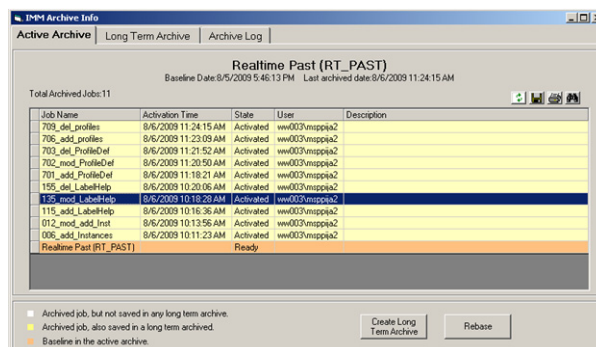
[RGIMM011-280411-enUS-01.tif]

Figure 2-17 Location of the Long Term Archives

Managing the Size of the Active Archive

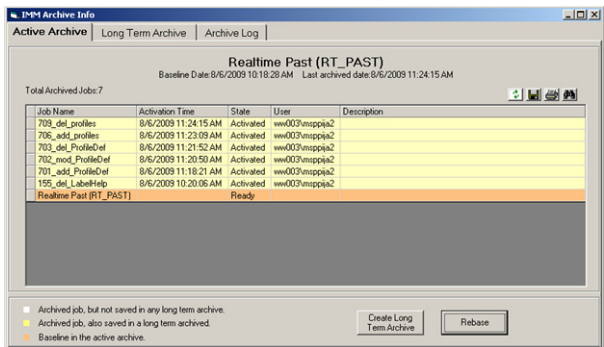
The size of the active archive is managed from the **Active Archive** tab. Here you can see the baseline date or, in other words, day 0 of the active archive. You can adjust this date to a later date in order to reduce the size of the active archive by right clicking on a job and selecting the Rebase menu option.

All jobs along with the base data between the initial baseline date up to and including the new baseline date will be compressed to form a new baseline. It is required that the jobs between and including the current baseline and the newly chosen baseline have been archived. In other words, they must be highlighted in yellow.



[RGIMM012-280411-enUS-01.tif]

Figure 2-18 Rebasing Realtime Past (RT_PAST)



[RGIMM013-280411-enUS-01.tif]

Figure 2-19 Realtime Past (RT_PAST) is Rebased

The Realtime Past (RT_PAST) dataset can contain 2 – 5 years of data depending on the number and size of jobs that are activated over time. The Realtime Past (RT_PAST) dataset is available at any time and is accessed similar to other datasets.

Periodically, the data in Realtime Past (RT_PAST) should be saved to a long term archive. Long-term archives are Oracle dump files that can be stored off site if required. The dump file can later be loaded into the long term archive dataset, Realtime Archive (RT_PAST_LT).

- Active Archive (about 2 – 5 years)
- Long Term Archive (multiple 2 – 5 year archives spanning several years)

As the active archive grows over time, you may want to save some of the data to a long-term archive and then reduce the size of the active archive by rebasing. As administrative user, you are responsible for when and what data is moved from the active archive to a long term archive. IMM provides you with the tools necessary to do this.

Loading a Long-Term Archive

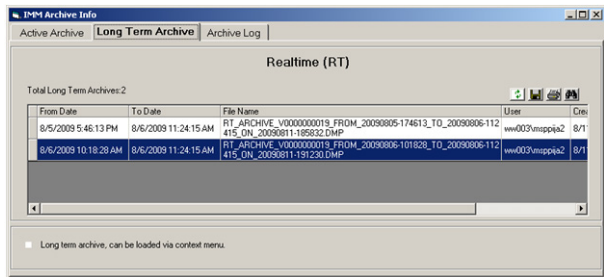
Loading a long-term archive is done from the **Long Term Archive** tab. You can see the date range of each long-term archive. To load a long-term archive into a dataset (RT_LT_ARC), right-click a file and select the **Load Long Term Archive** menu item.

The file must exist in the longterm_file_dir.



Note

Only one long-term archive can be loaded at a time. Multiple users, however, can view the long-term archive. When a new long-term archive is selected for loading, the previous one is unloaded.

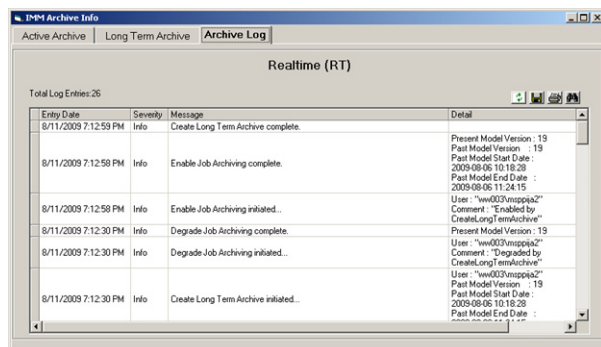


[RGIMM014-280411-enUS-01.tif]

Figure 2-20 Loading a Long-Term Archive

Viewing the Archive Log

Audit information is logged whenever actions related to **Job Archiving** occur such as enabling or disabling archiving, rebasing, and so on. This information can be viewed from the **Archive Log** tab.



[RGIMM015-280411-enUS-01.tif]

Figure 2-21 Archive Log

2.3.6 Troubleshooting Job Archiving Problems

When Job Archiving is Degraded

For any reason a job fails to be archived to the past dataset, the **Archive Status** will be changed from **Enabled** to **Degraded**. An **Archive Status** of **Degraded** means that the user will not be able to view the past dataset and that all subsequent archiving will fail, forcing the user to **Archive to File**, until the **Archive Status** returns to **Enabled**.

If you have administrative privileges, you can attempt to change the status from **Degraded** to **Enabled**. The **Archive Status** is only set to **Enabled** if all previously archived files are loaded into the system. The version number is used to determine if all activated jobs are loaded. You are forced to load all previously stored files (in order) into the past dataset before it can be viewed again. If this is not possible, a new base line must be created.

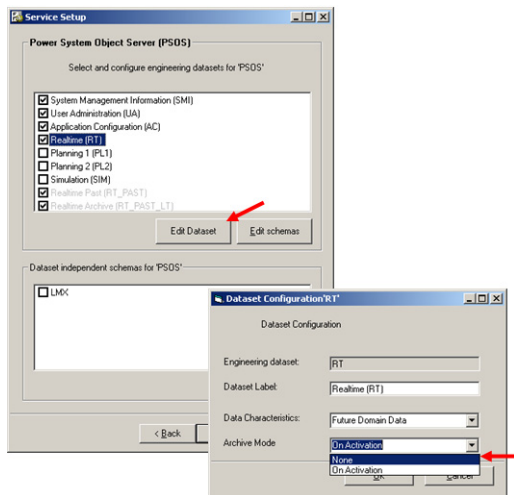
2.3.7 Removing the Archiving IMM Jobs Function

Configuring Job Archiving

By default the Realtime (RT) dataset is configured so that **Job Archiving** can be enabled once import of the bulk data into the root dataset is complete and you can edit the data. The structures is put in place and space is reserved for the Past dataset and long term archive.

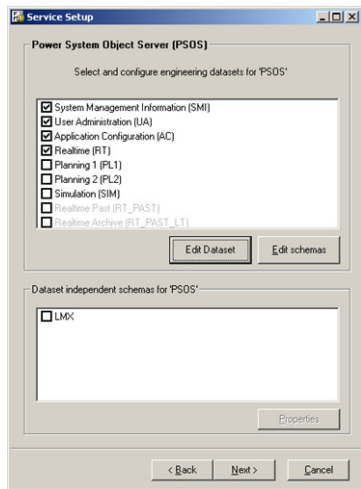
If **Job Archiving** is not desired, this can be specified during ServiceSetup.

ServiceSetup is run during Spectrum Power 5 installation and creates the IMM datasets. At this time, you can configure the **Archive Mode** for the Realtime (RT) dataset.



[RGIMM016-280411-enUS-01.tif]

Figure 2-22 ServiceSetup – Removing Job Archiving Function



[RGIMM017-280411-enUS-01.tif]

Figure 2-23 ServiceSetup – Job Archiving Function Removed



Note

These settings cannot be changed once the dataset is created.

2.4 Adding New Instances of an Existing Type

The goal of this task is to add a new power line to your data model. The line consists of several objects, such as, switches, power measurements, protection devices, and so on.

The first step is to add all the elements of the new power line that are based on existing types. You can do this

- By creating new instances from scratch, or
- By copying and pasting a similar power line and adapting the instances

Inserting a New Instance

Inserting a new instance is a 2-step procedure:

- Select the superordinate instance of the new instance within the instance tree.
- Select the type of the new instance.

In this example, you create a new bay below the substation **Net-E/Substations Generation/Sub Gen A**: browse to the instance **Net-E/Substations Generation/Sub Gen A** in the instance tree.

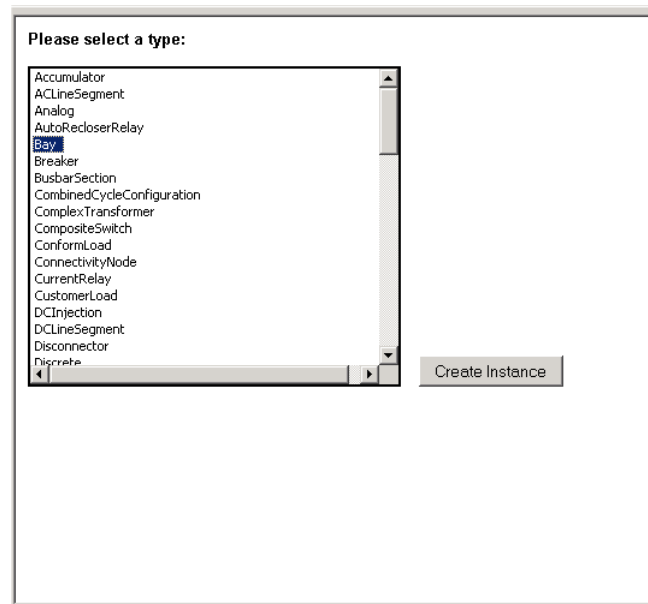
The instance tree can be used in the same way as a directory structure in the Windows Explorer. Click the '+' or '-' node at the right side of an instance name to expand or collapse subtrees.

- Right-click the instance **Sub Gen A**.

A context menu opens.

- Select the menu item **New** from the context menu.

The Type Selection Dialog opens on the working area.



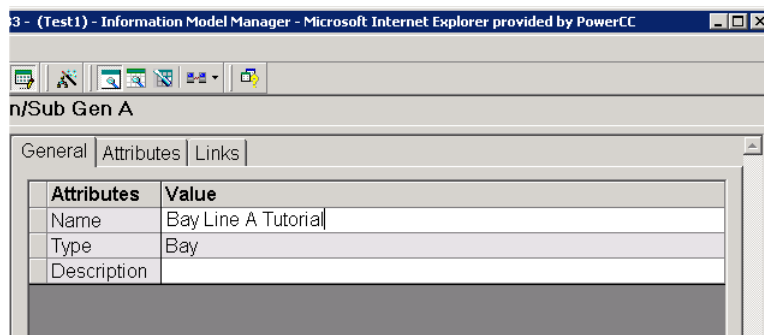
[IMMOP006-291008-enUS-01.tif]

Figure 2-24 Type Selection

- Select **Bay** and click the button **Create Instance** or double-click **Bay**.

The Single-Instance Editor opens in the working area. By default, the **General** tab is selected.

- Enter the name for the new instance: *Bay Line A Tutorial*.



[IMMOP007-301008-enUS-01.tif]

Figure 2-25 Single-Instance Editor Showing the General Tab

- Select **File > Save** from the menu bar or click the **Save** button on the toolbar to save the new instance. The new instance Bay Line A Tutorial is saved and automatically displayed in the tree. Sometimes it is necessary to refresh the tree to display new instances.
- Select **View > Tree Refresh** from the menu bar to refresh the content of the tree view. Now you can see the new instance **Bay Line A Tutorial** under the instance Sub A in the tree view. All subordinate instances (the switches, measurements, and so on) that belong to the bay of the new power line can now be entered manually under **Bay Line A Tutorial**.

**Note**

There is a faster and easier way to enter all subordinate instances of the new power line. Instead of entering all subordinate instances manually, you can copy and paste the instances of a similar line. For example, the existing power line instance **Bay A03 Line** is like **Bay Line A Tutorial**. It is faster to copy **Bay A03 Line** and paste it as Bay Line A Tutorial. However, if you try this, be aware that you must delete the manually created **Bay Line A Tutorial** before you copy and paste.

Deleting the New Instance

- Right-click the instance name **Bay Line A Tutorial** in the tree view. The context menu of the tree opens.
- Select **Delete** from the context menu. A confirm message window opens.
- Click the **OK** button in the message window. The instance **Bay Line A Tutorial** is deleted. A short message confirming the deletion is displayed in the status bar of the Information Model Management.
- Select **View > Tree Refresh** from the menu bar if necessary.

**Note**

If you delete an instance with existing links, the links to other instances are deleted as well. Errors occurred during the delete process are displayed in an error window.

Copying the Existing Elements of Line Sub A/ Bay Line A1

The easiest way to create a power line is to copy an existing one and paste it under a new name.

- In the Information Model Management right-click the instance name **Bay Line A1** in the tree view.
The context menu of the tree opens.
- Select **Copy** from the context menu of the tree.

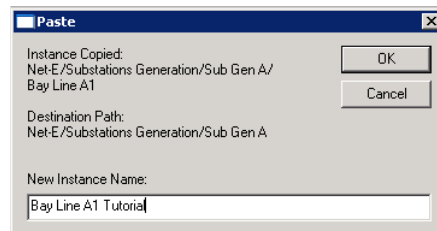


Note

All subordinate instances of bay **Bay Line A1** are copied.

Pasting the Elements as New Line Bay Line A Tutorial

- Click the instance name **Sub A** in the Overview section.
This sets the focus to **Sub A** and allows you to add the new line directly under this instance. This procedure defines the location (superordinate instance) of the new instances.
- Right-click the instance name **Sub A** in the tree view.
The context menu of the tree opens.
- Select **Paste** from the context menu. The **Paste** window opens.
- Enter a new name for the new power line: *Bay Line A1 Tutorial*



[IMMOP032-041208-enUS-01.tif]

Figure 2-26 Entering the New Instance Name

- Click the **OK** button in the **Paste** window.
A short message confirming the pasting is displayed in the IMM Status log of the Information Model Management.
Now you can see the new instances in the Overview section. They are all based on existing types. The instances are currently like the instances of the existing instance Bay Line A1.

2.5 Adding a New Curve

The goal of this task is to add a new curve. The curve consists of:

- A curve instance and
- Its child instances (interpolation points)

Creating a New Curve

You create a curve (instance) by the Curve Definition Editor in the following steps:

- Click the desired parent instance in the instance overview tree with the right mouse button.
- Click the menu item **New** in the opened tree view context menu.
The type selection dialog opens.
- Click the desired type, for example, `cfeCharacteristicCurve`.
- Click the button **Create Instance**.
- Enter the name of the new (curve) instance in the opened display (General tab).
- Click the button **Save** of the toolbar.
- Click the name of the new (curve) instance in the instance overview tree.
- Click the button **Add** in the opened Curve Definition Editor. Enter the attributes (for example, **Instance Name**, **Raw Value**, **Engineering Value**) for an interpolation point (child instance).
- Click the button **Add** for each interpolation point and then enter the attribute values of the interpolation point).
- After you have entered the parameters for all interpolation points, click the button **Apply**.

The new curve is displayed in the Curve Definition Editor. All attributes of an interpolation point are displayed in the Interpolation Point Editor.

2.6 Adding 2 New Attributes to the Existing Type Switch



Note

The following tasks "Add 2 New Attributes to the Existing Type Switch" and "Add the New Type Recloser and an Instance of this Type" can only be done if you have access rights for type modification. They cannot be done in a normal job created by you, but only in the real-time job. For additional information on user access rights, refer to the [Reference Guide System Software \(SYSW\)](#).

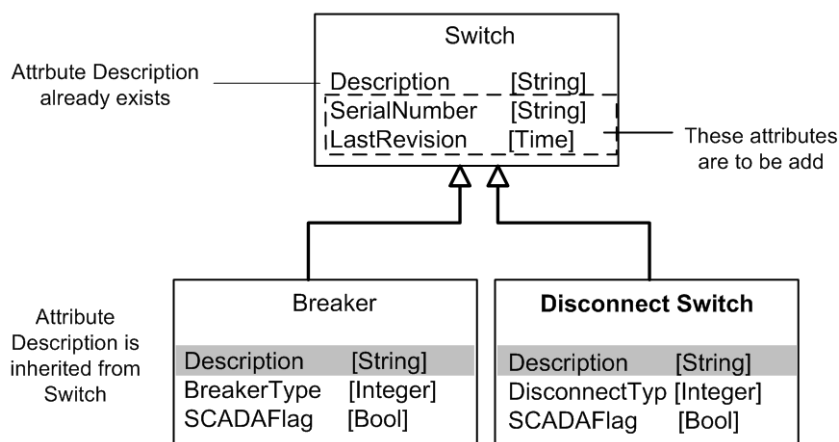
The goal of this task is to add 2 attributes to the existing type Switch. 2 subordinate types (or child types) are derived from the type Switch:

- Circuit breaker
- Disconnecter

Since these 2 types are children of the type switch, they inherited the attribute Description. In addition, they already have the attributes BreakerType or DisconnectType and ApplyMarker.

Suppose you want to add the following attributes to all types of switches:

- Serial Number
- Last Revision Date



[IMMOP009-160108-enUS-01.tif]

Figure 2-27 Inheritance of Attributes

Due to the object-oriented design of the data model, it is sufficient to add the new attributes to the type Switch. The child types automatically inherit them.

Opening the Type View

- Select **View > Types** in the menu bar or click the appropriate button in the toolbar.
The type tree opens in the tree view on the left side of the Information Model Management.
- Browse through the existing types. The types tree can be used just like the tree in the Windows Explorer. To expand or collapse subtrees click the symbols + or -, respectively.

- Browse to the type **Switch** and display its derived types by clicking the + symbol.

The path of the type Switch is: **Domain/IdentifiedObject/FieldResource/PowerSystemResource/Equipment/ConductingEquipment/Switch**

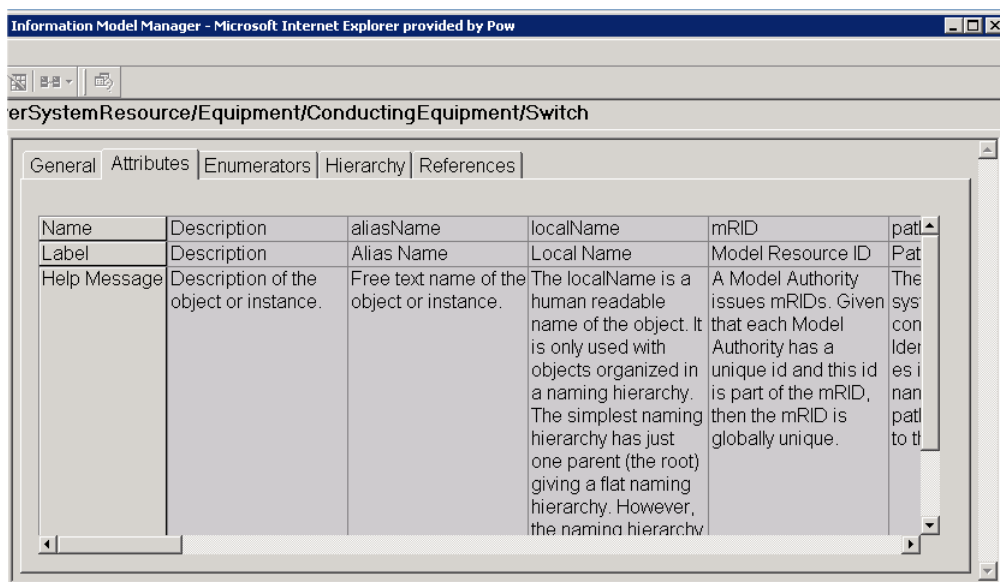
The type tree is organized differently than the instance tree. Their organizations follow different rules:

- The type tree indicates parent-child relationships. It is not domain-oriented and reflects the structure of the data model. A child type is derived from the parent type and inherits all characteristics, for example, attributes, enumerations, and so on.
- The instance tree is domain-oriented and indicates also parent-child relationships. You can group the instances to reflect the own domain (for example, the control and power system configuration).
- Another difference is that several instances can have the same name. Type names must be globally unique.

Viewing the Attributes of the Type Switch and Its Subordinate Types Breaker and DisconnectSwitch

- Click the type **Switch** in the tree view. The path of the type Switch is: **Domain/IdentifiedObject/FieldResource/PowerSystemResource/Equipment/ConductingEquipment/Switch**

The **Type Editor** opens in the working area. By default, the **Attributes** tab is selected. You can see all already defined attributes with their properties.



[IMMOP008-301008-enUS-01.tif]

Figure 2-28 Type Editor Displaying the Attributes of the Selected Type Switch

- Click the type **Breaker** in the tree view, view its attributes, and proceed with the type **Disconnecter**.



Note

The column showing the properties of the inherited attribute **Description** has a shaded background color. This means, the property values cannot be modified since **Description** is an inherited attribute. The property columns of the new attributes show a white background color. The property values of those attributes can be modified.

Adding 2 New Attributes to the Existing Type Switch



Note

Type changes are allowed only in the real-time dataset. Accessing the real-time dataset directly without a job requires appropriate access rights. In a job, type definitions are read only.

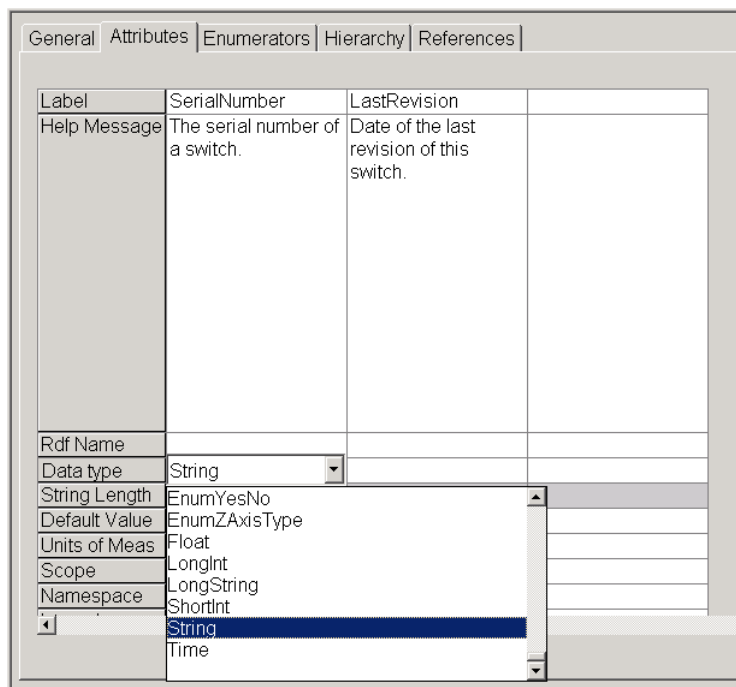
Now you can enter the new attributes to the type Switch.

- Click the type **Switch** in the tree view.
The **Type Editor** opens in the working area.
The Type Editor shows the **Attributes** tab.
- Click in the **Name** field of an empty white column in the **Attributes** tab.
This allows you to enter the new attribute name.
- Enter the following properties in the respective property fields:

Property	For Attribute Serial Number	For Attribute Last Revision
Name	SerialNumber	LastRevision
Label	Serial Number	Last Revision Date
Help Message	The serial number of a switch	Date of the last revision of this switch
Data type	String	DateTime
String Length	1000	
Scope	ALL	ALL
Name space	cim	cim
Internal	No	No
DoNotCopy	No	No
NercProfile	No	No

- Leave the other property fields empty.

- Sometimes you can access list boxes in a property field to get a selection list with valid values (for example, the property **Data Type** provides a list box with valid data types).



[IMMOP009-301008-enUS-01.tif]

Figure 2-29 Example of a List Box with a List of Valid Values

Saving Your Changes

- Select **File > Save** from the menu bar or click the toolbar button **Save** to save your changes.

The message **Switch saved successfully** is displayed in the status bar of the Information Model Management.



Note

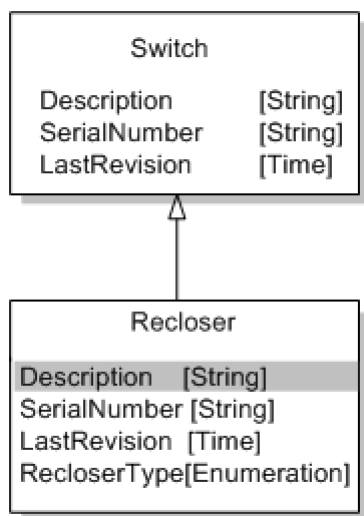
Existing instances of the type Switch do not get attribute values for the newly added attributes.

- View the derived types Breaker and DisconnectSwitch again.
You see that the new attributes you defined for the type Switch have been inherited automatically.

2.7 Adding the New Type Recloser and an Instance of This Type

The goal of this task is to enter the new type Recloser with all its attributes and references. The type Recloser is modeled as a derived type of the type Switch. This means that all attributes of the type Switch are inherited by the type Recloser.

The new additional attribute of the type Recloser is enumRecloserType. This attribute is modeled as an enumeration.



[IMMOP012-160108-enUS-01.tif]

Figure 2-30 Inheritance of Recloser

In a later step of this task, you also add an instance of this type to your new power line Bay Line A Tutorial.

Adding the New Type Recloser

- Select **View > Types** in the Menu Bar or click the appropriate button in the Toolbar to display the type tree in the tree view.
- Browse to the type **Switch**.
The path of the type Switch is:
Domain/Naming/FieldResource/PowerSystemResource/Equipment/ConductingEquipment/Switch
- Right-click the type name **Switch**. The context menu of the tree opens.
- Select **New Type** from the context menu.
The **General** tab of the Type Editor opens in the working area for the new type.
- Enter the following attribute values into the respective fields of the General tab:

Attribute	Value
Name	Recloser
Label	Recloser
Help Message	Circuit breaker, which is able to reclose after having tripped due to a fault.
Name space	sp5
Abstract Type	No
Nerc Profile	No

- Leave the other property fields empty.

**Note**

For additional information on the attributes, refer to [3.5.1.4 Attributes Tab](#).

Name	Recloser
Label	Recloser
Help Message	Circuit Breaker, which has the capability to reclose after having tripped due to a fault.
Rdf Name	
Namespace	cim
Scope	
Graphic Sheet	
Abstract Type	No
Rdf Default Name	
CIM children	
Nerc Profile	No

[IMMOP010-301008-enUS-01.tif]

Figure 2-31 General Tab of the Type Editor for a New Type

**Note**

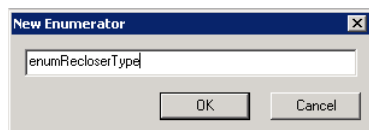
By default, the type name is used for the type label. Re-enter the type label only when it is different from the type name.

Creating a New Enumeration Called enumRecloserType

Enumerations can be used to predefine valid entries or values for a type attribute. An enumeration consists of a number and a related ID. The ID is a textual description of the entry or value.

- Select the **Enumerators** tab in the **Type Editor**.
The Enumerators tab opens.
- Select the **EnumBreakerType** in the **Inherited Enums** list.
- Click the button **New Enum** to create an enumeration.
The name entry dialog opens.

- Enter the name for your new enumeration (enumRecloser Type) in the name dialog.

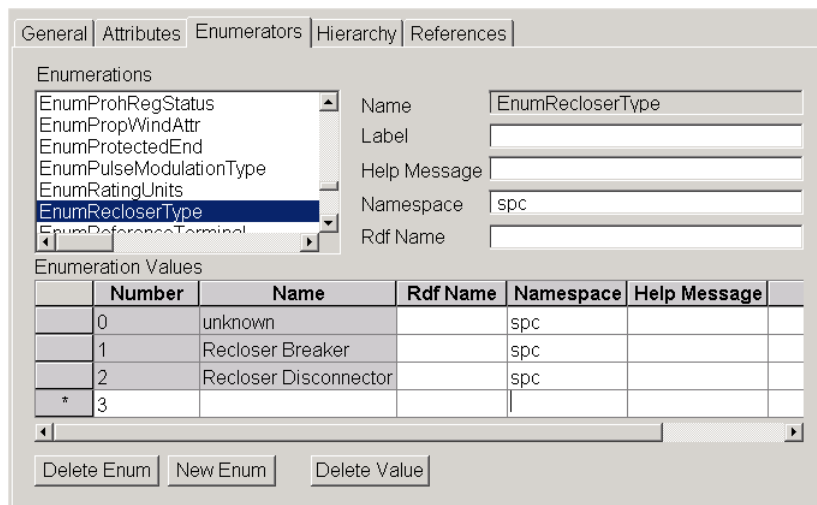


[IMMOP011-301008-enUS-01.tif]

Figure 2-32 Name Entry Dialog

- Click the **OK** button.
Your new enumeration is displayed in the Enums List.
- Enter a description for your new enumeration into the Description field on the Enumerators tab.
- Enter *One of several types of reclosers*.
- Enter the valid values for this enumeration into the **Enumeration** Entries Table on the **Enumerators** tab.
- The following table shows which values you must enter into the Name and name space fields of the Enumeration Entries Table. The **Number** fields representing the actual enumeration ID are filled in automatically.

Number	Name	Name space
0	unknown	sp5
1	Recloser Breaker	sp5
2	Recloser Disconnecter	sp5



[IMMOP012-301008-enUS-01.tif]

Figure 2-33 New Enumeration Entries

- Save the changes you made so far. Select **File > Save** from the menu bar or click the toolbar button **Save**.
A short message confirming the save is displayed in the IMM Status log.
The new type Recloser is now displayed in the type tree.

**Note**

You inserted the new type **Recloser** as a subtype of **Switch**. This defines a parent-child relationship between **Switch** and **Recloser** with **Switch** being the parent type and **Recloser** being the child type.

Adding Attributes to the New Type

- Click the type **Recloser** in the tree view.
The **Type Editor** opens in the working area.
- Select the **Attributes** tab.
The **Attributes** tab of the Type Editor displays. You can see the attributes inherited from the parent type **Switch**.
- Enter the new attribute *RecloserType*. Enter the property values shown in the following table:

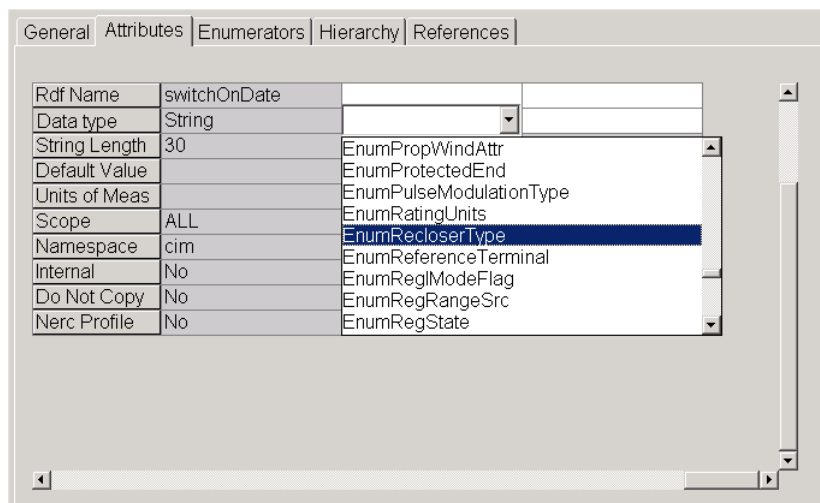
Property	Value
Name	RecloserType
Label	Recloser Type
Help Message	Indicates the type of switch that is operated as a Recloser
Data type	RecloserType ¹
Default Value	unknown
Scope	ALL
Name space	sp5
Internal	No
Do Not Copy	No
Nerc Profile	No

1. Your new enumeration.

- Leave the other property fields empty.

**Note**

Now there is a new data type available. The new data type is the enumeration **RecloserType** that you have created. Enter a default value, when you use an enumeration data type.



[IMMOP013-301008-enUS-01.tif]

Figure 2-34 New Data Type Recloser

- Save the changes you made so far. Select **File > Save** from the menu bar or click the toolbar button **Save**. A short message confirming the save is displayed in the status bar of the Information Model Management.

Opening the Instance View and the Previously Defined Job

- Select **View -> Instances** from the Menu Bar select or click the appropriate toolbar button to open the instance tree in the tree view.
- Open the previously defined job.

Deleting the Circuit-Breaker CB of the Instance Bay Line A Tutorial

Since your new power line Bay Line A Tutorial needs a recloser instead of a standard circuit-breaker, replace the circuit-breaker instance under Bay Line A Tutorial with a recloser instance.

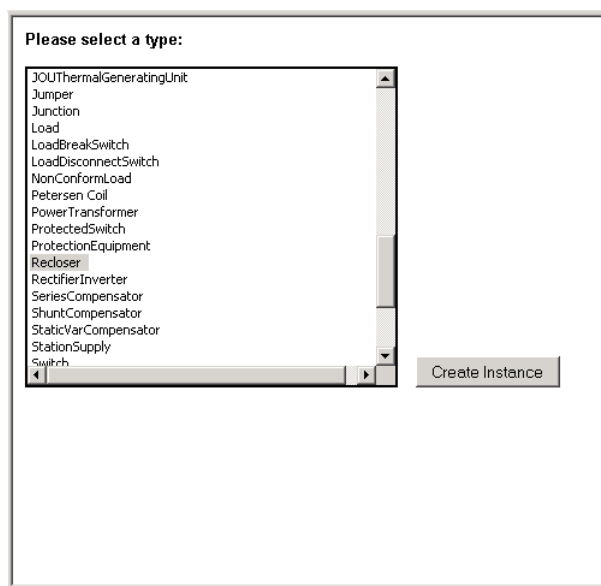
- In the instance tree, click the instance name **CB** under **Bay Line A Tutorial**. This is the circuit breaker instance of the bay you created previously.
The Single-Instance Editor opens in the working area. By default, the Single-Instance Editor shows the **Attributes** tab.
- Right-click the instance name **CB** on the tree view.
The context menu of the tree opens.
- Select **Delete** from the context menu.
A message window asking you to confirm the deletion opens.
- Click the button **OK**. The instance CB under Bay Line A Tutorial is deleted.
A message confirming the deletion is displayed in the status bar of the Information Model Management
- Refresh the instance tree if necessary.

Creating a New Instance of the Type Recloser

Now go ahead and add a new instance of the type Recloser.

- Right-click the instance name **Bay Line A Tutorial** on the tree view.
The context menu of the tree opens.
- Select **New** from the context menu of the tree.
The **Type Selection** dialog opens in the working area.

- Select the type **Recloser** from the type List on the Type Selection dialog.



[IMMOP014-301008-enUS-01.tif]

Figure 2-35 Type Selection Dialog

- Click the button **Create Instance** on the Type Selection Dialog.
This creates an instance of the selected type. The Single-Instance Editor opens in the working area showing the **General** tab.
- Enter the instance name *ReCI* into the **Name** field on the **General** tab.
- Select the **Attributes** tab.
The **Attributes** tab of the Single-Instance Editor opens.
- Enter the attribute values for Your new instance. Enter the values shown in the following table:

Attribute	Value
Description	New Recloser
Serial Number	3333
Last Revision Date	Click the field and enter the current date
Recloser Type	Recloser Breaker

**Note**

The description can only be modified in the **General** tab.

**Note**

By default, the value of the attribute **Recloser Type** is set unknown. Replace it with the value shown in the previous table.

Attributes	Value
In Transit Time	0
Message Configuration	Default
NA Flag	False
Network Mapping Sense	+ (1) - Map with normal sense of status
Nominal kV	
Normal Open	False
Phases	ABC
Protected End	Neither end is protected
Rating Units	MVA
Recloser Time	
RecloserType	Recloser Breaker
Reference Terminal	Recloser Breaker
Reliability Index Zone	Recloser Disconnecter
	unknown

Help Options

[IMMOP018-160108-enUS-01.tif]

Figure 2-36 Attribute Values of the New Instance

- Save the changes you made so far. Select **File > Save** from the menu bar or click the toolbar button **Save**. A short message confirming the save is displayed in the status bar of the Information Model Management.

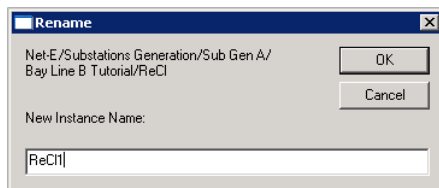
**Note**

In a real data entry session, the newly entered circuit breaker with automatic reclosing function would require additional subordinate instances. At least, a digital measurement for the switch position and an analog value must be added. Compare the new instance ReCI with an existing circuit breaker instance to see which subordinate instances are required. For example, view the circuit breaker with the instance path Network/Substations/Sub A/Bay Line A1/CB. For simplification reasons, the definition of further subordinate instances is omitted.

Renaming an Existing Instance

In some cases, it is necessary to rename an instance.

- Browse to the instance **ReCI** on the tree view.
The path of this instance is:
Network/Substations/Sub Gen A/Bay Line A Tutorial/ReCI
- Right-click the instance name **ReCI** on the tree view.
The context menu of the tree opens.
- Select **Rename** from the context menu.
A message window asking for confirmation opens.
- Click the button **OK**.
The **Rename** dialog opens.



[IMMOP015-301008-enUS-01.tif]

Figure 2-37 Rename Dialog

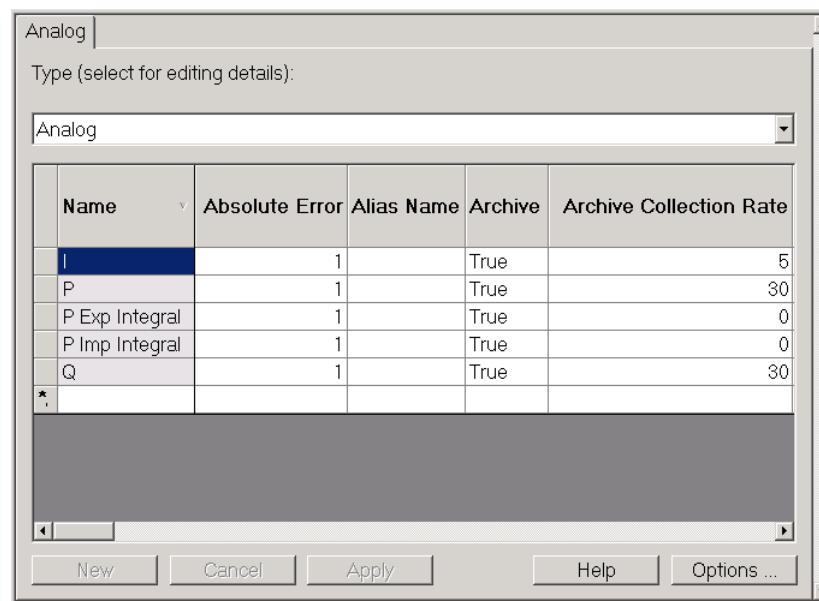
- Enter the new instance name *ReCI1* into the **New Instance Name** field on the **Rename** dialog.
- Click the button **OK**.

The instance ReCI is renamed to **ReCI1**.

2.8 Modifying Instances Using the Table Editor

Modifying Instances Using the Table Editor

- Select the instance **Bay Line A Tutorial** on the tree view.
The Single-Instance Editor opens.
 - Switch from single view (Single-Instance Editor) to table view (Table Editor) by clicking the appropriate toolbar button.
The Table Editor opens and displays all subordinated instances of the selected instance.
 - A type list box on top of the table allows you to select a type of subordinate instances. Based on this selection only instances of this type or a subtype of this type are shown. The columns are the attributes of the selected type.
- The following features are available to customize the appearance of the table:
- Right-click the column header of the table.
The Column menu opens. Select the names of the columns to be made unavailable. To make these columns visible, perform this procedure again.
 - Click the button **Options...** on the lower right corner of the Table Editor to open the Options dialog and set the respective display option.
 - Edit some attribute values.



[IMMOP016-301008-enUS-01.tif]

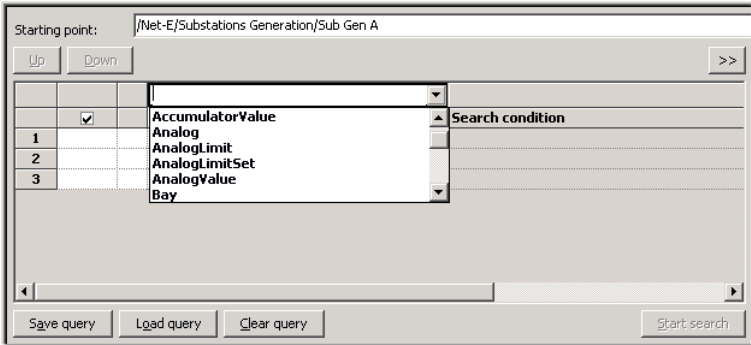
Figure 2-38 Table Editor Showing Attributes for Instances Based on a Particular Type

2.9 Modifying Instances Using the Bulk-Instance Editor

The goal of this task is to edit selected attributes of a set of instances at once. This set of instances is defined by a query.

Defining a Query

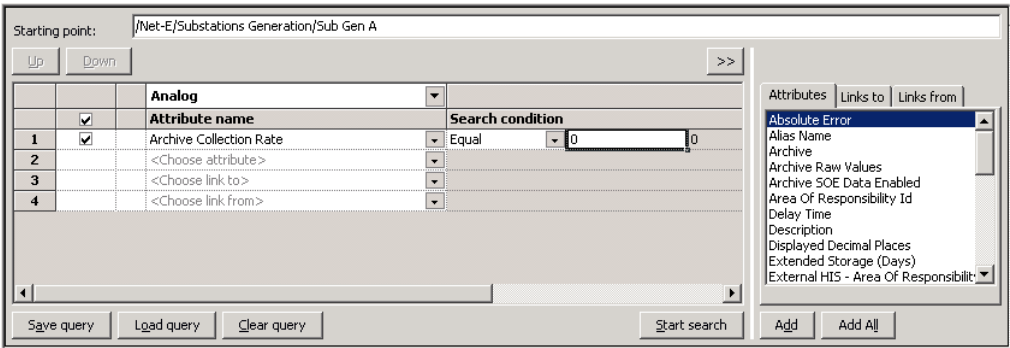
- Select the desired starting point for the search by drag and drop from the tree view, in this example: **Net-E/Substations Generation/Sub Gen A**.
- In the header of the **Query Designer** table select the desired type.



[IMMOP017-301008-enUS-01.tif]
Figure 2-39 Type Selection List Box

The **Property** table shows a list of all attributes for this type.

- Select an attribute and click the button **Add**.
The attribute is added to the **Query Designer** table.
- In the column **Search criteria** enter the search criterion.



[IMMOP018-301008-enUS-01.tif]
Figure 2-40 Query Definition

- Click the button **Start search** to display the search results in the **Result** Table.

Starting point: /Net-E/Substations Generation/Sub Gen A

Up Down >>

	Attribute name	Search condition
1	Archive Collection Rate	Equal 0 0
2	Extended Storage (Days)	0 0
3	Message Configuration	
4	Field Resource Subtype	
5	Measurement Type	Non-specific
6	Archive	
7	<Choose attribute>	
	<Choose link to>	

Save query Load query Clear query Start search Add Add All

Archive Collec...	Extended Stor...	Message Confi...	Field Resource Subtype	Measurement Type	Archive
1	0	Default	PowerSystemResource	Non-specific	☒
2	0	Default	PowerSystemResource	Non-specific	☒
3	0	Default	PowerSystemResource	Non-specific	☒
4	0	Default	PowerSystemResource	Non-specific	☒

Save

[IMMOP019-301008-enUS-01.tif]

Figure 2-41 Search Results

Saving a Query

- Define a query as described in the previous section.
- Click the button **Save Query**.

A display opens where you can enter a name and description for the query.

Save query

	Name	Description
1	Query1	

Save query Cancel

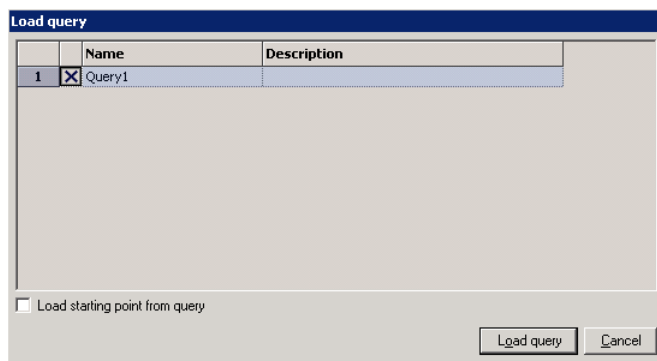
[IMMOP020-301008-enUS-01.tif]

Figure 2-42 Save Query

- Click the button **Save query** to save the query under the specified name.
This query is now available to all users on all clients.

Loading a Query

- Click the button **Load Query**.
- A display showing a list of all defined queries opens.



[IMMOP021-301008-enUS-01.tif]

Figure 2-43 Load Query

- Select a query with a mouse click.
- Select the option button **Load starting point from query** if you want to load also the starting point of the query. Click the button **Load query** to load the query data into the Query Designer table.

Modifying an Instance

- Define a query and execute it to display the desired data in the **Result** table as described.
- Click the field to be modified in the **Result** table. Depending on the data type of the selected field, you can either enter a new value using the keyboard, set a check box, or select from a range of values.
- If you want to enter the same value into several fields displayed below or beside each other, you can do so by entering the value into 1 cell, clicking the cell with the left mouse button, and then dragging the lower right corner across the fields, where the value shall be copied to. The same also applies to the setting of check boxes and the selections made in list boxes.

Analog				
	Archive Collect...	Extended Stor...	Message Confi...	File
1	0	0	Default	Poi
2	10	2	Default	Poi
3	10	2	Default	Poi
4	10	0	Default	Poi
5	10	0	Default	Poi
6	10	0	Default	Poi
7	10	0	Default	Poi
8	30	0	Default	Poi
9	10	0	Default	Poi
10	30	0	Default	Poi
11	30	0	Default	Poi

[IMMOP022-301008-enUS-01.tif]

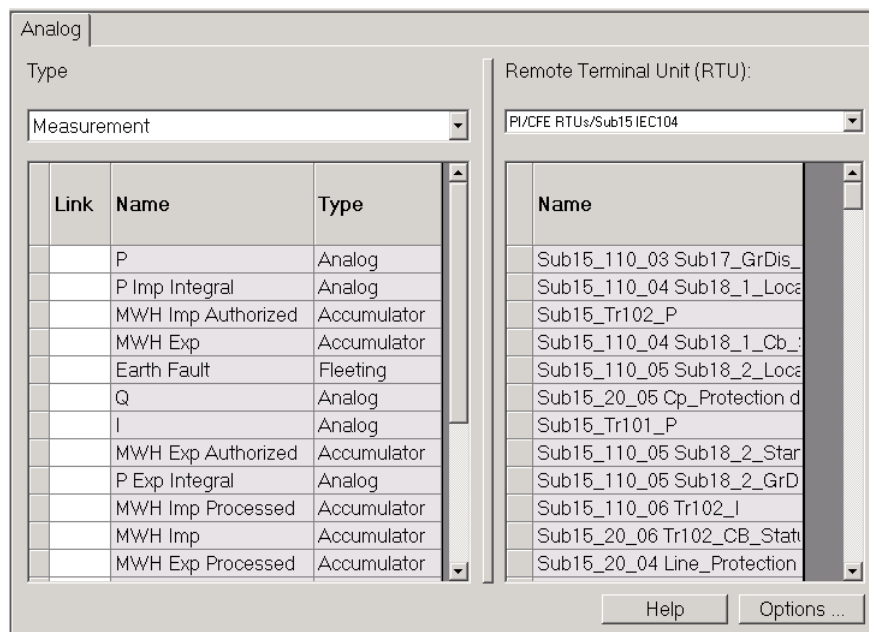
Figure 2-44 Modifying Several Instances Via Drag and Drop

2.10 Linking Instances to CFE Data Points

The goal of this task is to create a CFE data point and all necessary CFE links for your new power line automatically.

Linking Instances to CFE Data Points

- Select the instance **Bay Line A Tutorial** on the tree view.
- Select **View > Instance Editors > CFE** from the menu bar.
The CFE communication editor opens on the working area of the Information Model Management.
- After the network instances of the new bay were loaded into the left table, select the desired RTU in the Remote Terminal Unit (RTU) list.
- Select the network instance for which you want to create CFE data points and the necessary links.
- Right-click the selected network instance.
The row menu opens.
- Click the menu item **Create** on the row menu.
The CFE Communication Editor creates the following items in the RTU (right) table:
 - Instance name of the CFE data point.
The name is derived from the name of the network instance.
 - Link 1 (attribute column **SendsTo**) MeasurementValue to the network instance for the monitoring direction.
 - Link 2 (attribute column **ReceivesFrom**) MeasurementControl to the network instance for the control direction (if applicable).
- Maximize the Information Model Management to show all attribute columns.
If a connected data point exists, the connection is indicated in the column Link. For example, in case of data points connected to CFE, the column Link shows **CFE**. For data points connected to ICCP, the column Link shows **ICCP**. Calculated measurements cannot be connected to a data point. In that case, the column Link shows -.



[IMMOP023-301008-enUS-01.tif]

Figure 2-45 Creating Links between Network Instances and CFE Data Points

2.11 Configuring ICCP Data Objects

The goal of this task is to copy a substation and connect its measurements with a remote control center.



Note

This task is only an example for the typical use of the ICCP communication editor. For further information on the ICCP communication editor, refer to the [ICCP Reference Guide](#).

Configuring ICCP Data Objects

- Create a substation Sub F by Copy & Paste of an existing one (for example, Sub E).
- Select the new substation (Sub F).
- Select **View > Instance Editors > ICCPClientBlock1and2** from the menu bar.
The ICCP communication editor opens on the working area of the Information Model Management.
- After the Network instances of the new substation were loaded into the left table, select a Transfer Group from the **ICCP Client Block 1 And 2 Container** list box (for example, ICCP/RC East/Client Block1_2).
- Select the network instance for which you want to create a Transfer data point and the necessary link.
- Right-click the selected network instance.
The row menu opens.
- Select the menu item **Create** on the row menu.
The ICCP communication editor creates the following instances of the corresponding type under the selected Transfer Group:
 - Instance name of the Transfer data point. The name is derived from the name of the network instance. After the creation, you can rename it manually.
 - Link between the transfer data object and the network instance (measurement).



Note

For the ICCP client and server objects, a maximum name length of 32 characters is supported. Name mapping rules can be defined in a corresponding XML file (Path: ..\CurrentProject\UI\CommunicationEditor\mapping.xml):

- **GeneralPrefix**
Prefix for all new created names.
- **UseInstNameLength**
The path of the linked instance has a certain hierarchy. Each single instance of this hierarchy can have a name with a certain length. This property cuts the single names to a certain length using UseInstNameLength and uses the result to create the instance name (for example, UseInstNameLength = 3: Network\Substation\Sub A\Bay Gen 11 results in Net\Sub\Sub\Bay).
- **CutHierarchyLevel**
Cuts the path (for example, CutHierarchyLevel = 3: Network\Substation\Sub A\Bay Gen 11 results in Substation\Sub A\Bay Gen 11).
- **MaxLength**
For each automatically created name longer than the MaxLength, a dialog opens and asks you for the proper name. The name is not cut automatically. This maximum length includes the GeneralPrefix that is added to each name.

Analog

Type

Measurement

Link	Name	Type
	P	Analog
	P Imp Integral	Analog
	MWH Imp Authorized	Accumulator
	MWH Exp	Accumulator
	Earth Fault	Fleeting
	Q	Analog
	I	Analog
	MWH Exp Authorized	Accumulator
	P Exp Integral	Analog
	MWH Imp Processed	Accumulator
	MWH Imp	Accumulator
	MWH Exp Processed	Accumulator

ICCP Client Block 1 And 2 Container

ICCP/RC East/Client Block1_2

Name	ReceiveValueFor
Sub_A_Bay_11_I	Net-E/Substations Generat
Sub_A_Bay_11_P	Net-E/Substations Generat
Sub_A_Bay_11_Q	Net-E/Substations Generat
Sub_A_Bay_11_CB_S	Net-E/Substations Generat

Help Options ...

[IMMOP024-301008-enUS-01.tif]

Figure 2-46 ICCP Client Data for ICCP Block 1 and 2

2.12 Activating the Data Changes

The goal of this task is to merge your data modifications from the currently loaded job into the root dataset to make the data modifications available for the runtime system.



Note

The Power Control Center runtime system must be running before an activation can be performed.

In the time-based job model, activation is only possible in the scheduled order of all jobs. (This also applies for undo).

Activating the Data Changes

- Click the job management toolbar button.
The **Job Management** user interface opens.
- Select the created job on the job management user interface.
- Right-click the selected job.
The job context menu opens.
- Click the menu item **Validate** to start the validation process of the selected job (this step is optional).
For the possible validation modes see section [Validate](#) in [3.2.1.4 Job Management UI Tab - Job Context Menu](#).
After the validation process has been finished successfully, the job status is set to **Validated**, otherwise it is set to **Validation_Failed**. If there are any errors, you can correct the errors by editing the job. The validation messages of the validation process can be analyzed in the **Validation Log** window (see [3.2.11.3 Validation Log](#)).
- When the job status shows **Validated**, right-click the validated job.
The job context menu opens.
- Click the menu item **Prepare Activation** to prepare the activation of the selected job.
- When the job status shows **Prepared**, right-click the prepared job.
The job context menu opens.
- Click the menu item **Activate** to start the job activation process of the selected job.
A confirm message box opens. After clicking the button **YES**, the changes done in the selected job are loaded into the root dataset. If the activation process completes successfully, the job status is set to **Activated**.
- Now you can either undo the job or finish the job by the job context menu (see [3.2.1.4 Job Management UI Tab - Job Context Menu](#)).

After finishing the job, you have activated the data modifications. The data modifications are merged from the selected job to the root dataset to make them available on the runtime system.

The screenshot shows the 'Activation Log' tab in the IMM application. It displays a list of activation logs with columns for Type, User, and Date. Below this, a detailed log shows the steps of the activation process for 'Job: Test1 (RTJOB10)' on 10/29/2008, including postprocessing, committing, applying, preparing, preprocessing, and applying changes for SMMF and RDBMS based ASRs.

Type	User	Date
Do Activation	nbgdom\SPCAAdmin	10/29/2008 10:29:02 AM
Do Preparation	nbgdom\SPCAAdmin	10/29/2008 10:28:35 AM

Severity	Message	Time Stamp
Info1	Job: Test1 (RTJOB10) - Activation completed successful.	10/29/2008 10:29:18 AM
Info1	HOST: DEB21GKC - Changes postprocessed for SMMF based ASRs.	10/29/2008 10:29:18 AM
Info1	HOST: DEB21GKC - Changes committed for all ASRs.	10/29/2008 10:29:17 AM
Info1	HOST: DEB21GKC - Changes applied for SMMF based ASRs.	10/29/2008 10:29:15 AM
Info1	HOST: DEB21GKC - Changes prepared for SMMF based ASRs.	10/29/2008 10:29:14 AM
Info1	HOST: DEB21GKC - Changes preprocessed for SMMF based ASRs.	10/29/2008 10:29:13 AM
Info1	HOST: DEB21GKC - Changes applied to RDBMS based ASRs.	10/29/2008 10:29:12 AM

[IMMOP025-301008-enUS-20.tif]

Figure 2-47 Activation Log

2.13 Exporting and Importing Engineering Data

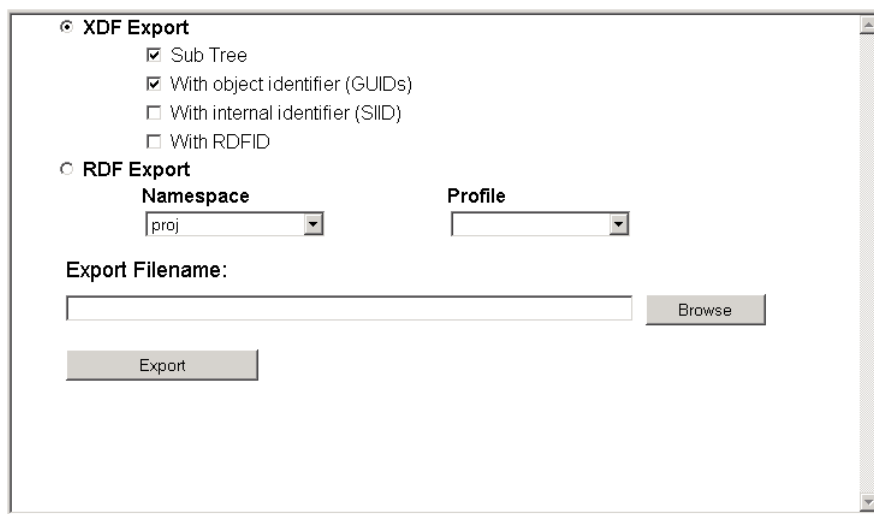
The goal of this task is to export the engineering data of the new power line to an external file. You modify the instances in this external file outside of the Information Model Management to simulate a file from another system. After the modifications, you import the modified file and see its content in the Information Model Management.

Selecting the Instance You Want to Export

- Right-click on the new power line **Bay Line A Tutorial** in the tree view to select the instance. The context menu of the tree opens.
- Select **Export** from the context menu of the tree. The **Export** dialog opens in the working area.

Selecting the Export File

- Enter the path name of a file, in which the export data are to be stored, into the text box **Export Filename** on the Export dialog. For example:
C:\Templates\tutorial.xml
- Alternatively, you can click the **Browse** button. A file selection dialog opens that allows you to select a file interactively.



[IMMRE029-041108-enUS-01.tif]

Figure 2-48 Export Display

- Click the button **Export** on the Export dialog. The export starts. When the export is finished, a message is displayed in the IMM Status log.

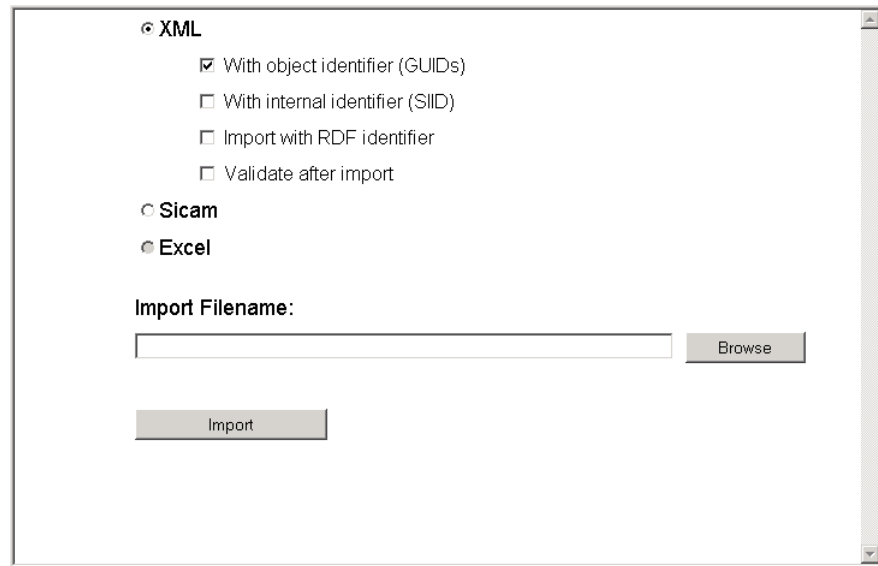
Modifying the XML File

- Open the exported XML file with an ASCII editor.
- Replace the name Bay Line A Tutorial with Bay Line B Tutorial.
- Save the XML file.

Importing the XML File

- If the Information Model Management is still open, switch to the Information Model Management. Otherwise open the Information Model Management.
- Select **File > Import** from the menu bar on the Information Model Management.

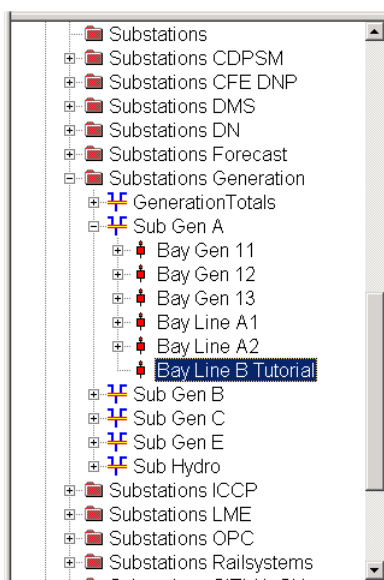
The **Import** dialog opens on the working area:



[IMMOP046-150909-enUS-01.tif]

Figure 2-49 Import Dialog

- Enter the path name of a file, in which the import data are stored, into the field **Import Filename** on the Import dialog. For example:
C:\Templates\tutorial.xml
- Alternatively, you can click the **Browse** button.
A file selection dialog opens that allows you to select a file interactively.
- Click **Import**.
The import process starts. When the import process was finished successfully, a message is displayed in the IMM Status log.
- Click the button **OK**.
A short message confirming the commitment is displayed in the status bar of the Information Model Management.
Now the new line Bay Line B Tutorial is displayed in the instance tree view and can be selected.



[IMMOP028-301008-enUS-01.tif]

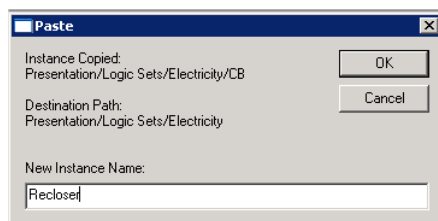
Figure 2-50 New Line Bay Line B

2.14 Modifying the Presentation Logic for the Circuit Breaker with Automatic Reclosing Function

The goal of this task is to distinguish the presentation of a circuit breaker from your new circuit breaker with automatic reclosing function and define a new presentation logic for the circuit breaker with automatic reclosing function. Circuit breakers are represented by squares, circuit breaker with automatic reclosing functions shall be represented by circles. However, most presentation rules of circuit breakers apply for both circuit breaker types. You create the presentation logic based on the presentation logic of the circuit breaker and then apply the necessary modifications.

Copying and Pasting the Presentation Logic of the Circuit-Breaker

- Right-click the instance name **Presentation/Logic Sets/Electricity/CB** on the tree view.
The context menu of the tree opens.
- Select **Copy** from the context menu.
- Right-click the instance **Presentation/Logic Sets/Electricity**.
The context menu of the tree opens.
- Select **Paste** from the context menu.
The **Paste** dialog opens.
- Enter an instance name (for example, *Recloser*) for the new presentation logic instance into the **New Instance Name** field on the Paste dialog.



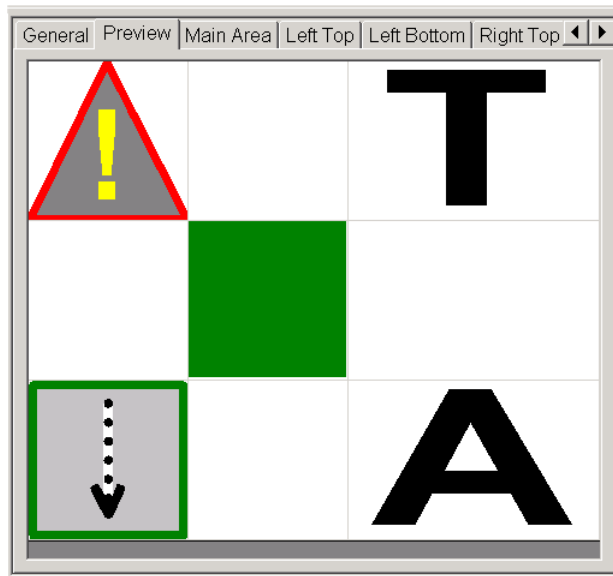
[IMMOP033-041208-enUS-01.tif]

Figure 2-51 Paste Dialog

- Click the button **OK**.
The new presentation logic instance **Recloser** is inserted.
The successful paste is indicated by a short message that is displayed in the IMM Status log of the Information Model Management. The new instance is displayed in the tree view.
- Perform a tree refresh if necessary.

Modifying the Presentation Logic of the Circuit Breaker with Automatic Reclosing Function

- Select the newly created instance **Recloser**.
The **Decision-Table Editor** opens in the working area. The Decision-Table Editor is a special purpose editor used to view and edit properties of presentation logic instances.
- Click the **Preview** tab.
The **Preview** tab of the Decision-Table Editor opens.



[IMMOP029-301008-enUS-01.tif]

Figure 2-52 Example of a Decision-Table Editor View

**Note**

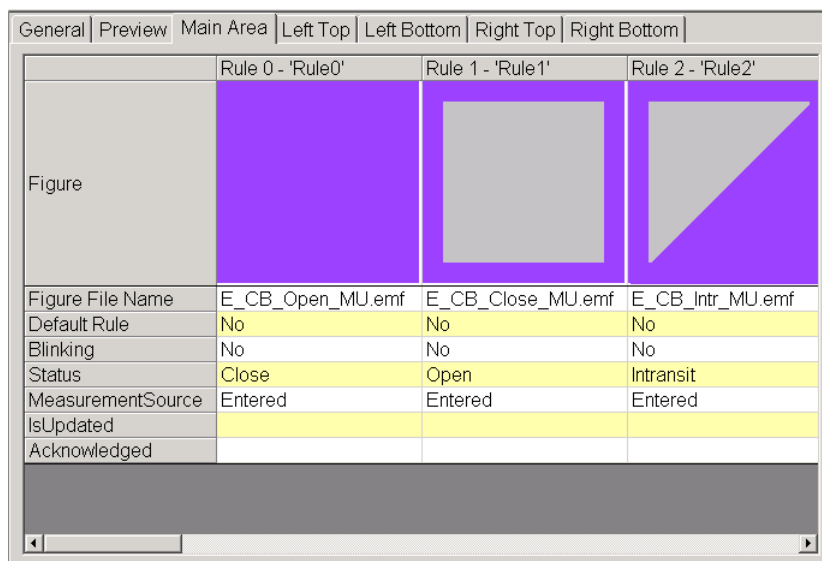
The Preview tab shows how the respective presentation logic instance would be displayed on the runtime UI. The display is divided in different presentation sections each of which shows the presentation of a certain information. Currently 5 presentation sections are used (Main Area, Left Top, Right Top, Left Bottom, Right Bottom).

The content of each presentation section is defined by a decision table. A decision table contains one or more rules to be evaluated for a certain purpose or a certain situation. The entire presentation logic is split into several decision tables. Otherwise too many rules (and therefore choices) would have to be considered at once.

Modifying the Decision Table for the Main Area

- Click the **Main Area** tab on the Decision-Table Editor.

The **Main Area** tab of the Decision-Table Editor opens, showing the decision table associated with the main presentation area.



[IMMOP030-311008-enUS-01.tif]

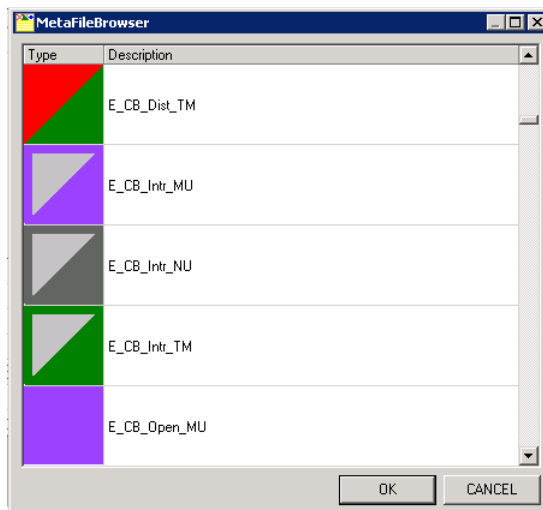
Figure 2-53 Main Area Tab

The **Main Area** tab shows the decision table associate with the main presentation section. Each column contains a rule defined for the respective presentation object. Each rule contains one or more conditions (logical expressions) and the representation options that are used when the particular rule is true. To determine a presentation, the rules and the contained logical expressions are evaluated and the results are combined. The first rule that is logically true (fulfilled), determines the representation of the respective instance. The top line of a rule shows which graphical symbol is used in cased the rule is fulfilled.

**Note**

The shown graphic symbols are determined by links to EMF graphic files. EMF means Enhanced Metafile. EMF is a vector-oriented graphic format that provides the possibility to use, for example, zooming capability in the runtime UI.

- Click the field **Figure File Name** in the column **Rule0**.
The button ... opens in the field **Figure File Name**.
- Click the button
The Graphic File Browser Dialog opens.



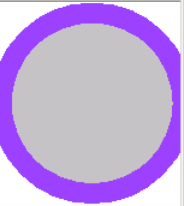
[IMMOP031-311008-enUS-01.tif]

Figure 2-54 Graphic File Browser Dialog

- Double-click the figure **DC_Open_Normal** in the Graphic File Browser dialog.
The Graphic File Browser dialog is closed. The square figure previously shown on the top line of **Rule0** is replaced by the round circle figure.
- Go ahead and repeat the last 2 steps for all rules to replace all squares with the corresponding circle presentations.

Choosing the Default Rule for the WinCC Generic Control Presentation

- Set one of the rules to be the default rule. This determines the presentation that you see on the Preview tab or when you create SCADA single-line diagrams.
- Set Rule3 to be your default rule. Click the **Default Rule** field of **Rule3** and select the value **Yes** from the list box. Make sure that only 1 rule is set to be the default rule.
- Save the changes you made so far. Select **File > Save** from the menu bar or click the toolbar button **Save**. A short message confirming the save is displayed in the status bar of the Information Model Management.

	Rule 0 - 'Rule0'	Rule 1 - 'Rule1'	Rule 2 - 'Rule2'
Figure			
Figure File Name	E_DIS_Open_MU.emf	E_DIS_InE_DIS_MU.e	E_DIS_Close_MU.em
Default Rule	Yes	No	No
Blinking	No	No	No
Status	Close	Open	Intransit
MeasurementSource	Entered	Entered	Entered
IsUpdated			
Acknowledged			

[IMMOP032-311008-enUS-01.tif]

Figure 2-55 Main Area Tab

2.15 Testing Jobs Using the Test System

The goal of this task is to copy the data model of the production system to the test system, to mark a job on the production system Ready to Test, and to transfer it to the test system. After testing, the state of this job shall be set to Tested on the test system and be transferred back to the production system.

Copying the Current Data Model to the Test System

- On the test system, right-click the **RT** root job in the **Job Management UI** tab.
- In the context menu select the item **Synchronize IMM**.

The data model of the connected production system is transferred to the test system.

Setting a Job Ready for Test

- On the production system, select the **Job Management UI** tab.
- Select the **Test State** cell in the line of the desired job.
- From the list box select the item **Ready for Test**.

Job Management UI Production System Change Request List						
Job Name	Planned Activation Time	Job Type	Job State	Reserved by	Test State	Change Rec
Realtime (RT)			View Only			Not Created
TestDoc1	11/26/2008 9:45:00 AM	Public	Editing	nbgdom\SPCAAdmin	Not Tested	Not Created
					Not Tested Ready for Test In Test Tested	

[IMMOP035-041208-enUS-01.tif]

Figure 2-56 Setting the Job Status to Ready for Test

Now this job can be transferred to the test system.

Transferring a Job to the Test System

- In the **Production System** tab of the test system, right-click the desired job.
- In the context menu select the item **Refresh View** to update the test state from the production system and then the item **Import from Production System**.

Job Management UI Production System Change Request List		
Job Name	State	Test State
a	Editing	Ready for Test
b	Editing	Not Tested
c	Editing	Not Tested
d	Editing	Not Tested
Realtime (RT)	View Only	Not Tested

[IMMOP036-041208-enUS-01.tif]

Figure 2-57 Transferring the Job to the Test System

Only jobs with state **Ready for Test** can be transferred.

A display opens where you can specify the target job at the test system.

- Either select an existing job at the test system.
Then the job is imported into the selected job.
Or enter the name for a new job.
Then a new job is created. After the job has been transferred successfully, the state of this job changes to In Test and the job can be tested. If errors occur during the import, these errors are displayed in the **Import Log**.

Marking a Job as Tested

- On the test system, select the **Job Management UI** tab.
- Select the **Test State** cell in the line of the desired job.
- From the list box select the item **Tested**.

Job Manangement UI Production System Change Request List							
Job Name	Planned Activation Time	Job Type	Job State	Reserved by	Test State	Change Req	
Realtime (RT)			View Only			Not Created	
TestDoc1	11/26/2008 9:45:00 AM	Public	Editing	nbgdom\SPCAdmin	In Test	Not Created	
					Not Tested Ready for Test In Test Tested		

[IMMOP037-041208-enUS-01.tif]

Figure 2-58 Setting the Job Status to Tested

Now this job can be transferred to the production system.

Transferring a Tested Job to the Production System

- In the **Test System** tab of the production system, right-click the desired job.
- In the context menu select the item **Refresh View** to update the test state from the test system and then the item **Import from Test System**.

Job Manangement UI Testing System Change Request List		
Job Name	State	Test State
TestDoc1	Editing	Tested
Realtime (RT)	View Only	Not T
		Import from Testing System Refresh View

[IMMOP038-041208-enUS-01.tif]

Figure 2-59 Transferring the Job to the Production System

Only jobs with state **Tested** can be transferred.

A display is opened where you can specify the target job at the production system.

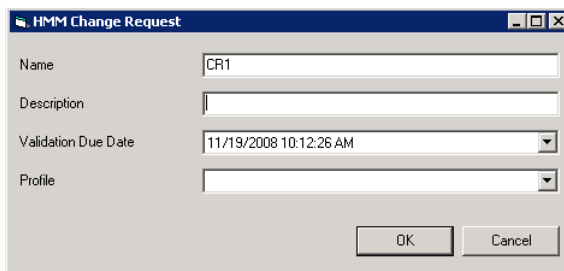
- Either select an existing job at the Production System.
Then the job is imported into the selected job.
Or enter the name for a new job.
Then a new job is created and the data of the selected job of the test system are imported into the new job. If errors occur during the import, these errors are displayed in the **Import Log**.

2.16 Handling Change Requests Within a Hierarchically Organized Control Center System

The goal of this task is to create and send change requests between different control centers, to change the Validation Due Date and the Commissioning date, and then to download and accept, reject or withdraw a change request.

Creating and Sending a Change Request (Second-Level Control Center)

- After you have created and validated a job on a second-level control center, select the item **Is Change Request Required** from the job context menu.
A message in the **IMM Status Log** shows whether a change request is required.
- If a change request is required, select the item **Create Change Request** from the job context menu.
A display opens where you can enter **Name**, **Description**, and **Validation Due Date**, and select an export profile.



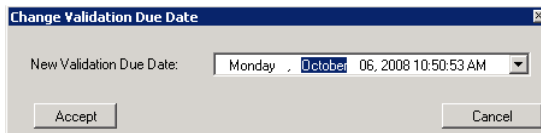
[IMMOP040-041208-enUS-01.tif]

Figure 2-60 Creating a Change Request

- Click the button **OK**.
- Select the **Change Request List**.
The new change request has been added.
- Right-click the new change request and select the item **Send Change Request** from the context menu.
The change request is sent to the first-level control center.

Changing the Validation Due Date

- In the **Change Request List** tab, right-click the desired change request.
- Select the item **Change Validation Due Date** from the context menu.
A display opens where you can select the new date.



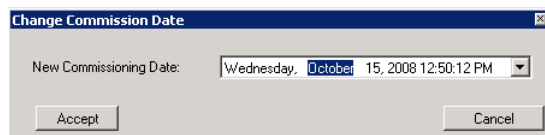
[IMMOP041-041208-enUS-01.tif]

Figure 2-61 Changing the Validation Due Date

- Click the button **Accept** to save the new date and close the display.
The new date is displayed in the respective column.

Changing the Commissioning Date

- In the **Change Request List** tab, right-click the desired change request.
 - Select the item **Change Commissioning Date** from the context menu.
- A display opens where you can select the new date.



[IMMOP042-041208-enUS-01.tif]

Figure 2-62 Changing the Commissioning Date

- Click the button **Accept** to save the new date and close the display.
- The new date is displayed in the respective column.

Downloading a Change Request (Second-Level Control Center)

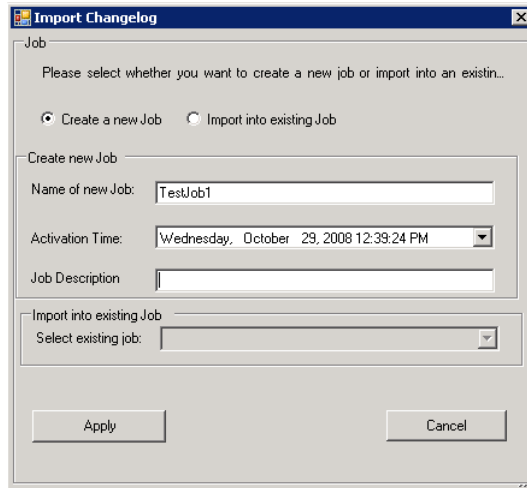
- In the **Change Request List** tab, right-click the desired change request.
 - Select the item **Send Download Request** from the context menu.
- The change request is downloaded to the second-level control center.

Accepting a Change Request (Second-Level Control Center)

- In the **Change Request List** tab of a second-level control center, right-click the desired change request.
 - Select the item **Send Change Request Response > Accept Change Request** from the context menu.
- The state of the change request changes to **Accepted** and this decision is sent to the first-level control center.

Importing a Change Request (Second-Level Control Center)

- In the **Change Request List** tab, right-click the desired change request.
 - Select the item **Import Change Request** from the context menu.
- A display opens where you can select the job, into which the change request shall be imported.



[IMMOP043-041208-enUS-01.tif]

Figure 2-63 Importing the Change Request

- If you select the option button **Create a new Job**, enter name, activation time, and a description for the new job. If you select the option button **Import into existing Job**, select a job from the list box **Select existing job**.
- Click **Apply** to start the import.
The data changes are imported.

■

3 User Interface Description

3.1	Information Model Management	76
3.2	General Engineering Applications	104
3.3	Instance Editors	139
3.4	Special Editors	164
3.5	General Editors	172
3.6	Wizards	200

3.1 Information Model Management

3.1.1 Structure of the Information Model Management

The Spectrum Power Information Model Management allows you to view, enter, and customize types and instances of power system components.

IMM represents an application framework for engineering applications. It is composed of 6 basic areas of distinct functionality for all interactive elements provided.

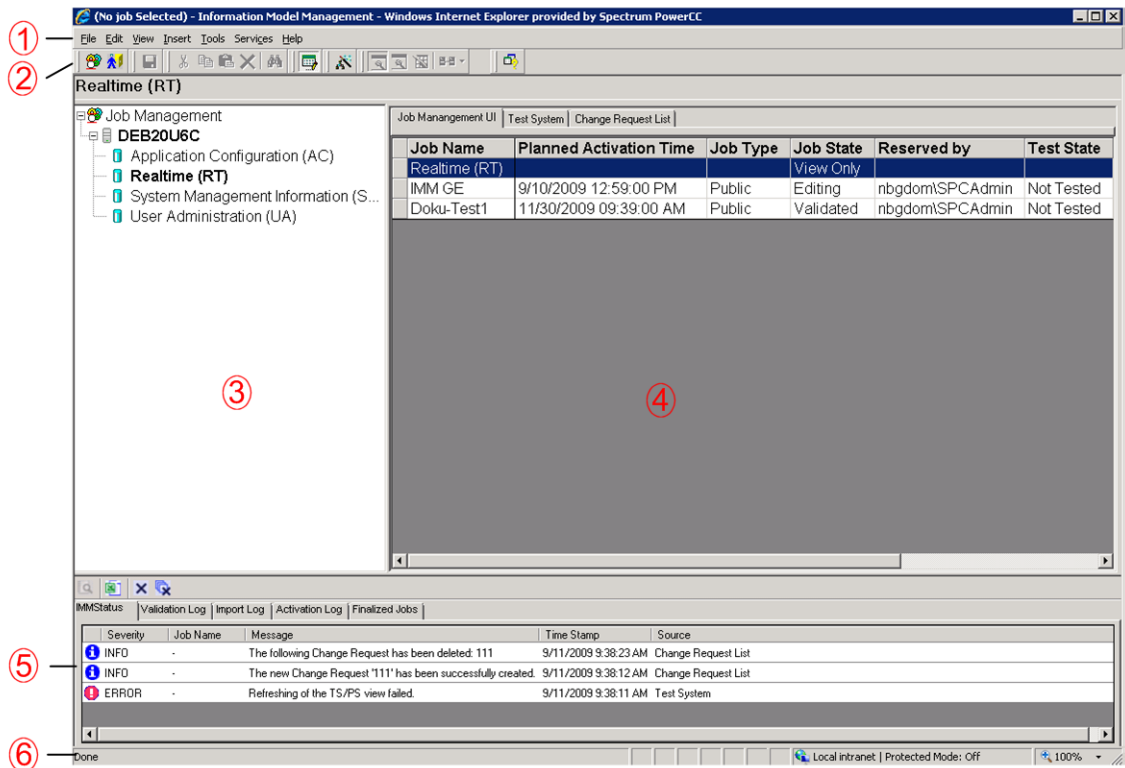
These areas are:

- Menu bar
- Toolbar
- Tree view
- Working area
- Log section
- Status bar

Starting the Information Model Management

The Information Model Management is started in the following way:

- Selecting the **Spectrum PowerCC > Information Model Management** from the **Start** menu of the Windows task bar, or by
- Selecting **IMM > Information Model Management** from the Power 5 Application Toolbar (PAT).



[IMMRE050-210909-enUS-01.tif]

Figure 3-1 Information Model Management Framework

- (1) Menu bar
- (2) Toolbar
- (3) Tree view
- (4) Working area
- (5) Log section
- (6) Status bar

3.1.2 Menu Bar

3.1.2.1 Overview

Menu items are enabled or disabled depending on various conditions imposed by the user role and the current operational situation.

The following menus are included in the menu bar of the Information Model Management:

- File
- Edit
- View
- Insert
- Tools
- Services
- Help

3.1.2.2 File

The **File** menu of the Information Model Management is the interface by which you apply data modifications, starts data import/export processes, and receive information about the session.

The File menu provides the following menu items:

- **Save**
- **Import**
- **Export**
- **Session Properties**
- **Exit**

Detailed descriptions of the menu items are provided in the subsequent paragraphs.

Save

Applies data modifications performed to the currently selected job. The data modifications are stored into the current job. After the data save is completed without errors, an appropriate message is displayed in the status bar.

Import

Opens the **Import** dialog in the working area (see [3.2.6 Import Dialog](#)).

Export

Opens the **Export** dialog in the working area (see [3.2.7 Export Dialog](#)).

Session Properties

Opens the **Session Properties** tab in the working area to view information about the currently loaded job.

Exit

Closes the Information Model Management.

3.1.2.3 Edit

The **Edit** menu provides editing tools for types and instances. Due to the nature of these tools, the included menu items are enabled only when a job was loaded and the corresponding modify permissions are set respectively. Also, different menu items are provided depending on the selected data view.

The Edit menu provides the following menu items:

- **Cut Instance**
- **Copy Instance**
- **Paste**
- **Delete Instance**
- **Rename Instance**
- **Search**

Detailed descriptions of the menu items are provided in the subsequent paragraphs.

Cut Instance

Marks the selected instance for cut.

After the menu item **Cut Instance** was selected, the Cut Instance dialog pops up to notify you of the impending marking operation. When you proceed, the selected instance and all its subordinate instances are marked for a subsequent paste operation.

When the paste operation is invoked, the marked instances are relocated within the instance overview tree. For further information on this procedure, refer to the description of the menu item **Paste**.



Note

When the marked instance has subordinate instances, they are implicitly marked, so that the entire subtree is relocated.

Copy Instance

Marks the selected instance for copy.

The selected instance is marked for a subsequent paste operation. When the paste operation is invoked, a duplicate of the marked instance is inserted into the database and into the instance overview tree at the selected destination.



Note

When the marked instance has subordinate instances, they are implicitly marked, so that the entire subtree is duplicated.

For further information on this procedure, refer to the description of the menu item **Paste**.

Paste

Relocates or duplicates a marked instance and all its subordinate instances and retains all links between subordinate instances or the marked instance and subordinate instances.

Restoring is performed when an instance was marked for cut. During restoring, the marked instance (and, if available, all subordinate instances) are deleted and reinserted under another instance. Duplication is performed when an instance was marked for copy. In this case, the marked instance (and all subordinate instances) remain and a duplicate of them is inserted under another instance.

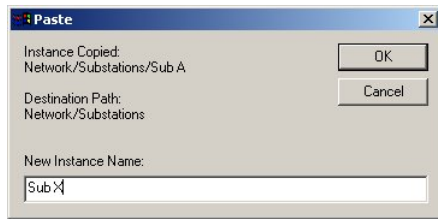
For further information on marking operations, refer to the description of the menu items **Cut Instance** and **Copy Instance**.

In both cases (restoring and duplication), enter a name for the reinserted instance. This is done in the Paste dialog. The Paste dialog pops up after the menu item **Paste** was selected.



Note

The entered name applies only to the reinserted instance itself. Subordinate instances of that instance retain their original names.



[IMMRE003-160108-enUS-01.tif]

Figure 3-2 Paste Dialog

After all concerned instances were reinserted, the instance overview tree is updated to show the new instance hierarchy.



Note

Paste after a Copy operation duplicates all concerned instances, all links between duplicated instances and attributive links between duplicated and not duplicated instances. None-attribute links between duplicated and not duplicated instance are not copied.

Paste after a Cut operation relocates all concerned instances and all their links

After the Paste function is completed, an appropriate message is displayed in the status bar.

Delete Instance

Deletes a selected instance.

After the menu item **Delete** was selected, the Delete Instance dialog pops up to notify you of the impending activity. When you proceed, the selected instance and its subordinate instances are deleted and the instance overview tree is updated.

After the deletion is completed, an appropriate message is displayed in the status bar.

Rename Instance

Renames a selected instance.

After the menu item **Rename** was selected, the Rename dialog opens. In this dialog, you can enter a new instance name. After clicking the button **OK**, a short message confirming the renaming is displayed in the status bar of the Information Model Management.

Search

Searching for instances in the instance tree. This function supports you in managing the large amount of data that is manipulated in the Information Model Management.

After the menu item **Search** was selected, the Search Instance dialog is opened in the working area.

3.1.2.4 View

The menu items provided by the View menu change your view of the dataset. The commands related with the menu items can affect both the availability of types and instances (for example: changing from types to instances or the other way round) and the scope of the selected types or instances (for example: by applying a filter). Due to the nature of these menu items, some of them are enabled only when a dataset was loaded.

The **View** menu provides the following menu items:

- **Instances**
- **Instance Editors**
- **Types**
- **Tree Refresh**
- **Set Tree From Here**
- **Reset Tree**
- **Logs**
- **Change Log**

Detailed descriptions of the menu items are provided in the subsequent paragraphs.

Instances

Changes to the instance view.

This data view allows you to view, create, and edit instances.

Instance Editors

The menu **Instance Editors** contains the following submenu items:

- **Single View**
Shows the Single-Instance Editor on the working area.
- **Table View**
Shows the Table Editor on the working area.
- **Bulk Editor**
Shows the Bulk Editor in the working area.
- **CFE**
Opens a table editor in the working area. This table editor allows you to enter CFE data points.
- **ICCPClientBlock1and2**
Opens a table editor in the working area. This table editor allows you to enter ICCP data objects for the ICCP Client Blocks 1 and 2.
- **ICCPClientBlock5**
Opens a table editor in the working area. This table editor allows you to enter ICCP data objects for the ICCP Client Block 5.
- **ICCPServerBlock1and2**
Opens a table editor in the working area. This table editor allows you to enter ICCP data objects for the ICCP Server Blocks 1 and 2.
- **ICCPServerBlock5**
Opens a table editor in the working area. This table editor allows you to enter ICCP data objects for the ICCP Server Block 5.

Types

Changes to the type view.

This data view allows you to view, create, and edit power system component types. A type tree is displayed in the tree view.

Tree Refresh

This item refreshes the content of the tree view.

In some cases, a user data view is not up-to-date. For example, when more than 1 user is working with the same dataset or when a new instance is created and saved, it may happen the notification of all Information Model Management clients takes a while. This may be the cause that the shown tree is outdated. Clicking **Tree Refresh** causes the content of the tree view to be updated.

Set Tree from Here

Clears the tree view temporarily. Only the selected type and its entire subtree are displayed in the tree view.

Reset Tree

Undo the action Set Tree From Here.

Logs

Opens a submenu with items for selecting the different log types.

For additional information on the available log types refer to [3.1.6 Log Section](#)

Change Log

This item opens a window where the change log of a job can either be viewed or exported (see [3.2.3 Change Log](#)).

3.1.2.5 Insert

The Insert menu provides menu items for creating new types and instances. The provided menu items may be disabled depending on the selected data view (refer to the previous chapter for further information).



Note

It is possible to edit in the root dataset (for example, real-time dataset) when there was no job loaded yet.

The **Insert** menu provides the following menu items:

- **New Type**
- **New Instance**

Detailed descriptions of the menu items are provided in the subsequent paragraphs.

New Type

Opens the **Type Editor** on the working area to create a type. This menu item is enabled only if the type view has been selected earlier.

New Instance

Opens the Type Selection on the working area to select a type for creating an instance. This menu item is enabled only if the instance view has been selected earlier.

3.1.2.6 Tools

The Tools menu provides general-purpose tools for editing, validating, and obtaining engineering data and information on engineering data. Due to the nature of these tools, the included menu items are only enabled if the corresponding modify permissions are set.

The Tools menu provides the following menu items:

- **Graphics Editor**
- **Template Editor**
- **EA Wizard**
- **Script Definition Wizard**
- **Calculation Wizard**
- **Script Wizard**
- **HTML Reports**
- **Excel Reports**

Graphics Editor

Opens the Graphics Editor.



Note

To be able to run the Graphics Editor, WinCC must be installed on the computer on which the Graphics Editor was invoked. For additional information on the functionality and the handling of the Graphics Editor, refer to the [Reference Guide Graphics Editor](#).

Template Editor

Opens the **Template Editor** showing the available library templates.

EA Wizard

Opens the **Energy Accounting Wizard**.

Script Definition Wizard

Opens the **Script Definition Wizard**.

Calculation Wizard

Opens the **Calculation Wizard**.

Script Wizard

Opens the **Script Wizard**.

HTML Reports

Opens the **Report Dialog** in the working area. If an error occurs, the **Error Window While Reporting** is opened. This window lists all failed actions.

EXCEL Reports

Opens the **Report Generator** dialog to create predefined or user-defined reports in MS EXCEL format.

3.1.2.7 Help

The menu items on the **Help** menu provide access to online help information of the Information Model Management.

The **Help** menu provides the following menu items:

- **Help Topics**
- **Working Area Content**
- **About Information Model Management**

Help Topics

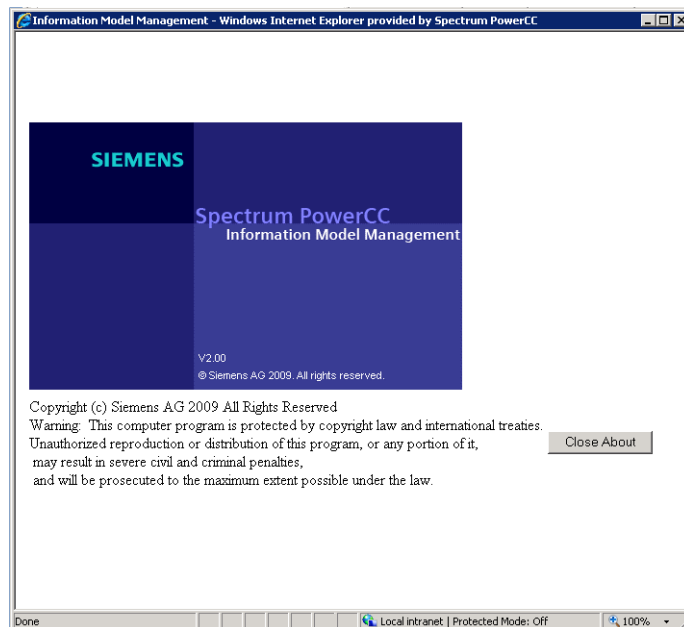
Opens the help topics browser showing the introduction tab of the Information Model Management Online Help.

Working Area Content

Context-sensitive help. Opens the help topics browser showing help information for the current working area content.

About Information Model Management

Opens the About dialog of the Information Model Management. This dialog displays copyright and version information.

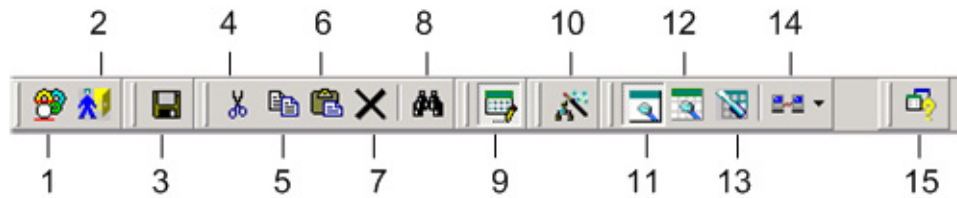


[IMMRE047-150909-enUS-01.tif]

Figure 3-3 About Dialog

3.1.3 Toolbar

The toolbar contains icon buttons that invoke frequently used menu actions. Each action executed by a toolbar button is also available as a menu item on the menu bar. Thus, toolbar buttons can be considered desktop shortcuts for related menu items. Toolbar buttons are enabled or disabled depending on your authorities, the loaded dataset, and on the current operational situation.



[IMMRE053-250909-xxXX-01.tif]

Figure 3-4 IMM Toolbar

When the mouse pointer is moved over one of the toolbar buttons, the corresponding tooltip is displayed.

The toolbar provides the following buttons (numbers refer to the callouts shown in the previous figure):

- Job Management (1)
- Log on (2)
- Save (3)
- Cut Instance (4)
- Copy Instance (5)
- Paste Instance with Internal Links (6)
- Delete Instance (7)
- Search (8)
- Single-Instance Editor (9)
- Type Editor (10)
- Single View of the selected instance (11)
- Table View of the selected instance (12)
- Bulk Editor (13)
- Communication Editors (14)
- Help (15)

Detailed descriptions of the toolbar buttons are provided in the subsequent paragraphs.

Job Management

This button is used to invoke the Job Management.

Log On

Invokes the Log on window

Save

This button saves the changes made in IMM. Invokes the menu action **File > Save**.

Cut Instance

The selected instance is marked for cut. Invokes the menu action **Edit > Cut Instance**.

Copy Instance

The selected instance is marked for copy. Invokes the menu action **Edit > Copy Instance**.

Paste Instance with Internal Links

Paste instance with internal links. Invokes the menu action **Edit > Paste Instance with Internal Links**.

Delete Instance

The selected item is deleted. Invokes the menu action **Edit > Delete Instance**.

Search

Search for instances. Invokes the menu action **Edit > Search**.

Single-Instance Editor

This button is clicked to view instances. Invokes the menu action **View > Instances**.

Type Editor

This button is used for types. Invokes the menu action **View > Types**.

Single View

Invokes the menu action **View > Instance Editors > Single View**.

Table View

Invokes the menu action **View > Instance Editors > Table View**.

Bulk Editor

Invokes the menu action **View > Instance Editors > Bulk Editor**.

Communication Editors

This button is used to invoke the communication editor s. It displays further submenus for data entry of CFE and CFE data points as well as ICCP data objects.

Help

This button is used to get help information about the current working area content. It invokes the menu action **Help > Work Pane Content**.



Note

Depending on the active application, an additional Help button is displayed on the working area of the Information Model Management.

3.1.4 Tree View

3.1.4.1 Introduction

Depending on the selected job, the selected view and the particular situation, the tree view is empty or contains a tree that shows datasets, types, or instances.

3.1.4.2 Tree

Datasets, Types, and Instances

Depending on the selected view, the tree view shows a list of datasets, types, or instances. The outline of this list is based on the hierarchical structure of the shown items. Depending on whether types or instances have been selected previously, it is referred as dataset tree or as type tree or as Instance Overview Tree.

The tree is used to select a dataset, a type, or an instance. After a dataset was selected, the corresponding job list is loaded into the working area. A type or an instance is opened automatically on the tree view after a certain view was selected.

When the view has changed from types to instances or the other way round, the content of the tree changes accordingly.

The dataset tree shows a list of the available datasets. The datasets can be clicked to load the corresponding job list into the working area.

The type tree and the instance overview tree show a list of types and instances sorted by name. Visual indicators, such as, lines and indents illustrate relationships between entries and distinct levels. Each tree entry represents a type or an instance and can be clicked to open the Type Editor or an Instance Editor in the working area.

The type tree and the instance overview tree also provide buttons for expanding or collapsing the tree outline. Expand/collapse buttons are provided for each entry with related subordinate entries. Expand/collapse buttons can be used to make a branch of subordinate entries a so-called subtree - visible (expanding) or invisible (collapsing).

Scroll bars are automatically attached at the bottom edge or at the right edge of the tree to enable scrolling when the tree exceeds the provided space.

Context Menu of the Tree

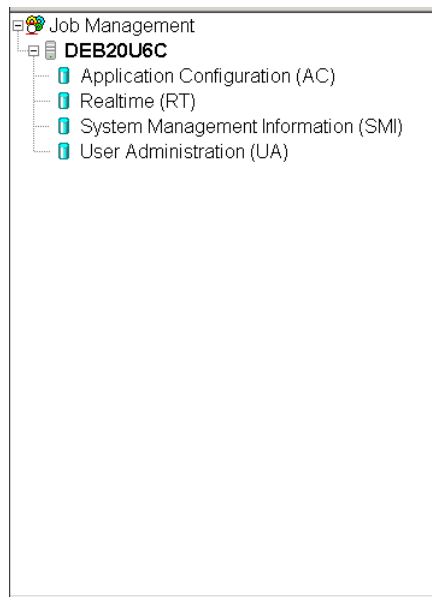
The context menu of the tree is a content-sensitive menu of the tree associated with tree entries. It displays only on demand (when the right mouse button is clicked) and is used to invoke operations that apply to a selected dataset, type, or instance.

The menu items provided by the context menu of the tree as well as their accessibility depend on the selected tree entry and on the current operational situation. They change from selection to selection. Operations invoked by menu items in the context menu of the tree can also be accessed via the menu bar or the toolbar.

The context menu of the tree is opened by clicking a tree entry on the tree with the right mouse button. For further information on the provided menu items, refer to the following chapters.

3.1.4.3 Dataset Tree

The dataset tree lists all available datasets. Each tree entry represents a single dataset.



[IMMRE044-140909-enUS-01.tif]

Figure 3-5 Dataset Tree

Context Menu of the Tree

The context menu of dataset tree provides different menu items (see [3.2.1.4 Job Management UI Tab - Job Context Menu](#)).

The menu item **New Job** invokes the **Create New Job** window that allows you to create a job of the corresponding dataset. When the System Configuration dataset is selected, this menu item is disabled for all datasets except the RT dataset.

3.1.4.4 Type Tree

The type tree lists all types that are available in the currently selected job. Each tree entry represents a single type and shows the type name of the respective type. The outline of the type tree represents the hierarchical structure of the shown types.

The entire structure is a sequence of parent-child relationships established during type definition. Each type to be defined has a parent type. It is the first type, this would be the root type which is the top level of the structure. So, the first type is a subordinate (child) type of the root type. The next type can again be a subordinate type of the root type, or can be a subordinate type of the first type, and so on. A subordinate type automatically inherits all attributes, methods, enumerations, and association declarations from its parent type.

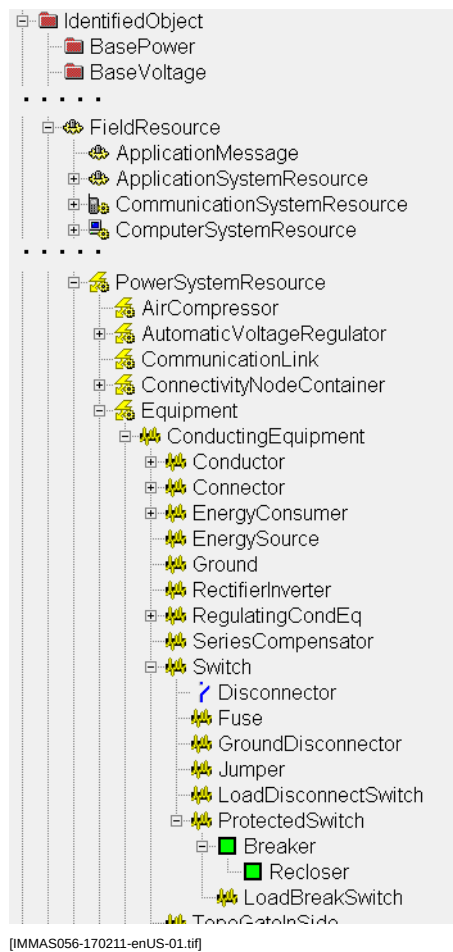


Figure 3-6 Type Tree

Context Menu of the Tree

The context menu of the tree of the type tree provides menu items that allow you to invoke operations that apply to types in general or to specific entries of the type tree.

The subsequent bulleted list shows all menu items provided on the context menu of the type tree. The list covers all menu items that can be displayed on the context menu of the tree of the type tree regardless of the selected tree entry and other restrictions that can apply.

- New Type
Invokes the menu action **Insert > New Type**.
- Delete
Invokes the menu action **Edit > Delete**.
- Export
Invokes the menu action **File > Export**.
- HTML Reports
Invokes the menu action **Tools > HTML Reports**.
- Tree Refresh
Invokes the menu action **View > Tree Refresh**.

Again, be aware that some of the described menu items are not displayed or are disabled depending on the previously described conditions. For example, when the root dataset is selected, some of the menu items, like New Type or Delete, are disabled as the root dataset is read only.

3.1.4.5 Instance Overview Tree

The instance overview tree lists all instances that are available in the currently selected job. The instance data are organized in a hierarchical parent/child structure. Each tree entry represents a single instance and shows the instance label of the respective instance. The outline of the instance overview tree represents a hierarchical structure that arranges the instances in a logical, domain-oriented order.



[IMMAS058-170211-enUS-01.tif]

Figure 3-7 Instance Overview Tree

Because the structure of the instance overview tree is domain-oriented, the instance hierarchy follows the rules to stay consistent in the sense of the power system domain (for example, a switch can be defined as a subordinate instance of a bay, but cannot be defined as a subordinate instance of a measurement).

These rules are defined by the hierarchy associations of the types that are the basis of the instances in the tree. They are automatically enforced when an instance is created. For example, when a new instance is created, you can only use those types determined by the hierarchy associations of the parent instance.

For additional information on hierarchy associations, refer to [3.5.1.6 Hierarchy Tab](#).

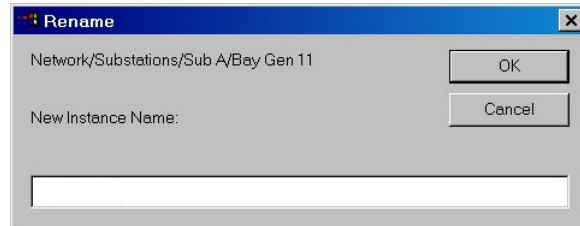
Context Menu of the Tree

The context menu of the tree of the instance overview tree provides menu items that allow you to invoke operations that apply to instances in general or to specific entries of the instance overview tree.

The subsequent bulleted list shows the menu items provided on the context menu of the tree of the instance overview tree. The list covers all menu items that can be displayed in the context menu of the instance overview tree regardless of the selected tree entry and other restrictions.

- New
Invokes the menu action **Insert > New Instance**.
- Cut
Invokes the menu action **Edit > Cut Instance**.
- Copy
Invokes the menu action **Edit > Copy Instance**.
- Paste
Invokes the menu action **Edit > Paste Instance with Internal Links**.
- Delete
Invokes the menu action **Edit > Delete Instance**.

- **Rename**
Invokes the menu action **Edit > Rename Instance**.
After the context menu item **Rename** was selected, the **Rename** dialog is displayed. Enter a new instance name.



[IMMRE007-160108-enUS-01.tif]

Figure 3-8 Rename Dialog

- **Template Editor**
Invokes the menu action **Tools > Template Editor**.
- **Accounting Wizard**
Invokes the **Accounting Wizard**.
- **Script Definition Wizard**
Invokes the **Script Definition Wizard**.
- **Calculation Wizard**
Invokes the **Calculation Wizard**.
- **Script Wizard**
Invokes the **Script Wizard**.
- **Common Calculation**
Starts the **Common Calculation Editor**.
- **Export**
Invokes the menu action **File > Export**.
- **HTML Reports**
Invokes the menu action **Tools > HTML Reports**.
- **EXCEL Reports**
Invokes the menu action **Tools > EXCEL Reports**.
- **Tree Refresh**
Invokes the menu action **View > Tree Refresh**.
- **Set Tree From Here**
Clears the tree view temporarily. Only the selected instance and its entire subtree are displayed in the tree view.
- **Reset Tree**
Undo the action **Set Tree From Here**.
- **Help**
Invokes the menu action **Help > Help Topics** to describe the instance overview tree.

Some of the described menu items are not displayed or are disabled depending on the previously described conditions. For example, when the root dataset is loaded, some of the menu items, like **New** or **Rename**, are disabled since the root dataset is read only.

3.1.5 Working Area

3.1.5.1 Introduction

The right section of the Information Model Management is called working area. Depending on the currently selected dataset and on the particular situation, it can either be empty or can contain interactive elements that provide access to the Information Model Management functionality.

Editors, Dialogs, and Information Tabs

The applications that are used in the working area can be classified into:

- Editors
- Dialogs
- Wizards
- Information Tabs

Editors, dialogs, wizards, and information tabs can be accessed by various means, for example, by clicking a menu item in a menu, by clicking a button on the toolbar, and so on. When a new application like an editor is started in the working area, the currently running application in the working area is closed automatically.



Note

Data modifications which have not been saved are lost when an editor is closed.

3.1.5.2 Editors

Editors allow you to

- View and edit types, instances, and validation scripts and rules
- View, create, modify and delete attributes, and
- View and modify properties of attributes

Attributes	Value
Adaptive Forecast Flag	False
Alias Name	
Area Of Responsibility Id	Area A
Breaker Type	Non-specific
Continuous Rating	0
Control Enable	True
Current Flow	
DMS Flag	False
Disconnects two Voltage Levels	False
Display Order	
Fault Level Rating	0
Flow Breaker Flag	False
In Transit Time	0
Local Name	
Message Configuration	Default

[IMMRE010-031108-enUS-01.tif]

Figure 3-9 Single-Instance Editor

For detailed information on a particular editor, refer to its description.

3.1.5.3 Tabs

Engineering activities regarding type and instance attributes and properties normally involve a rather large amount of information and, therefore, interactive elements. This requires the provided elements to be categorized and grouped into logical sets to maintain clarity and easy usage. Editors use tabs to bundle the provided interactive elements according to their intended purpose.

General Attributes Enumerators Hierarchy References	
Name	PowerSystemResource
Label	PowerSystemResource
Help Message	A power system resource can be an item of equipment such as a Switch, an EquipmentContainer containing many individual items of equipment such as a Substation, or an organisational entity such as Company or SubControlArea. This provides for the nesting of collections of PowerSystemResources within other PowerSystemResources. For example, a Switch could be a member of a Substation and a Substation could be a member of a division of a Company.
Rdf Name	PowerSystemResource
Namespace	cim
Scope	ALL
Graphic Sheet	
Abstract Type	Yes
Rdf Default Name	
CIM children	No
Nerc Profile	Yes

[IMMRE011-031108-enUS-01.tif]

Figure 3-10 Example of Tabs

3.1.5.4 Table Editor

The Table Editor shows all information of the child instances of the selected instance in one comprehensive view. This view provides context menus for data viewing and data modification. The content and the provided context menus depend on the selected editor and the current operational situation. For detailed information on a specific editor and the provided views, refer to the description of the respective editor.

In the following section, the general features of the table editor and the provided context menus are described.

Name	Type	Description
CB	Breaker	
Dis A	Disconnector	
Dis B	Disconnector	
Dis Ground	GroundDisconnector	
Earth Fault	FleetingMeasurement	
I	AnalogMeasurement	
MWH Exp	AccumulatorMeasurement	

[IMMRE010-160108-enUS-01.tif]

Figure 3-11 Example of a Table Editor

General Table Editor Features

A table editor provides the following general features:

- Each instance is shown in 1 row.
- The types of already available instances cannot be changed. If they are shown they are read only.
- When you create an instance, the available types for the new parent-child associations are shown in a type list box.
- Internal attributes are read only.
- Read-only cells have a shaded background.
- Some cells provide list boxes.
- Ascending or descending sort is done in columns by clicking the header of a particular column.
- The table provides an empty row for entering a new instance. The empty row is always at the bottom of the table. New instances can only be added to the end of the list of instances.
- An instance can have multiple links for an association. A multiple link column representation is collapsed by default and expanded by command.
- The Cell Editor for an attribute depends on the attribute data type. If the attribute is defined with an enumeration type, the cell provides a list box. Otherwise it provides a Text Editor, a number editor, or a date/time editor for the different formats in the data model.
- References can be edited with Drag & Drop from an instance tree. There are some specially assigned links to instances, which can be selected from a list box.
- The column width is automatically adjusted to the maximum cell width of the column. But the width is restricted to a maximum (optionally). Strings that are longer are truncated by default (unwrapped text). They can be displayed in multiple lines (wrapped) by command.
- The tables provide context menus depending on the clicked area.

Instance Editor Context Menu

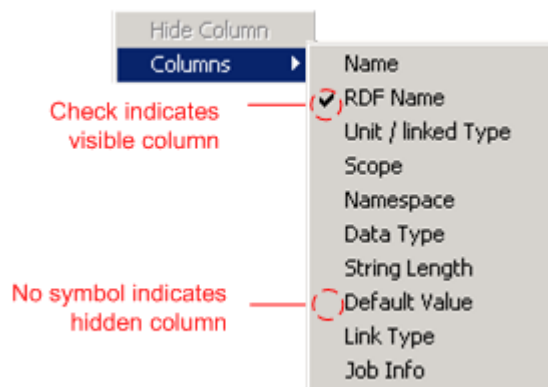
The Instance Editor context menu is opened by clicking the right mouse button on the body area. Depending on the clicked area and the selected view a different menu appears. The following figure shows the areas for the different menus. The following tables describe the possible commands and the required input for activation.

Table 3-1 Context Menu for Single View

Menu Name	Command	Enabled if	Action
Column Menu ¹	Hide Column	Not a default column	Makes this column invisible. The column is temporarily removed from the table.
	Columns	Always	Opens the submenu Columns.
Row Menu ²	Hide row	there is more than 1 row	Makes this row invisible.
	Expand row	an association field contains more than 1 link	Splits the association field into separate rows for each link so that the links can be edited.
	Rows	Always	Opens the submenu Rows.
Cell Menu ³	Edit	Column not read only	Changes grid cell to edit mode.
	Cut	Always	Creates a copy from the grid cell content in the clipboard and removes the content from the selected cell.
	Copy	Always	Creates a copy from the grid cell content in the clipboard.
	Paste	Column not read only	Paste content from clipboard to grid cell content.
	Delete	Column not read only	Deletes the grid cell content.

Menu Name	Command	Enabled if	Action
	Add Link	No expanded link column	Inserts empty cell for a new link.
	Propagate Value	Always	Propagates the selected attribute values to the subordinate instances.
Tab Menu	New	Always	Creates a tab with the content of the selected tab.
	Rename	It is a new tab	Opens the Rename dialog for editing the tab name.
	Delete	It is a new tab	Deletes the tab.
	Read only	Always	Set tab to read only.

1. The submenu Columns contains the names of the attributes currently shown on the table grid. The data shown in the submenu Columns are attribute properties defined with the Type Editor. These names are represented on the table grid as attribute column headings that are read only. A symbol displayed to the left side of a menu item represents the visibility status of the related column. A check is displayed when the related column is currently visible. No symbol is displayed when the related column is currently invisible. The visibility of an attribute column can be toggled by clicking its related menu item on the submenu Columns or by clicking a column heading to select a column and clicking the menu item Hide Columns on the instance editor context menu. However, some attribute columns are mandatory and remain visible at all times. Menu items related with mandatory attribute columns are disabled.
2. The submenu Rows contains the names of the rows currently shown on the table grid. Rows can be displayed or unavailable like the submenu Columns as described before.
3. Right-clicking the corresponding cell in Edit mode opens a context menu with the usual edit commands (Undo, Cut, Copy, Paste, Delete, and Select All)



[IMMRE012-160108-enUS-01.tif]

Figure 3-12 Example of the Submenu Columns (Populated with Sample Entries)

Table 3-2 Context Menu for Table Editor

Menu Name	Command	Enabled if	Action
Column Menu ¹	Expand Column	Cell count > 1	Expands all instances with cell count > 1 in column.
	Hide Column	Not first column	Makes this column invisible.
	Columns	Always	Opens the submenu Columns.
Row Menu	Duplicate	Always	Appends a copy of this row as a new row.
	Delete	Always	Deletes the instance.
Cell Menu ²	Edit	Column not read only	Opens the Single view, if the Link column is selected; else changes cell to edit mode.

Menu Name	Command	Enabled if	Action
	Copy	Always	Creates a copy from the grid cell content in the clipboard.
	Paste	Column not read only, not new line	Paste content from clipboard to grid cell content.
	Clear	Column not read only	Deletes the cell content.
Table Menu	Hide Row	Not first row	Makes this row invisible.
	Rows	Always	Opens the submenu Rows.
Tab Menu (only available in detailed view)	New	Always	Creates a tab with the content of the selected tab.
	Rename	Always	Opens the Rename dialog for editing the tab name.
	Delete	It is a new tab	Deletes the tab.
	Read only	Always	Set tab to read only.

1. The submenu Columns contains the names of the attributes currently shown on the table grid. The data shown in the submenu Columns are attribute properties defined with the Type Editor. These names are represented on the table grid as attribute column headings that are read only. A symbol displayed to the left side of a menu item represents the visibility status of the related column. A check is displayed when the related column is currently visible. No symbol is displayed when the related column is currently invisible. The visibility of an attribute column can be toggled by clicking its related menu item on the submenu Columns or by clicking a column heading to select a column and clicking the menu item Hide Columns on the instance editor context menu. However, some attribute columns are mandatory and remain visible at all times. Menu items related with mandatory attribute columns are disabled.
2. Right-clicking the corresponding cell in Edit mode opens a context menu with the usual edit commands (Undo, Cut, Copy, Paste, Delete, and Select All).

Table 3-3 Context Menu for Communication View (Selected Instances)

Menu Name	Command	Enabled if	Action
Row Menu	Edit	Always	Opens the Single Editor with the selected instance.
	Create	Not linked	Creates a new linked instance in the Communication Editor.

Table 3-4 Context Menu for Communication View (Linked Instances)

Menu Name	Command	Enabled if	Action
Row Menu	Delete	Always	Deletes instance.
	Edit	Always	Opens the Single Editor with the selected instance.
	Rename	Enabled if Name -column	Changes cell in Name -column to Edit mode.
	Unlink	Enabled if linked	Deletes link from data point.



Note

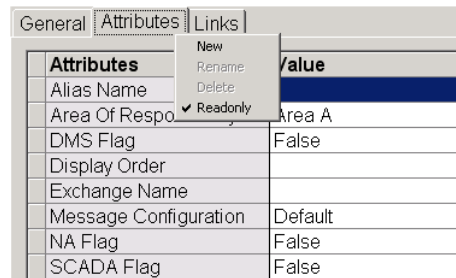
For linking existing measurements to existing communication objects Drag & Drop is enabled. You can select a measurement in the left table or the tree section and drops the measurement into a row of the right table that contains an unlinked communication object. After the measurement is dropped, the editor validates whether the instance type of the measurement and the instance of the CFE data point and the ICCP data type of an ICCP object, respectively represent a valid combination.

3.1.5.5 Handling of Views

New views can be created and the content of existing views can be modified using the context menus of the view. The configuration of views is stored in the database and are available for all users.

For defining and formatting a new view, proceed as described in the following example:

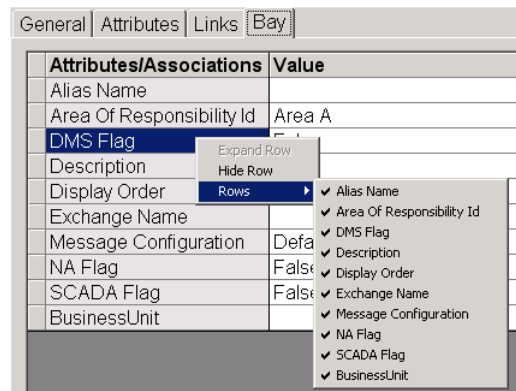
- Click the name of the tab with the right mouse button.



[IMMRE013-170108-enUS-01.tif]

Figure 3-13 Tab Context Menu

- Select the item **New** to create a tab.
- Click the left side of the tab with the right mouse button.

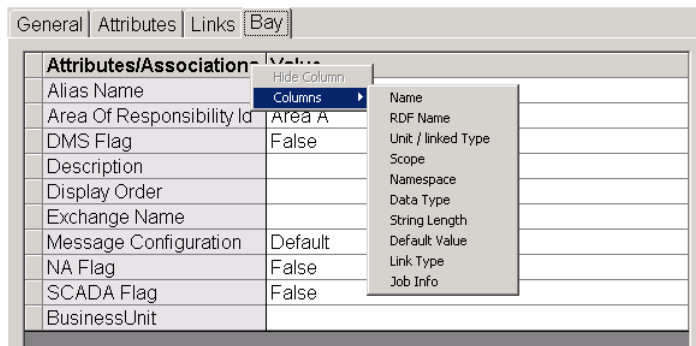


[IMMRE014-170108-enUS-01.tif]

Figure 3-14 Row Context Menu

- In this context menu, you can select the attributes (rows) to be displayed.

- Click the header of the table with the right mouse button.



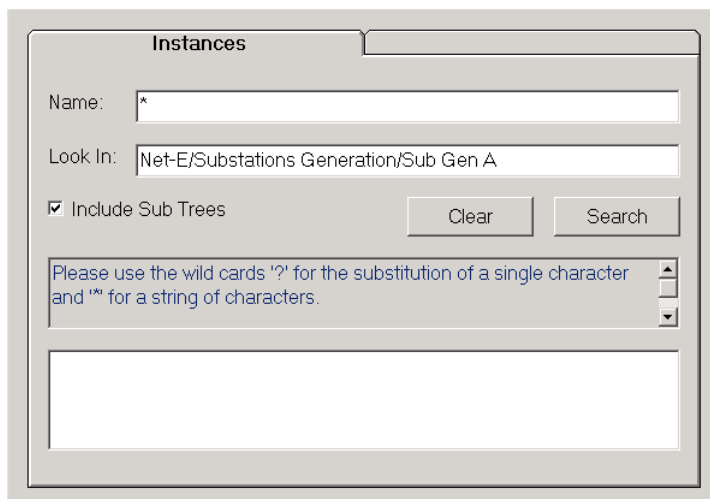
[IMMRE015-170108-enUS-01.tif]

Figure 3-15 Column Context Menu

- In this context menu, you can select the data (columns) to be displayed for the selected attributes.

3.1.5.6 Dialogs

Dialogs provide access to general Information Model Management functions, such as, data import and export, report tools, and so on. The amount of information involved in this interaction is rather small compared to editors. This fact reduces the amount of involved interactive elements.



[IMMRE016-170108-enUS-01.tif]

Figure 3-16 Example of a Dialog

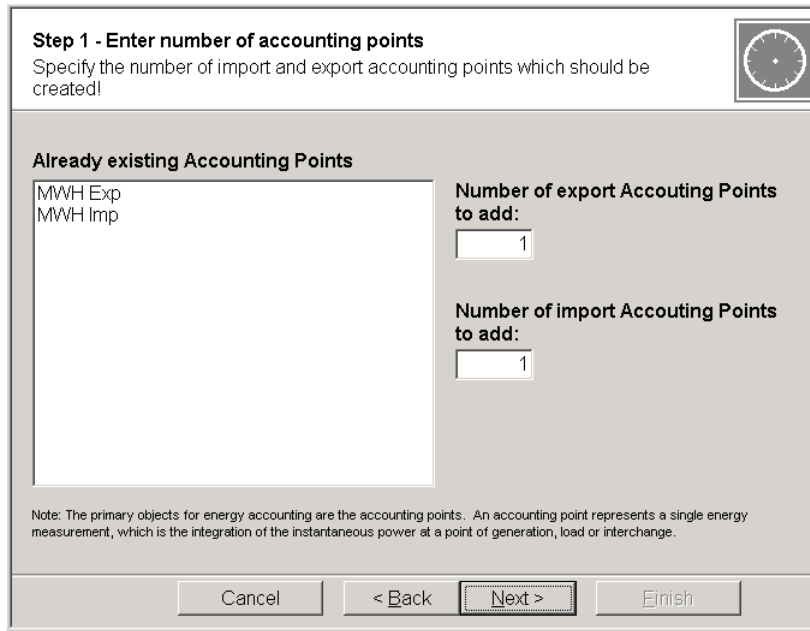
For detailed information on how to access a particular dialog, refer to its description. The available dialogs include:

- Instance Creation
- Report dialog
- Report Generator dialog
- Export dialog

- Import dialog
- Instance Search dialog

3.1.5.7 Wizards

Wizards support you when creating new instances to make sure that all required attributes of the generated instances are set according to the corresponding instance requirements.



Step 1 - Enter number of accounting points
Specify the number of import and export accounting points which should be created!

Already existing Accounting Points

MWH Exp
MWH Imp

Number of export Accounting Points to add:
1

Number of import Accounting Points to add:
1

Note: The primary objects for energy accounting are the accounting points. An accounting point represents a single energy measurement, which is the integration of the instantaneous power at a point of generation, load or interchange.

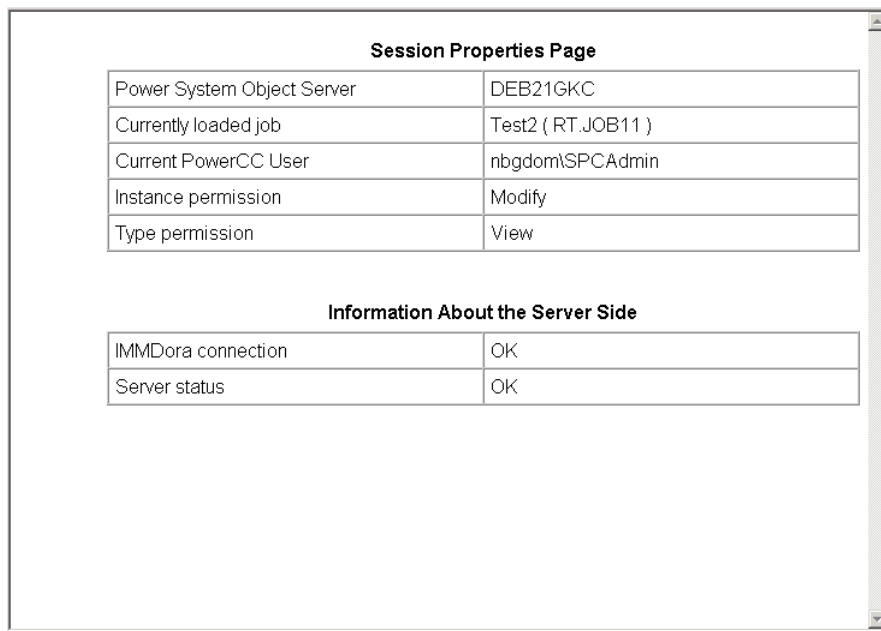
Cancel < Back Next > Finish

[IMMRE012-031108-enUS-01.tif]

Figure 3-17 Example of a Wizard Step

3.1.5.8 Information Tabs

Information tabs provide status information about a particular situation, activity, or, condition. All status information presented on an information tab is read only.



[IMMRE013-031108-enUS-01.tif]

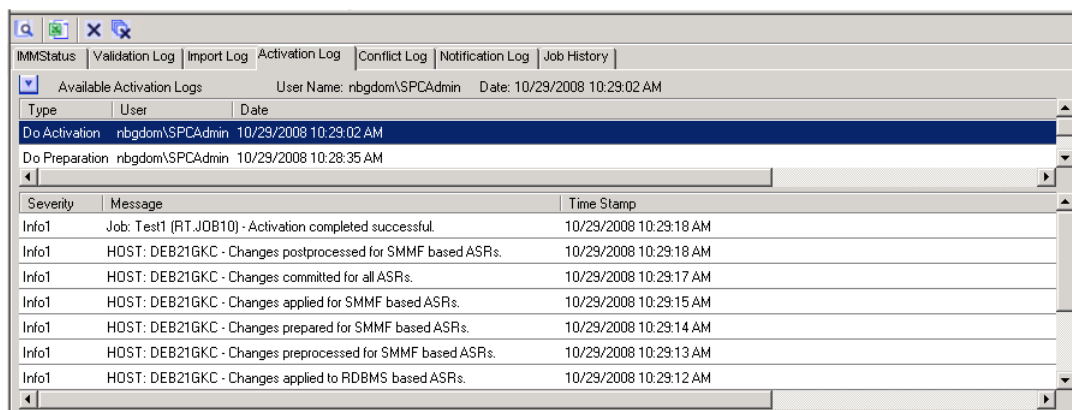
Figure 3-18 Example of an Information Tab

Information tabs can open automatically to indicate the result of a previously performed engineering activity. For detailed information on a particular information tab, refer to its description. The available information tabs include:

- Session Properties tab (described in [3.2.2 Session Properties Tab](#))

3.1.6 Log Section

The Log section contains all logs of the selected dataset or job. Each log can be selected via the respective tab. The logs shown in the log section depend on the current situation (selected dataset, job, and so on).



[IMMRE001-311008-enUS-01.tif]

Figure 3-19 Log Section

The log section consists of the following elements:

- Toolbar

- Selection tabs
- Log overview
- Detail section

Toolbar

The toolbar provides icons for handling logs.



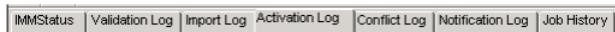
[IMMRE002-311008-enUS-01.tif]

Figure 3-20 Log Section Toolbar

The following icons are provided:

- Show
Displays the content of the selected log in the detail section.
- Export
Exports the selected log to an Excel file. A file selection dialog opens for selecting a file name and path where to store the Excel file.
- Delete
Deletes the selected log. Only validation, import, activation, and notification/conflict logs can be deleted. For the other log types, this button is not available.
- Delete all
Deletes all logs of the currently selected type. Only validation, import, activation, notification, and conflict logs can be deleted. For the other log types, this button is not available.

Selection Tabs



[IMMRE003-311008-enUS-01.tif]

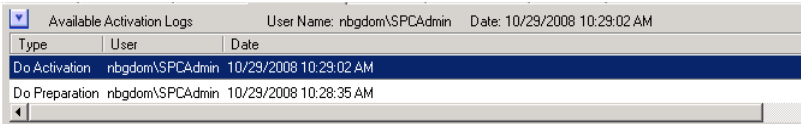
Figure 3-21 Log Selection Tabs

The selection tabs provide access to the different log types. Depending on the selected dataset or job, different log types are available.

- IMM Status
The IMM Status tab contains all messages that occurred during work with IMM. For additional information on content and structure of the IMM Status list, refer to [3.2.11.1 IMM Status](#).
- Task Log
Contains all messages that occur during an IMM UI task such as copy/paste, cut/paste, delete subtree, graphic editor processing, modify instance data, modify type data, etc. Where applicable, the object path is a hyperlink that can be used to navigate to the instance in question.
- Validation Log
The Validation tab contains all logs that were created during job validation or global validation. This log type is always available. For additional information on content and structure of the validation log, refer to [3.2.11.3 Validation Log](#).

- **Activation Log**
The Activation tab contains all logs that were created during the preparation or activation of a dataset or job. This log type is not available when the UI is in editing mode (the tree view shows an overview of all datasets and jobs, no job has been selected for editing). For additional information on content and structure of the activation log, refer to [3.2.11.4 Activation Log](#).
- **Import Log**
The Import tab contains all logs that were created when a user initiated import was done within a dataset or job. Depending on whether a dataset or a job has been selected in the job tree, either all import logs of the selected dataset or only the job-specific import logs are displayed. For additional information on content and structure of the import log, refer to [3.2.11.5 Import Log](#).
- **Job History**
The Job History log contains all status changes of the currently selected job (as long as the job has not been finished). This log type is only available when a job has been selected in the tree view (with a mouse click), but the UI is not in edit mode. For additional information on content and structure of the job history log, refer to [3.2.11.6 Job History](#).
- **Finalized Jobs**
The Finalized Jobs tab shows all jobs that have been activated and finalized. This log type is only available when the Realtime dataset has been selected in the tree view. For additional information on content and structure of the finalized jobs log, refer to [3.2.11.7 Finalized Jobs](#).
- **Notification Log**
This log is only available if the time-based job model is used.
As in a time-based job model several jobs can modify the same instances, the modifications made in different jobs may come in conflict with each other. The Notification Log tab shows these conflicts. For additional information on content and structure of the notification log, refer to [3.2.11.9 Notification Log](#).
- **Conflict Log**
This log is only available if the time-based job model is used.
When a job is published, deleted, or rescheduled via the Job Management, this tab shows the integrity conflicts that occur. For additional information on content and structure of the conflict log, refer to [3.2.11.8 Conflict Log](#).

Log Overview



The screenshot shows a window titled 'Available Activation Logs'. At the top, it displays 'User Name: nbgdom\SPCAAdmin' and 'Date: 10/29/2008 10:29:02 AM'. Below this is a table with three columns: 'Type', 'User', and 'Date'. The first row is highlighted in blue and contains 'Do Activation', 'nbgdom\SPCAAdmin', and '10/29/2008 10:29:02 AM'. The second row contains 'Do Preparation', 'nbgdom\SPCAAdmin', and '10/29/2008 10:28:35 AM'. At the bottom left of the window, there is a small icon and the text '[IMMRE004-311008-enUS-01.tif]'.

Type	User	Date
Do Activation	nbgdom\SPCAAdmin	10/29/2008 10:29:02 AM
Do Preparation	nbgdom\SPCAAdmin	10/29/2008 10:28:35 AM

Figure 3-22 Log Overview

The Log Overview section contains all logs of the selected type or all status messages, respectively. To display the entries, the arrow button in the top left corner must be clicked. Clicking a column header with the right mouse button opens a context menu, where individual columns can be set visible or unavailable.

Selecting 1 row and clicking the toolbar button Show opens the selected log in the Detail Section (the same can be achieved by double-clicking the respective row). For some log types, this section may not be available.

Detail Section

Severity	Message	Time Stamp
Info1	Job: Test1 (RTJOB10) - Activation completed successful.	10/29/2008 10:29:18 AM
Info1	HOST: DEB21GKC - Changes postprocessed for SMMF based ASRs.	10/29/2008 10:29:18 AM
Info1	HOST: DEB21GKC - Changes committed for all ASRs.	10/29/2008 10:29:17 AM
Info1	HOST: DEB21GKC - Changes applied for SMMF based ASRs.	10/29/2008 10:29:15 AM
Info1	HOST: DEB21GKC - Changes prepared for SMMF based ASRs.	10/29/2008 10:29:14 AM
Info1	HOST: DEB21GKC - Changes preprocessed for SMMF based ASRs.	10/29/2008 10:29:13 AM
Info1	HOST: DEB21GKC - Changes applied to RDBMS based ASRs.	10/29/2008 10:29:12 AM

[IMMRE005-311008-enUS-01.tif]

Figure 3-23 Log Detail

The Detail Section shows the content of the currently selected log. Clicking a column header with the right mouse button opens a context menu, where individual columns can be set visible or unavailable.

The content of the detail area depends on the log type. For more information refer to the description of the respective log.

3.1.7 Status Bar

The Status Bar shows messages that reflect the status of a previously performed or an ongoing engineering activity. For example: load data, pasting instance, and so on.

3.1.8 Progress Indication Window

Depending on the amount of information processed by the Information Model Management, some actions like import/export of data, and so on, can result in lengthy operations. To keep you informed of an ongoing operation, a Progress Indication Window is provided.

The Progress Indication Window is a visual feedback of the Information Model Management that reflects the progress of an extensive operation. It pops up automatically after such an operation has been initiated and remains open until the operation has been completed. Then, it is closed automatically. It does not contain interactive elements nor provides it access to Information Model Management functionality. All information provided by the Progress Indication Window is read only and is intended to keep you informed of the ongoing operation.



Note

In some cases it can happen, that the Progress Indication Window is not closed automatically. For example, when a time-out or an error has occurred. In such a case, close the Progress Indication Window.

3.2 General Engineering Applications

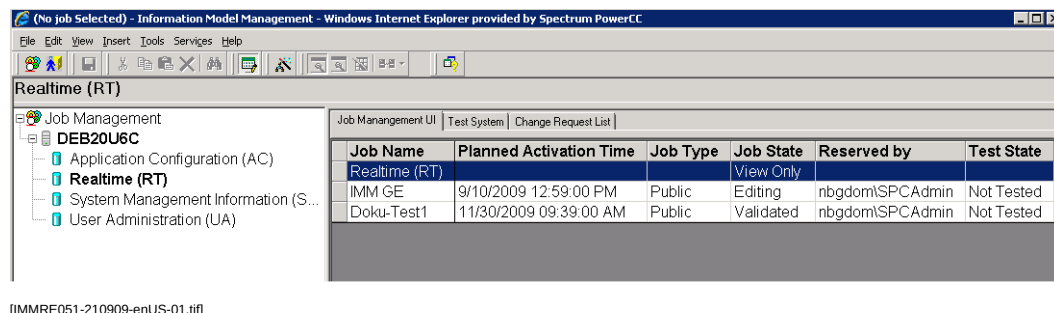
3.2.1 Job Management User Interface

3.2.1.1 Overview

The Information Model Management provides the Job Management user interface to control all data entry and engineering activities of the system. The Job Management is the method by which changes to a (root) dataset, for example, real-time dataset are grouped and controlled. A job contains a set of modifications belonging together. Those changes are activated together independently of other jobs. If you wish to define a new substation and propagate this station to the runtime system, create a job, edit, validate, and activate the job. Activation of the job makes the changes available in the online system.

Selecting the Job Management User Interface

The Job Management UI is automatically opened in the working area when the Information Model Management is opened, but it can also be accessed by clicking the corresponding button in the toolbar.



[IMMRE051-210909-enUS-01.tif]

Figure 3-24 Job Management User Interface

3.2.1.2 Dataset Tree

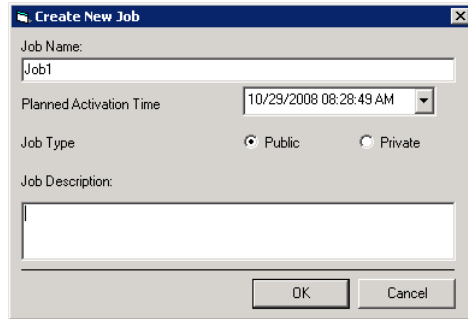
The following dataset types are mandatory:

- Realtime
- System Management Information
- User Administration
- Application Configuration

Realtime

Clicking this dataset loads the assigned jobs into the job list. When the Job Management user interface opens, all jobs of this dataset are automatically displayed in the job list.

Right-clicking the real-time dataset item opens the context menu for creating a job. Left-clicking the context menu item **Create Job** opens the Create New Job window. The new job is added to the job list in the body by clicking the button **OK**.



[IMMRE015-031108-enUS-01.tif]

Figure 3-25 Create New Job Window (Time-based Model)

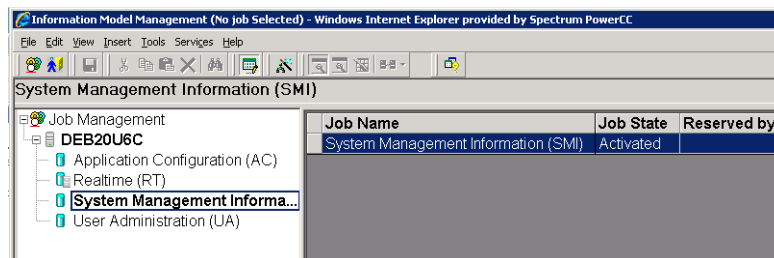
System Management Information

Clicking this dataset loads the assigned System Management Information-job into the job list.



Note

A new job cannot be created for this dataset.



[IMMRE052-210909-enUS-01.tif]

Figure 3-26 System Management Information Dataset

User Administration

Clicking this dataset loads the assigned User Administration-job into the job list.



Note

A new job cannot be created for this dataset.

Application Configuration

Clicking this dataset loads the assigned Application Configuration-job into the job list.



Note

A new job cannot be created for this dataset.

The following dataset types are optional:

- Planning 1
- Planning 2

Planning

Clicking this dataset loads the assigned planning jobs into the job list.

Right-clicking the Planning dataset item opens the context menu for creating a job. For creating a job, refer to the description of the Realtime dataset.

3.2.1.3 Job Management UI Tab - Job List

The job list is displayed when a root dataset is selected in the tree view. In the job list, all available jobs of the corresponding dataset are displayed. It consists of the following attributes:

Job Name

The name of the job.

Planned Activation Time

This column is only displayed if the time-based job model is used.

It shows the date and time, when the job is planned to be activated.

Job Type

This column is only displayed if the time-based model is used.

The following types are possible:

- Public
- Private

For additional information on the time-based model, refer to the [Technical Specification Information Model Management \(IMM\)](#).

Job Status

The job status identifies the state of the job. The valid values of the job statuses are listed in the following table:

Table 3-5 Valid Job Statuses

Job Status	Description
Aborting Activating DOR	The DOR job activation is being canceled.
Aborting Activating System	The System job activation is being canceled.
Aborting Undoing DOR	The DOR undo job is being canceled.
Aborting Undoing System	The System undo job is being canceled.
Activated	Job activation has been completed successfully.
Activated_Dor	The Dor job activation has been completed successfully
Activating_Dor	The Dor job is being activated.
Activating_System	The System job is being activated.
Activating System Failed	The activation of the System job failed.
Activating_Dor_Failed	The activation of the Dor job failed.
Activation_Failed	Job activation has failed.
Archived	Activated job has been archived.

Job Status	Description
Archiving	Activated job is being archived.
Archiving_Failed	Archiving of activated job has failed.
Canceling Validation	Validation of the job is being canceled.
Create_Original_Job_Failed	The creation of the do job failed. The do job contains all changes which were undone.
Create_Undo_Job_Failed	Creation of the undo job failed.
Creating_Original_Job	The do job is being created.
Creating_Undo_job	The undo job is being created.
Editing	Job is being edited.
Erased	The job has been erased.
Erase Failed	Erasing of the job has failed.
Erasing	The job is being erased.
Finalized	The job has been finished.
Finalizing	Job is being finished.
Finalize_Failed	The finalization of the job failed.
Preparation Failed	Preparation of job data has failed.
Prepared	Job data have been prepared for insertion into the real-time system
Preparing	Job data are being prepared for insertion into the real-time system.
Preparing Undo	Data of the undo job are being prepared for insertion into the real-time system.
Preparing Undo Failed	Preparation of undo job data has failed.
Ready	Job is ready for users to work with it.
Undoing_DOR	The DOR job is being undone.
Undoing_DOR_Failed	The DOR job could not be undone.
Undoing_System	The System job is being undone.
Undoing_System_Failed	The System job could not be undone.
Undo_Job_Created	The undo job is successfully created.
Undone	Undo activation of Job has been completed successfully.
Undone_Dor	Undo activation of Dor Job has been completed successfully.
Undone_System	Undo activation of System Job has been completed successfully
Undo Prepared	Data of the undo job have been prepared for insertion into the real-time system.
U_Archived	Job with state undone has been archived.
U_Archiving	Job with state undone is being archived.
U_Archiving_Failed	Archiving of job with state undone has failed.
Validating	Job is being validated.
Validated	Validation has completed successfully.
Validated with Errors	Validation has finished with error messages. In this state, a job cannot be activated.
Validated with Warnings	Validation has finished with warning messages.
Validation Canceled	Validation has been canceled.
Validation Failed	Validation could not complete.
View_Only	The job can only be viewed.



Note

If the job activation fails, the job status changes to `activation_failed`. Details about the error can be retrieved from the Activation Log.



Note

Failed job statuses can be recovered by selecting the menu item **Reset** from the Job context menu.

Reserved By

The name of the user, who has reserved the job. The user for the Realtime (RT)-job and the System Configuration (SC)-job is default.

Test Status

Test status of the job. For information on the possible states, refer to [3.2.1.5 Test System Tab / Production System Tab](#).

This column is only provided if a test system is configured.

Change Request

Change request state of the job:

- **Undefined**
Initial state of a job or after the change request has been withdrawn.
- **Required**
Change request is required.
- **Not Required**
Change request is not required.
- **Accepted**
The sent change request was accepted from the first level control center.
- **Rejected**
The sent change request was not accepted from the first level control center and other affected second-level control centers.
- **Sent**
Change request has been sent but there is no response.

This column is only provided if the Hierarchical Model Maintenance functionality is installed.

Description

A description of the job (optional).

3.2.1.4 Job Management UI Tab - Job Context Menu

You open the Job context menu by right-clicking a job in the tree view.

The Job context menu contains the following menu items:

- **Open**

- **Validation**
 - **All**
 - **Job**
- **Activation**
 - **Prepare Activation**
 - **Activate**
 - **Prepare Undo**
 - **Undo**
- **Administration**
 - **Release**
 - **Finalize**
 - **Erase**
 - **Reschedule**
 - **Publish**
- **Reset**
- **Create Change Request**
- **Is Change Request Required**
- **Synchronize IMM**

Detailed descriptions of the menu items are provided in the subsequent paragraphs.

Open

Displays the corresponding instances of the selected job in the tree view of the Information Model Management. Depending on your access rights, you can view and modify instance data. The instance data can be also displayed by double-clicking the corresponding job on the job list.

Validation

By selecting one of the following sub items, you start a definite validation process of the selected job:

- **Job**

This item starts the validation process for all instances changed in the currently selected job.
- **All**

This item starts the validation process for all instances in the database.

During the validation process, the job status changes to **Validating**. After the validation process has been finished successfully, the job status is set to **Validated**. If the validation could not complete, it is set to **Validation_Failed**. If the validation has finished with one or more warning messages/error messages, the job status is set to **Validated_With_Warnings** or **Validated_With_Errors**. If there are any errors, you can correct the errors by editing the job.

Activation

- **Prepare Activation**

Prepares the job data offline for insertion into the real-time system. During the preparation, the job status is set to **Preparing**. After successful preparation the job status is set to **Prepared**, in case of errors it is set to **Preparation Failed**.
- **Activate**

Starts the activation process of the selected job. By clicking this menu item a confirm message box opens. After clicking the button **OK**, the Job Activation window opens and the changes done in the selected job are loaded into the root dataset. Before the activation process starts, the job status is set to **Activating**. If

the activation process completes successfully, the job status is set to **Activated**. The entry of State Modified shows the date and time at which the job was activated.

If there are any errors, the job status is set to **Activation_Failed** and the job is not activated. To correct the errors, reset the status of the job by clicking the menu item **Reset**. The job status is set to **Ready**. After editing and validating the job, the activation process can be started once again.

After a successful activation, data changes in the job are no longer possible. You can either finish the job and make the changes permanent on the runtime system by clicking the menu item **Finalize** or can undo the activation by clicking the menu item **Undo** and continue to edit data in the corresponding job.

- **Prepare Undo**

Prepares the data of the undo job offline for insertion into the real-time system. During the preparation, the job status is set to **Preparing Undo**. After successful preparation the job status is set to **Undo Prepared**, in case of errors it is set to **Preparing Undo Failed**.

- **Undo**

An undo activation process can be performed only on an activated job. In some cases, the activated changes can cause problems even if they have been validated and tested thoroughly. In these cases, it is necessary to undo the changes. The undo activation process of the job can be started by clicking this menu item. The changes done to the root dataset in the previous activation are rolled back and the job status is set to **Undone**.

Administration

- **Release**

Releases the lock of the selected job. The job status changes to **Ready**. In this case, the job is not reserved and another user can modify data of this job.

- **Finalize**

Makes the data changes performed in the selected job permanent on the runtime system. During the finalization process of a job, the job status is set to **Finalizing**. After the finalization process, the job is deleted from the job list. All locks held by this job are released.

- **Erase**

Removes the selected job from the job list. A job can be erased as long as it was not activated. During the erasing process, the job status is set to **Erasing**. After a successful erasing process, the job is removed from the job list and job interlockings are released. If there are any errors during the erasing process, the job status is set to **Erase_Failed** and the job is not deleted.

- **Reschedule**

Opens a display for entering a new planned activation date for the selected job.

- **Publish**

This item is only available for private jobs. Changes a private job to a public job.

Reset

Unlock a reserved job of another user. If you select this menu item, a confirm message box opens, which shows the selected job and the user name for which the job is reserved. After clicking the button **Yes**, the job status changes to **Ready** and the job is locked by the new user. Now this user can modify data of this job.

Additionally, failed job statuses can be recovered by selecting this menu item. In this case, the failed job status changes to the previous job status.

Create Change Request (Only If Hierarchical Model Maintenance Is Configured)

Opens a display for creating a change request for the selected job.

Is Change Request Required (Only If Hierarchical Model Maintenance Is Configured)

Checks, whether a change request must be created for the data changes made in the selected job.

Synchronize IMM

This menu item is only available on a test system when the **Realtime** root job is selected. Transfers the data model of the production system to the test system.

Job Actions and Job Statuses

All actions, which can be performed on a job, depend on the status of the job. The allowed job actions for the corresponding job statuses are listed in the following table. It is assumed that you have modify permission and you have reserved the job.

Table 3-6 Allowed Job Actions for the Corresponding Job Statuses

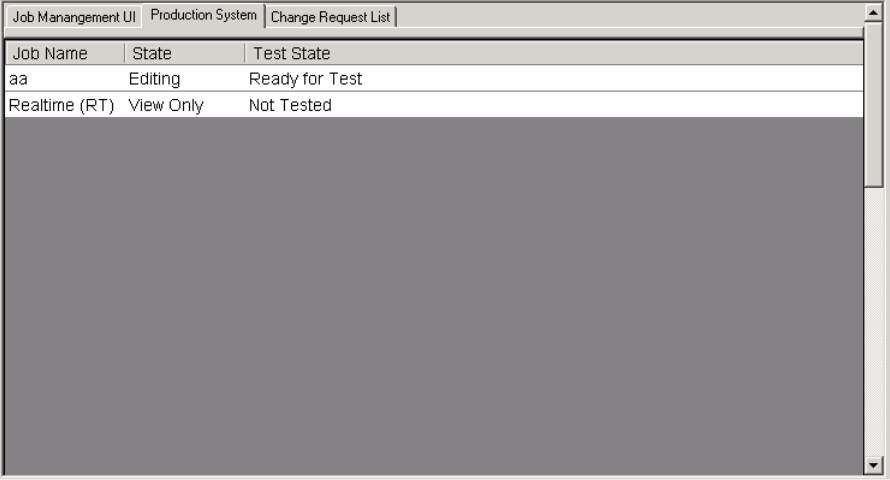
Job Action	Job Statuses
Open	All states
Release	Ready, Validated, Activated, Editing, Undone, Validation_Failed, Validated_With_Errors, Validated_With_Warnings, Activation_Failed, Create_Undo_Job_Failed, Create_Original_Job_Failed, Finalize_Failed, Erase_Failed, Validation_Canceled
Validate	Ready, Validated, Editing, Undone, Validated_With_Errors, Validated_With_Warnings, Validation_Canceled
Prepare	Ready, Validated, Editing, Undone, Validated_With_Warnings
Activate	Ready, Editing, Undone, Validated_With_Warnings
Finalize	Activated, Create_Undo_Job_Failed, Create_Original_Job_Failed
Erase	Ready, Validated, Editing, Validating, Undoing_System_Failed, Validation_Failed, Activation_Failed, Validated_With_Errors, Validated_With_Warnings, Validation_Canceled
Reset	Ready, Validated, Activated, Editing, Validating, Finalize, Finalizing, Erased, Erasing, Creating_Undo_Job, Undone, Creating_Original_Job, Validation_Failed, Activation_Failed, Create_Undo_Job_Failed, Create_Original_Job_Failed, Finalize_Failed, Erase_Failed, Preparation_Failed, Preparing_Undo_Failed, Undo_Prepared, Validated_With_Errors, Validated_With_Warnings, Validation_Canceled
Undo	Activated, Finalize_Failed

3.2.1.5 Test System Tab / Production System Tab

If a test system is configured, the job management window on both the test system and the production system contains a second tab besides the **Job Management UI** tab to handle the transfer of jobs between the 2 systems:

- On the production system, an additional tab named **Test System** is provided. This tab shows a list of all jobs available at the test system.
- On the test system, an additional tab named **Production System** is provided. This tab shows a list of all job available at the production system,

The structure of both tabs is similar.



Job Name	State	Test State
aa	Editing	Ready for Test
Realtime (RT)	View Only	Not Tested

[IMMRE026-041108-enUS-01.tif]

Figure 3-27 Production System Tab

A job can either be created, edited, and tested on the test system and then transferred to the production system. Or it can be created and edited on the production system, transferred to the test system for testing, and then transferred back to the production system. The transfer of a job is always initiated on the target system, where the job shall be imported. Before you can test a job, the complete data model of the production system must be transferred to the test system. This transfer is also initiated at the test system.

Job List

The job list provides an overview of all jobs at the respective system (test system or production system). It provides the following information for each job:

- **Job Name**
Name of the job as shown in the Job Management UI tab.
- **State**
State of the job as shown in the Job Management UI tab.
- **Test Status**
One of the following statuses:
 - **Not Tested**
This is the initial state when a job is created at edited.
 - **Ready for Test**
The job is ready for being tested. In this status, the job can be transferred to the test system.
 - **In Test**
The job is currently being tested on the test system.
 - **Tested**
The job has been tested successfully at the test system and can be transferred to the production system.

Job Context Menu

When right-clicking a job, the job context menu is opened. It provides the following entries:

- **Import from Test System**

This item is only provided on the production system when selecting a job in the **Test System** tab.

Imports the selected job from the test system into the production system. For this purpose, a display is opened where a job name can be entered manually or selected from a list. The data of the job selected in the Test System tab are imported into this job on the production system

- **Import from Production System**

This item is only provided on the test system when selecting a job in the **Production System** tab.

Imports the selected job from the production system into the test system. For this purpose, a display is opened where a job name can be entered manually or selected from a list.

- **Refresh View**

Updates the list with the most recent data.



Note

Depending on the test status of the selected job, the import option may not be selectable.

3.2.1.6 Change Request List

If Hierarchical Model Maintenance is configured in the system, the Job Management user interface contains an additional tab besides the **Job Management UI** tab to handle change requests:

Job Management UI		Production System		Change Request List		
Change Request Name	State	Control Center Name	Validation Due Date	Commissioning Date	Created By	Export Profile
t6	Finally Rejected	SLCCdeb21cjc	10/29/2008 4:28:27 PM	10/29/2008 5:27:20 PM	nbgdom\SPCAdmin	NERC
t7	Finally Accepted	SLCCdeb21cjc	10/30/2008 4:31:34 PM	10/29/2008 5:27:20 PM	nbgdom\SPCAdmin	NERC
t99	Finally Rejected	SLCCdeb21cjc	10/28/2008 4:38:20 PM	10/31/2008 5:37:40 PM	nbgdom\SPCAdmin	NERC
t5	Accepted	SLCCdeb21cjc	10/28/2008 3:35:41 PM	10/29/2008 4:35:06 PM	nbgdom\SPCAdmin	NERC
t123	Finally Withdrawn	SLCCdeb21cjc	10/28/2008 5:09:00 PM	11/18/2008 6:08:13 PM	nbgdom\SPCAdmin	NERC

[IMMRE027-041108-enUS-01.tif]

Figure 3-28 Change Request List

Change Request List

This list provides an overview of all changes requests that currently exist in the system. For each change request, it provides the following information:

- **Change Request Name**

Name of the change request.

- **State**
Current state of the change request.
- **Control Center Name**
Name of the control center where the change request has been created.
- **Validation Due Date**
Date when the validation when validation should be completed, that is, the deadline to accept or reject the CR.
- **Commissioning Date**
Planned commissioning date of the change request.
- **Created By**
Name of the user who created the change request.
- **Export Profile**
Profile used for data export.
- **Description**
Description of the change request.

CR Context Menu

Depending on the status of the control center, where IMM has been started (first level control center or second-level control center), different menu items are provided in the CR context menu.

The CR context menu of the first level control center provides the following items:

- **Save Change Request**
Saves the change request as an XML file.
- **Delete Change Request**
Deletes the change request on the local control center.
- **Change Validation Due Date**
Opens a display for changing the **Validation Due Date**.
- **Change Commissioning Date**
Opens a display for changing the **Commissioning Date**.
- **Submit to Control Centers**
Sends the change request to the affected second-level control centers.
- **Import Change Request**
Opens a display for importing the change request into an existing or new job.
- **Send Final Decision**
 - **Finally Accept Change Request**
Accepts the change request and sends the decision to the affected second-level control centers.
 - **Finally Reject Change Request**
Rejects the change request and sends the decision to the affected second-level control centers.
 - **Finally Withdraw Change Request**
Sends this decision to the affected second-level control centers.

The CR context menu of the second-level control center provides the following items:

- **Save Change Request**
Saves the change request as an XML file.
- **Delete Change Request**
Deletes the change request on the local control center.

- **Change Validation Due Date**
Opens a display for changing the **Validation Due Date**.
- **Change Commissioning Date**
Opens a display for changing the **Commissioning Date**.
- **Send Change Request**
Opens a display asking for a validation due date and a description and sends the change request to the first-level control center.
- **Import Change Request**
Opens a display for importing the change request into an existing or new job.
- **Send Download Request**
Downloads the change request.
- **Send Change Request Response**
 - **Accept Change Request**
Accepts the change request and sends the decision to the first-level control center.
 - **Reject Change Request**
Rejects the change request and sends the decision to the first-level control center.
 - **Withdraw Change Request**
Sends this decision to the first-level control center.

Depending on the status of the selected change request and the current situation, not all items are selectable.

Control Center List

This list shows the status of the selected CR at the other affected control centers.

3.2.1.7 Emergency-Backup Control Center Tab

If at least 1 Emergency-Backup Control Center is configured in the system, the Job Management user interface contains an additional tab next to the **Job Management UI** tab, showing the state of the control centers.

Job Management UI Backup Control Center Change Request List Test System			
Control Center Name	State	Database Version	Database Version Date

[IMMRE054-041109-enUS-01.tif]

Figure 3-29 Emergency-Backup Control Center Tab

Emergency-Backup Control Center Table

This table provides the following information for each control center:

- **Control Center Name**
Shows the name of the control center: for example, main control center, emergency-backup control center 1, and so on.
- **State**
Shows the current state of each control center. It indicates whether the control center state is alright, has a version that is lower than the main control center and thus needs attention, or the connection cannot be established.

- **Database Version**

Shows the database version of the control center. If a connection error occurs, an error description is shown instead of the version number.

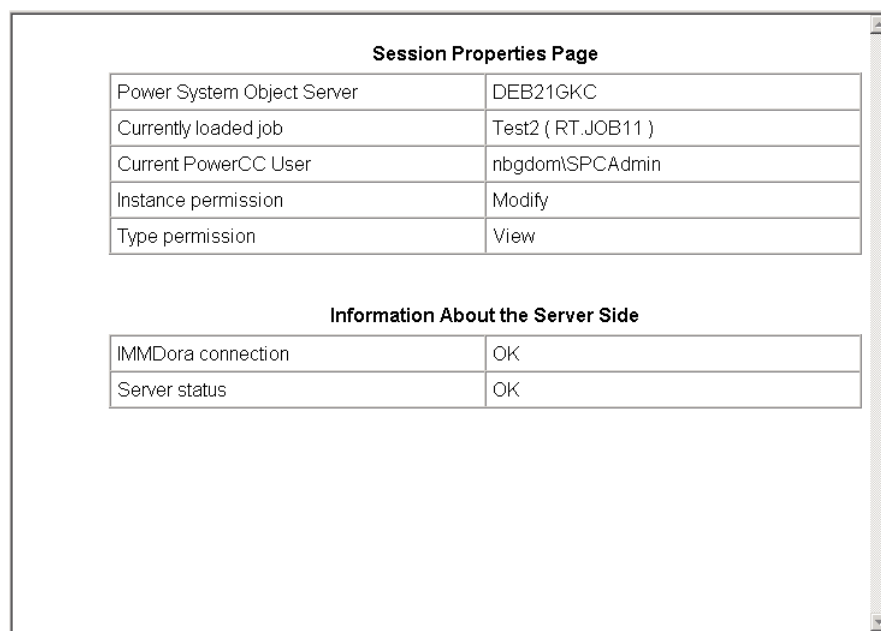
- **Database Version Date**

Shows the date of the respective database version.

3.2.2 Session Properties Tab

The **Session Properties** Tab provides information on the currently loaded dataset and general status information on the current Information Model Management session. All information presented by the Session Properties tab is read only.

The Session Properties Tab can be accessed by clicking **Session Properties** on the **File** menu.



[IMMRE013-031108-enUS-01.tif]

Figure 3-30 Session Properties Tab

The Session Properties tab shows information about the currently loaded job as well as general status information related with the current engineering session.

Power System Object Server

Name of the database server (PSOS).

Currently Loaded Job

Name of the currently loaded job.

Current Power 5 User

User name of the currently logged on Spectrum Power 5 user.

IMMDora Connection

A short status message about the current database connection status.

Server Status

The current server status of the Power System Object Server.

3.2.3 Change Log

Changes executed by the Instance Editors, are displayed in the so-called **Change Log**.

Selecting the Change Log

The Change Log Dialog is invoked by clicking the menu item **Change Log** from the **View** menu. The Change Log dialog is loaded into the working area of the Information Model Management.

☒ **Change Log Report:**

- ☒ Detailed Report
- ☐ Summary Report

☐ **XDF Change Log Export**

- ☒ With object identifier (GUIDs)
- ☐ With internal identifier (SIID)
- ☐ With RDFID

Export Filename:

☐ **RDF Change Log Export**

Namespace: Profile:

Export Filename:

[IMMRE048-150909-enUS-01.tif]

Figure 3-31 Change Log

Change Log Report

This check box determines whether the Change Log is presented in report format for viewing.

Summary Report

This option button determines the scope of the report content.

When checked, only the name of the selected instance and the name of the type of the instance is reported. Otherwise, all attributes and links of the selected instance are reported, too.

Detailed Report

This option button determines whether all details of the selected instance are included in the report.

When checked, all attributes and links of the selected instance are reported. Otherwise, only the name of the selected instance and the name of the type of the instance are reported.



Note

The settings of the option buttons **Summary Report** and **Detailed Report** are mutually exclusive.

XDF Change Log Export

This check box determines whether the Change Log is exported to an XDF file.

With Object Identifier (GUIDs)

Types and instances are exported with object identifier (technological address).

With Internal Identifier (SIID)

Types and instances are exported with the system-internal identification (SIID).

With RDFID

Types and instances are exported with the CIM-RDF identification.

Export File Name

You define the name of the file to which the Change Log is exported by:

- Entering the name in this field or
- Clicking the button **Browse**

Browse

This button opens the File Browser dialog to select the XDF output file.

RDF Change Log Export

This check box determines whether the Change Log is exported to an CIM-RDF file.

Name Space

Instances can belong to the cim, sp5, sp3 or a project scope. Selecting an entry of this list box determines the scope whose instances are exported.

Profile

A profile may be assigned to instances. Selecting an entry of this list box determines that only instances, attributes, and associations which are part of the profile are exported. If no profile is selected (the list box field is empty), all instances are exported, independent of their profile.

Export File Name

You define the name of the file to which the Change Log is exported by:

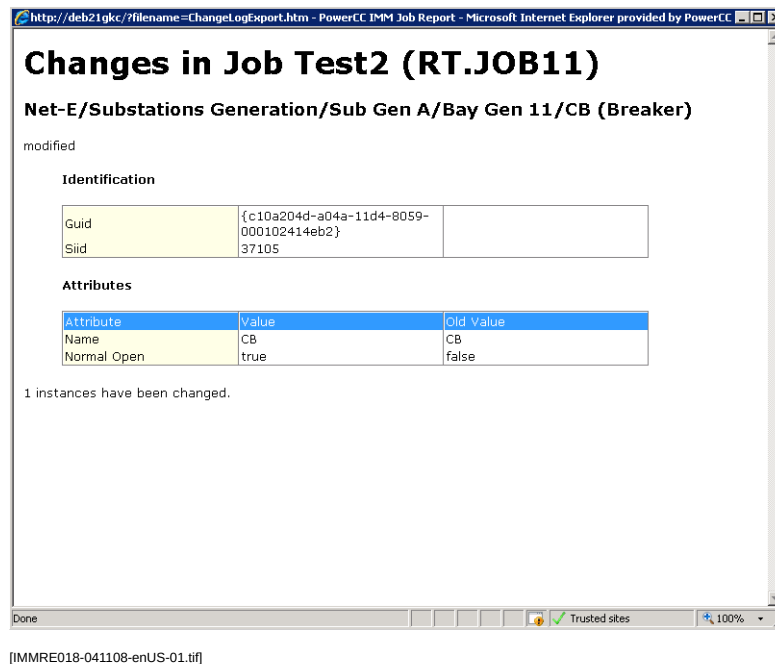
- Entering the name in this field or
- Clicking the button **Browse**

Browse

This button opens the File Browser dialog to select the CIM-RDF output file.

Execute

This button displays the Change Log and/or exports the changes.



[IMMRE018-041108-enUS-01.tif]

Figure 3-32 Change Log Report

3.2.4 Report Dialog

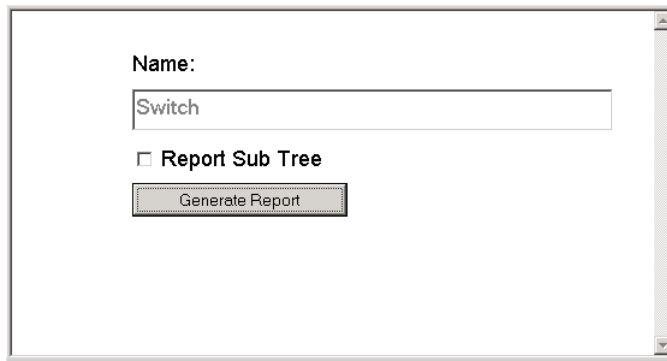
3.2.4.1 Overview

The **Report** Dialog provides tools for creating and viewing reports. A report is a summary that shows attributes and properties of the selected type or instance. Reports are pre-configured in terms of formatting and style, but you determine the amount of information shown in a report. Depending on the selected report property settings, you can decide to report only the selected type or instance or the selected type or instance and all subordinate types or instances.

The Report Dialog is accessed either by clicking **HTML Reports** on the **Tools** menu or by opening the context menu of the tree for a certain entry in the tree and clicking the menu item **HTML Reports**.

3.2.4.2 Report Dialog for Types

The Report dialog for types is used for creation of a report containing all information for the selected type. The report is then displayed in the Report Viewing window.



[IMMRE019-041108-enUS-01.tif]

Figure 3-33 Report Dialog for Types

The body of the Report Dialog for Types provides interactive elements which allow you to create a type report and to set options that influence the report creation.

Name

Contains the name of the type to be reported. By default (right after the Report Dialog was opened) it shows the name of the latest selection made in the tree view.

Selecting another type for reporting can be done either by entering the name of the respective type in the text box **Name** or by right-clicking a tree entry on the tree and clicking the menu item **HTML Reports** in the context menu of the tree.

Valid values: The name of an existing type.

Report Subtree

You determine by this check box whether all subordinated types are included in the report. When checked, the selected type itself and all subordinate types are reported. Otherwise, only the selected type itself are reported.

Detailed Report

The setting of the option button **Detailed Report** determines whether all details of the selected type are included in the report. When checked, all attributes and links of the selected type are reported. Otherwise, only the name of the selected type is reported.



Note

The settings of the option buttons **Summary Report** and **Detailed Report** are mutually exclusive.

Summary Report

The setting of the option button **Summary Report** determines the scope of the report content. When checked, only the name of the selected type is reported. Otherwise, all attributes and links of the selected type are reported, too.

Generate Report

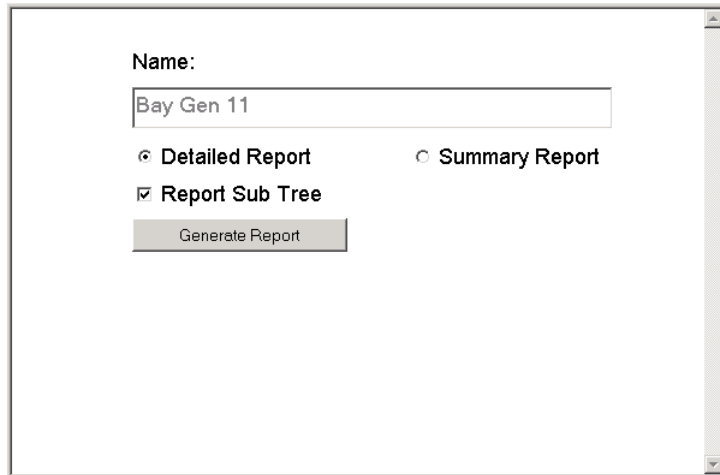
Generates the report.

The requested information is collected, formatted, and presented in a separate Report Viewing window. The Report Viewing window is opened automatically after the generation is completed.

For an example of a type report, refer to [Figure 3-35](#).

3.2.4.3 Report Dialog for Instances

The Report dialog for instances is used for creation of a report containing all information for the selected instance. The report is then displayed in the Report Viewing window.



[IMMRE020-041108-enUS-01.tif]

Figure 3-34 Report Dialog for Instances

The body of the Report Dialog for Instances provides interactive elements which allow you to create a report and to set options to influence the report creation.

Name

Contains the name of the instance to be reported. By default (right after the Report Dialog is opened) it shows the name of the latest selection made in the tree view.

Valid values: The name of an existing instance.

Detailed Report

The setting of the option button **Detailed Report** determines whether all details of the selected instance are included in the report. When checked, all attributes and links of the selected instance are reported. Otherwise, only the name of the selected instance and the name of the type of the instance are reported.



Note

The settings of the option buttons **Summary Report** and **Detailed Report** are mutually exclusive.

Summary Report

The setting of the option button **Summary Report** determines the scope of the report content. When checked, only the name of the selected instance and the name of the type of the instance are reported. Otherwise, all attributes and links of the selected instance are reported, too.

Report Subtree

The setting of the check box **Report Subtree** determines if all subordinated instances are included in the report. When checked, the selected instance itself and all subordinate instances are reported. Otherwise, only the selected instance itself are reported.

Generate Report

Generates the report.

The requested information is collected, formatted, and presented in a separate Report Viewing window. The Report Viewing window is opened automatically after the generation is completed.

For an example of an instance report, refer to [Figure 3-36](#).

3.2.4.4 Report Viewing Window

The Report Viewing window displays the content of a report.

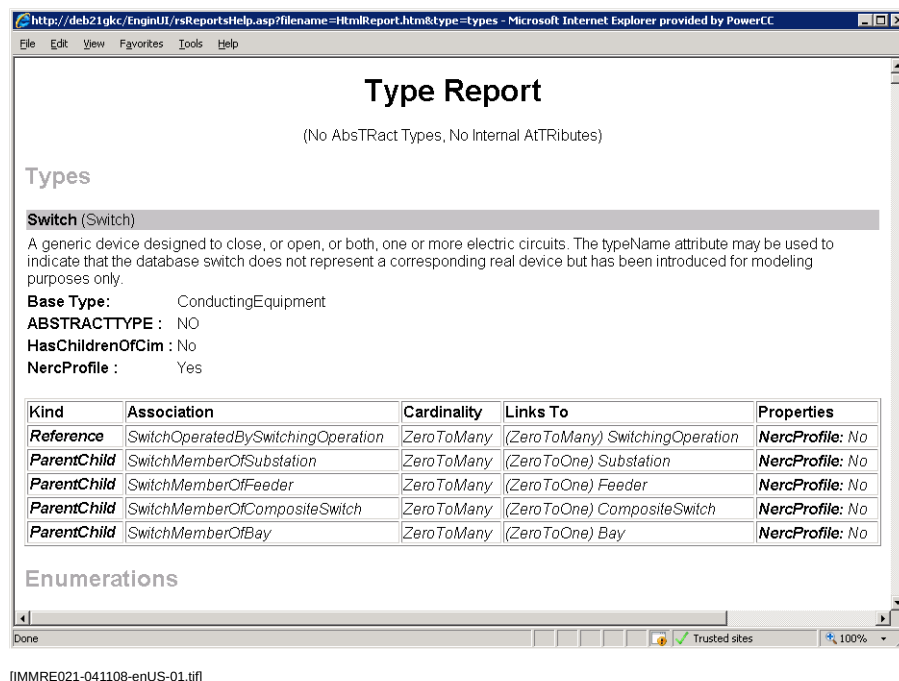


Figure 3-35 Report Viewing Window Containing a Type Report

Net-E/Substations Generation/Sub Gen A/Bay Gen 11 (Bay)

Attribute	Value
Guid	{c10a2031-a04a-11d4-8059-000102414eb2}
busBarConfiguration	
breakerConfiguration	
bayPowerMeasFlag	false
bayEnergyMeasFlag	false
SCADAFlag	false
NAFlag	false
DMSFlag	false
DisplayOrder	
MessageConfiguration	1
AreaOfResponsibilityId	3
pathName	
mRID	
localName	
Description	
aliasName	

[IMMRE022-041108-enUS-01.tif]

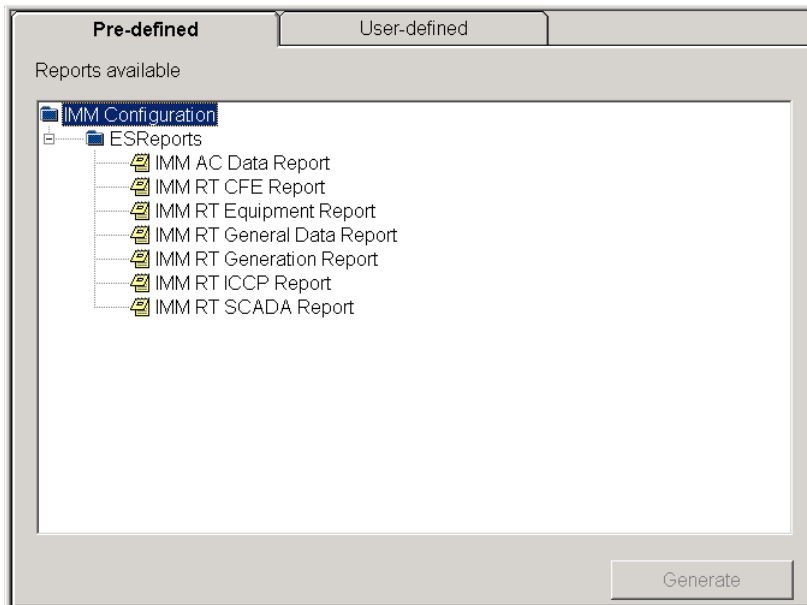
Figure 3-36 Report Viewing Window Containing an Instance Report

3.2.5 Report Generator Dialog

3.2.5.1 Overview

An Information Model Management Report formats instance data and displays it in MS EXCEL format. These reports can be pre-defined and delivered with the Spectrum Power Information Model Management or defined by you after the installation of the Spectrum Power Information Model Management (user-defined).

The Report Generator Dialog provides the generation of pre-defined and user-defined reports. It is accessed either by clicking the menu item **EXCEL Reports** on the **Tools** menu or by opening the context menu for a certain entry in the tree and clicking the menu item **EXCEL Reports**.

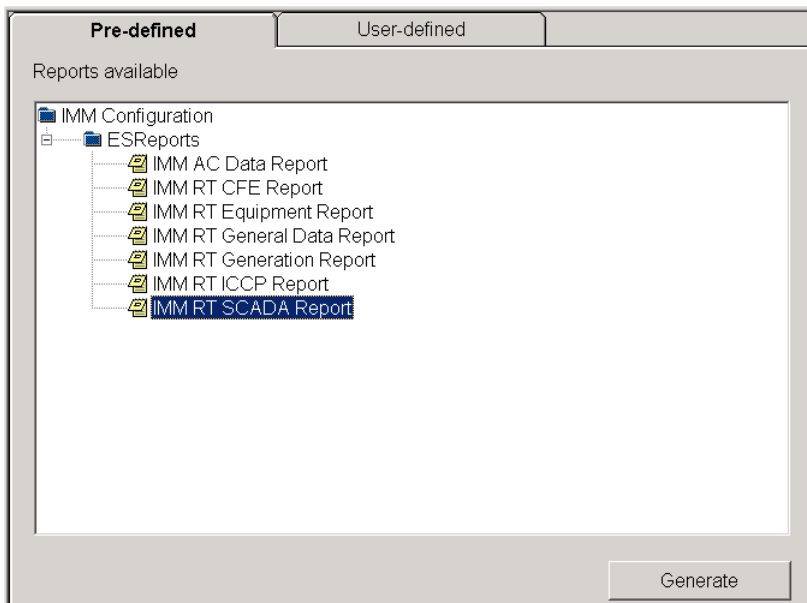


[IMMRE023-041108-enUS-01.tif]

Figure 3-37 Report Generator

The body of the Report Generator dialog provides 2 tabbed tabs for specifying pre-defined and user-defined reports.

3.2.5.2 Predefined Reports



[IMMRE024-041108-enUS-01.tif]

Figure 3-38 Predefined Reports

All available instances of the type ViewGroup where the attribute type has the value Report are displayed. You can select an instance to generate an appropriate report.

The following predefined reports are provided:

Reports Available

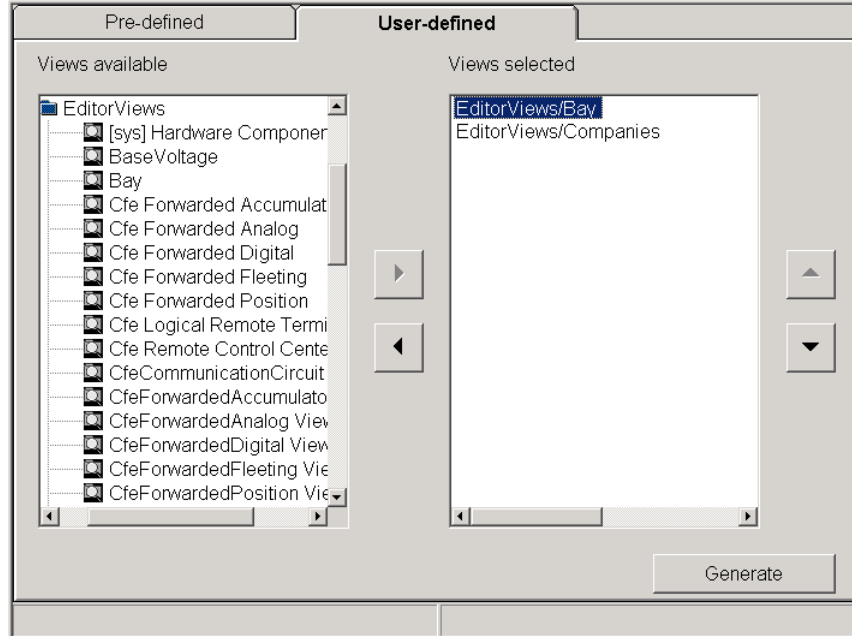
- IMM AC Data Report
- IMM RT CFE Report
- IMM RT Equipment Report
- IMM RT General Data Report
- IMM RT Generation Report
- IMM RT ICCP Report
- IMM RT Presentation
- IMM RT SCADA Report

Generate

The data is transferred to an MS EXCEL spreadsheet. You can format, print, or save this spreadsheet. No control to the Excel spreadsheet is passed to you. If you want to access the generated file, open it with Excel directly.

3.2.5.3 User-Defined Reports

In addition to the predefined reports, you can also create user-defined reports. For this purpose a list of all available ViewGroups is displayed, from which you can select the desired one.



[IMMRE025-041108-enUS-01.tif]

Figure 3-39 User-Defined Reports

Views Available

All available instances of the type ViewGroup are displayed. You can select multiple instances and move them to the right list by clicking the Move right arrow button.

Views Selected

The selected instances are displayed in this list. You can browse through the entries by clicking the corresponding Up and Down button. Selected entries can be removed from this list by clicking the Move left arrow button. Removed entries are inserted into the left list.

A user-defined report is generated with the selected instances. For each selected instance, an EXCEL sheet is created in the result EXCEL file.

Left/Right Buttons

Moves the selected instance entries between the 2 lists. By double-checking an entry of the left list, this entry can be moved also into the list views selected.

Up/Down Buttons

Browses through the instance entries in the list views selected.

Save as Report

The current instance selection is saved as a pre-defined report. You are prompted to enter a name under which the report is saved. After the save is completed, the report is displayed in the **Reports Available** list on the **Pre-defined** Reports tab.

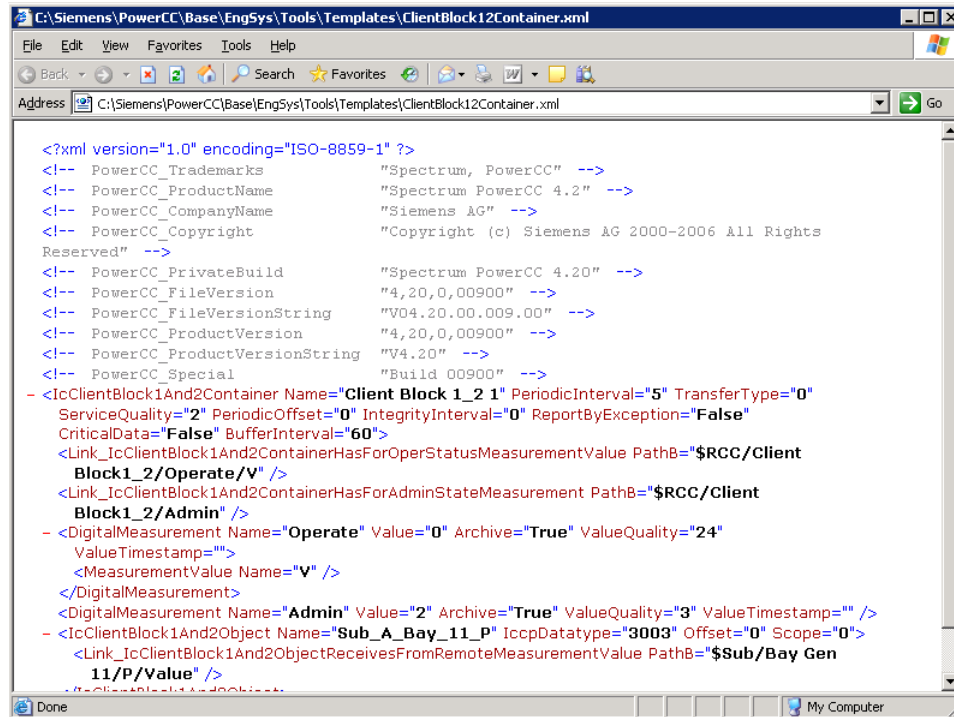
Generate

Described in the previous chapter.

3.2.6 Import Dialog

The **Import** dialog allows you to populate the currently loaded dataset from an XDF or CIM-RDF file. XDF files are ASCII files that contain import data definitions of types or instances in XML format. CIM-RDF files are files in XML format that contain import data definitions of instances.

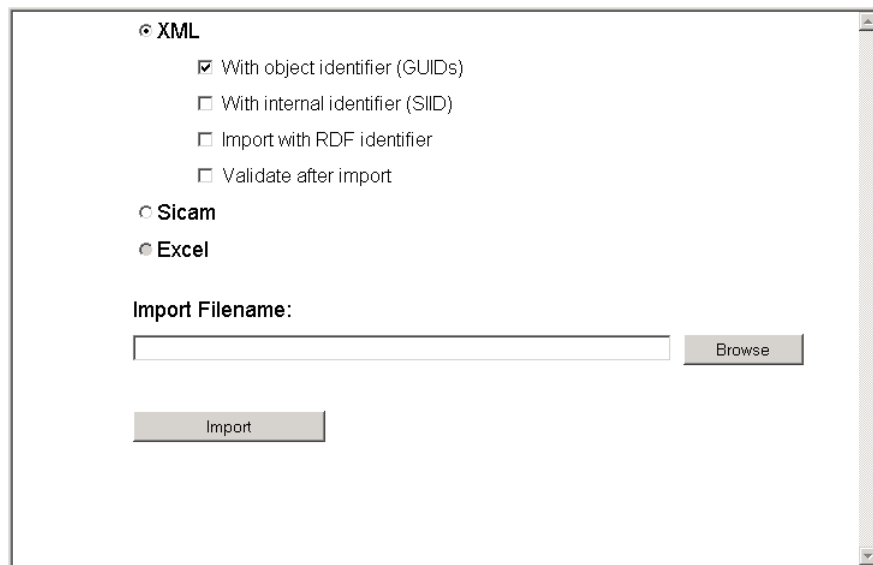
The Import dialog is accessed by selecting the menu item **Import** from the **File** menu.



[IMMRE042-180108-enUS-01.tif]

Figure 3-40 Example of an XDF File

Import Dialog



[IMMOP046-150909-enUS-01.tif]

Figure 3-41 Import Dialog

Interactive elements provided on the body of the Import dialog allow you to specify an XDF or CIM-RDF file type and to invoke data import.

XML

Depending on the file content, XML imports data either in XDF format or CIM-RDF format.

Import with Object Identifier (GUIDs)

If this check box is selected, objects are imported with object identifier (technological address).

Import with Internal Identifier (SIID)

If this check box is selected, objects are imported with their system-internal object identifier (SIID).

Import with RDF Identifier

If this check box is selected, objects are imported with their RDF identifier.

Validate after Import

If this check box is selected, the imported data are validated automatically after import.

Sicam PAS

Imports XML files generated from SICAM PAS.

Import File Name

The text box **Import File Name** contains the full path name of the file to be imported.

Specifying an XDF or CIM-RDF file can be done either by entering its full path name in the text box **Import File Name** or by clicking the button **Browse** and selecting a file from the File Browser dialog. The full path name of the selected file is automatically displayed in the text box **Import File Name** after the File Browser dialog has been closed.

Valid values: The entire path name of an existing XDF or CIM-RDF file.



Note

The text box **Import File Name** can hold only one full path name. Multiple path name entries are not allowed.

Template File Name



Note

This text box is only available if Excel import has been selected.

Specifying a template file can be done either by entering its full path name in the text box **Import File Name** or by clicking the button **Browse** and selecting a file from the File Browser dialog. The full path name of the selected file is automatically displayed in the text box **Template File Name** after the File Browser dialog has been closed.

Browse

Opens a File Browser dialog from which you can select a file for import. After selection, the full path name of the selected file is displayed in the text box **Import File Name**.

Import

Invokes the import process.

The specified file is scanned, its data records are converted and the resulting type or instance data are stored.

When errors occurred during the data import, then a message is displayed in the Import Log.

3.2.7 Export Dialog

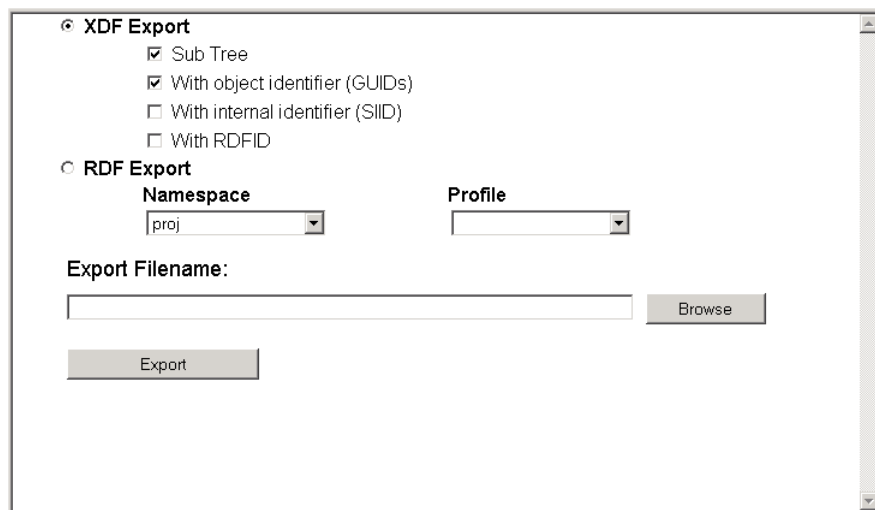
The **Export** dialog provides tools to create an ASCII file of data definitions of a selected type or instance as XDF file, instance data can also be exported as CIM-RDF file.



Note

By default, the entire subtree of the selected type or instance is exported. This includes the selected type or instance itself, the parent, and all subordinated types or instances with the internal attributes.

The Export dialog is accessed either by clicking **Export** on the **File** menu or by opening the context menu for a certain entry in the tree and clicking the menu item **Export**.



[IMMRE029-041108-enUS-01.tif]

Figure 3-42 Export Dialog

Header

The header of the Export dialog shows the name of the selected type or instance.

XDF Change Log Export

This check box determines whether the data are exported to an XDF file.

Subtree

All descendant instances under the selected node are exported.

With Object Identifier (GUIDs)

Types and instances are exported with object identifier (technological address).

With Internal Identifier (SIID)

Types and instances are exported with the system-internal identification (SIID).

With RDFID

Types and instances are exported with the RDF identification.

RDF Change Log Export

This check box determines whether the data shall be exported to an CIM-RDF file.

Name Space

Instances can belong to the cim, sp5, sp3 or a project scope. Selecting an entry of this list box determines the scope whose instances are exported.

Profile

Instances may be assigned to a profile. Selecting an entry of this list box determines that only instances with this profile are exported. If no profile is selected (the list box field is empty), all instances are exported, independent of their profile.

Export File Name

Path name of the XDF/CIM-RDF file to be created.

Browse

Opens a File Browser dialog from which you can select an XDF/CIM-RDF file. After selection, the full path name of the selected file is displayed in the text box **Export File Name**.

Export

Activates the data export.

All information related with the selected type or instance and all its children is collected, converted into XML and written into the specified XDF or CIM-RDF file.

When the data export was completed without errors a message is displayed in the status bar.

When errors occurred during the data export, the Export Error Message box pops up to notify you.



Note

The Export dialog remains open in the working area after a data export is completed.

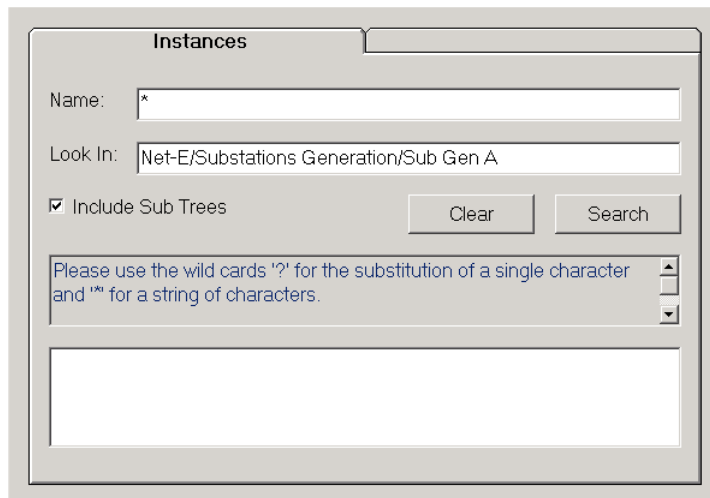
3.2.8 Instance Search Dialog

The **Instance Search** dialog provides tools that allow you to search for specific instances.

Features of the Instance Search Dialog

- Search for instance names.
- You can select to search the entire instance tree (starting with the root instance), or to search within a specific selected subtree (for example, look in: /Net-E/Substations Generation/Sub Gen A), or to search with the tree defined by the applied filter.
- Supports placeholders (asterisk).
- The search results list provides entries for immediate selection.
The entries of the search result list can be selected. When an entry is double-clicked, the tree navigates to the selected instance and selects it.

The Instance Search dialog is accessed either by clicking **Search** on the **Edit** menu or by clicking the appropriate toolbar button.



[IMMRE045-180108-enUS-01.tif]

Figure 3-43 Instance Search Dialog

Name

Contains the search string.

The search string can contain placeholders (asterisk symbol).

Valid values: Any character string.

Possible placeholders are:

- *
Stands for any set of characters
- ?
Stands for a single character

Look In

Specifies the part of the instance tree that is searched.

If this field is left empty, the search starts at the root instance and searches the entire instance tree.

Valid values: An instance path name.

Include Subtrees

Specifies whether the subtrees are searched, too.

If this check box is checked, all subtrees of the specified part of the instance tree are searched, too. Otherwise, only the specified subtree are searched and the contained subtrees are not considered.

Clear

Clears all text boxes.

Search

Starts the search.

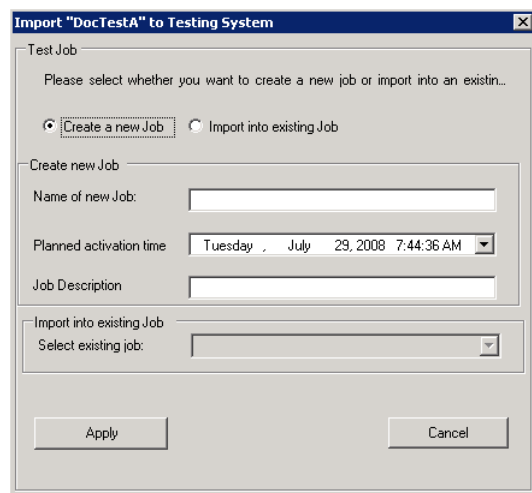
Search Results List

Shows the path names of all instances matching the specified search string.

The list entries can be clicked to select the respective instances immediately.

3.2.9 Import to Test System

This display is opened after the item **Import from Production System** has been selected from the job context menu in the **Test System** tab.



[IMMRE030-041108-enUS-01.tif]

Figure 3-44 Import Job to Test System

Create a New Job

Selecting this option button determines that the selected job shall be imported into a new job at the Test System.

Import Into Existing Job

Selecting this option button determines that the selected job shall be imported into an existing job at the Test System.

Name of New Job

When the option button **Create a New Job** has been selected, a name for the new job must be entered here.

Planned Activation Time

When the option button **Create a New Job** has been selected, a planned activation date for the new job must be entered here. This field is only available if the time-based job model is used.

Job Description

When the option button **Create a New Job** has been selected, a description for the new job can be entered here.

Select Existing Job

When the option button **Import into Existing Job** has been selected, this list box provides a list of all available jobs at the Test System. One of these jobs must be selected for importing the data of the job to be tested.

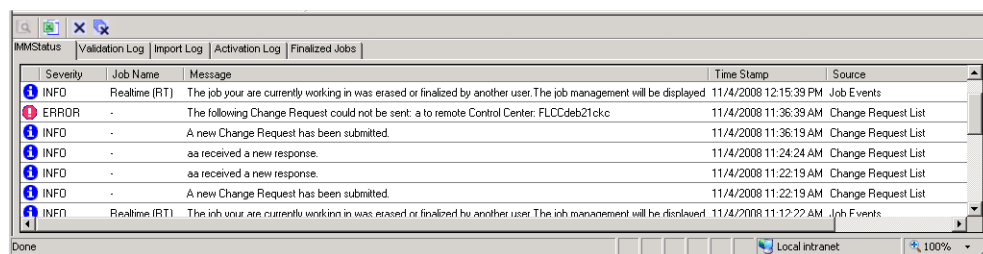
3.2.10 Import to Production System

This display is opened after the item **Import from Test System** has been selected from the job context menu in the **Production System** tab. It has the same structure and similar functionality as the Import to Test System display (see [3.2.9 Import to Test System](#)).

3.2.11 Logs

3.2.11.1 IMM Status

The IMM Status tab shows the messages that occurred during work with IMM.



[IMMRE031-041108-enUS-01.tif]

Figure 3-45 IMM Status Log

The IMM Status log only consists of 1 area, which provides the following information for each message:

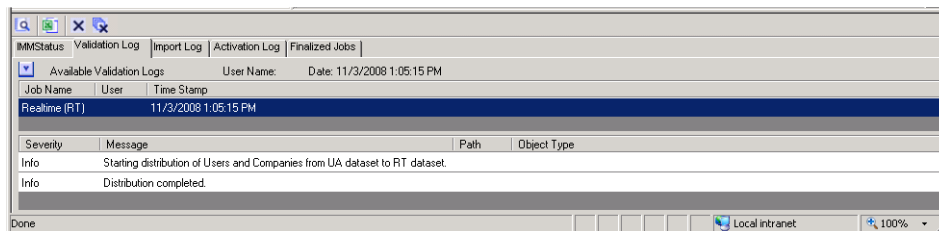
- Severity
- Job Name
- Message
- Time Stamp
- Source

3.2.11.2 Task Log

The Task Log contains all messages that occur during an IMM UI task such as copy/paste, cut/past, delete sub-tree, graphic editor processing, modify instance and type data etc. Where applicable, the object path is a hyperlink that can be used to navigate to the instance in question.

3.2.11.3 Validation Log

The **Validation Log** contains a summary of all validation messages generated during the validation of a job or when an instance change was saved.



[IMMRE032-041108-enUS-01.tif]

Figure 3-46 Validation Log

The log overview section shows the following information for each validation log:

- Job Name
- User
- Time Stamp

When selecting a row and clicking the button Show or selecting a row with a double-click, the content of the selected validation log is displayed in the detail section.

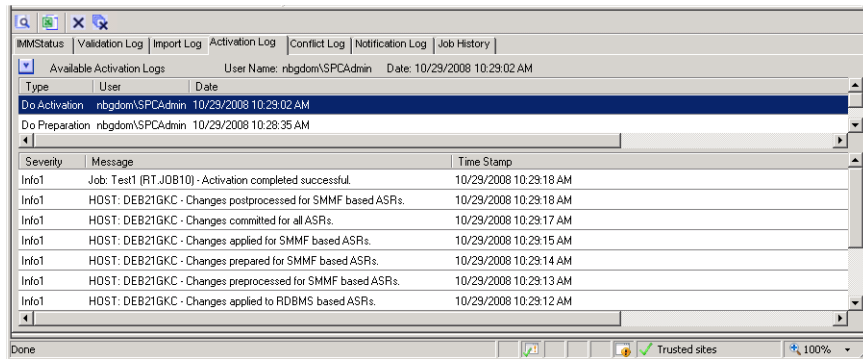
The following information is displayed for each entry (only a subset of these fields are displayed by default, the rest (for example, error id,...) can be selected):

- Error ID (internal ID for the error)
- Severity
- Path
- Detail Description
- Short Identifier (internal identifier)
- Time Stamp
- Message

Double-clicking a line in the detail section displays the affected instance in the working area. If you are within a dataset or job, the instance is displayed in the currently selected editor. If you are not in a job, the Single-Instance Editor is started.

3.2.11.4 Activation Log

The **Activation Log** contains all logs that were created during the preparation or activation of a dataset or job.



[IMMRE033-041108-enUS-01.tif]

Figure 3-47 Activation Log

The log overview section shows the following information for each activation log:

- Type
- User
- Date

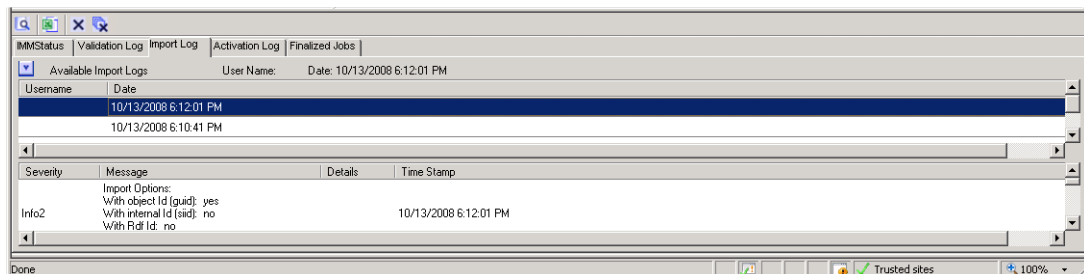
When selecting a row and clicking the button Show or selecting a row with a double-click, the content of the selected activation log is displayed in the detail section.

The following information is displayed for each entry:

- Severity
- Message
- Time Stamp

3.2.11.5 Import Log

The **Import Log** contains all logs that were created when an import was done. Depending on whether a dataset or a job is selected in the tree view, either all import logs of the dataset or only the job-specific import logs are displayed.



[IMMRE034-041108-enUS-01.tif]

Figure 3-48 Import Log

The log overview section shows the following information for each import log:

- User Name
- Date

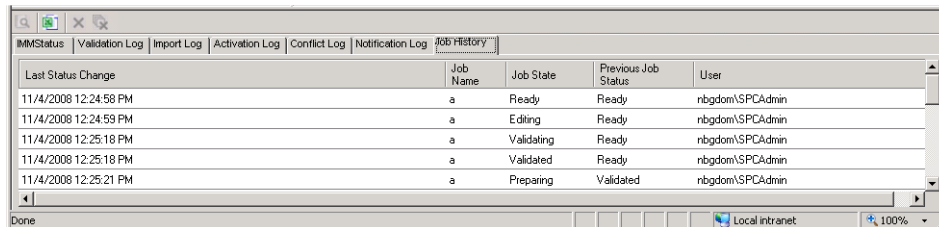
When selecting a row and clicking the button Show or selecting a row with a double-click, the content of the selected import log is displayed in the detail section.

The following information is displayed for each entry:

- Severity
- Message
- Time Stamp

3.2.11.6 Job History

The **Job History** shows all status changes of the currently selected job.



Last Status Change	Job Name	Job State	Previous Job Status	User
11/4/2008 12:24:58 PM	a	Ready	Ready	nbgdom\SPCAAdmin
11/4/2008 12:24:59 PM	a	Editing	Ready	nbgdom\SPCAAdmin
11/4/2008 12:25:18 PM	a	Validating	Ready	nbgdom\SPCAAdmin
11/4/2008 12:25:18 PM	a	Validated	Ready	nbgdom\SPCAAdmin
11/4/2008 12:25:21 PM	a	Preparing	Validated	nbgdom\SPCAAdmin

[IMMRE035-041108-enUS-01.tif]

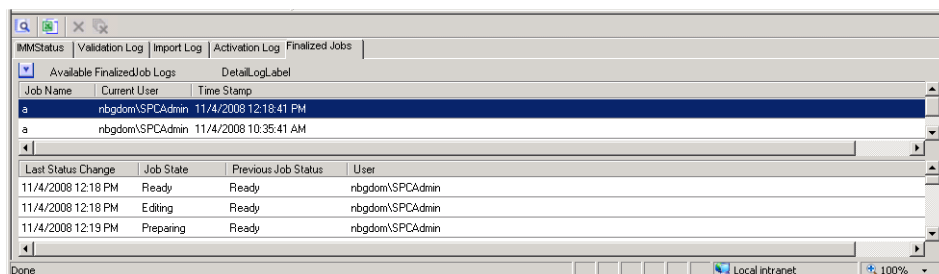
Figure 3-49 Job History

The Job History log only consists of 1 area, which provides the following information for each status change:

- Last Status Change
- Job Name
- Job Status
- Previous Job Status
- User

3.2.11.7 Finalized Jobs

The Finalized Jobs tab contains an overview of all finished job logs.



Job Name	Current User	Time Stamp
a	nbgdom\SPCAAdmin	11/4/2008 12:18:41 PM
a	nbgdom\SPCAAdmin	11/4/2008 10:35:41 AM

[IMMRE036-041108-enUS-01.tif]

Figure 3-50 Finalized Jobs

The log overview section shows the following information for each log:

- Job Name
- Current User
- Time Stamp

When selecting a row and clicking the button Show or selecting a row with a double-click, the content of the selected log is displayed in the detail section.

The following information is displayed for each entry:

- Last Status Change
- Job Status
- Previous Job Status
- User

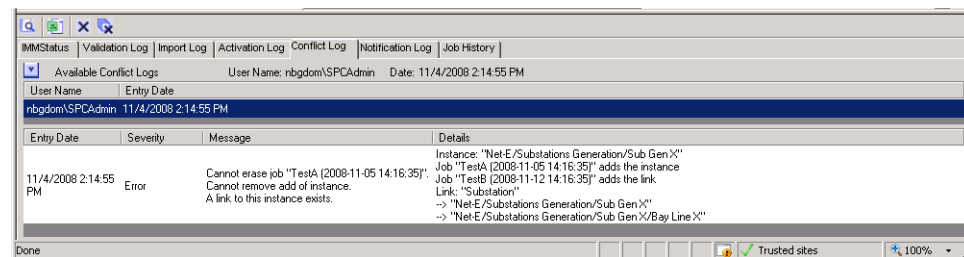
3.2.11.8 Conflict Log



Note

This log is only available if the time-based job model is used.

The Conflict Log tab contains an overview of all integrity conflicts that occurred when rescheduling, deleting, or publishing a time-based job model.



[IMMRE037-041108-enUS-01.tif]

Figure 3-51 Conflict Log

The log overview section shows the following information for each log:

- User Name
- Entry Date

When selecting a row and clicking button Show or selecting a row with a double-click, the content of the selected log is displayed in the detail section.

The following information is displayed for each entry:

- Entry Date
- Severity
- Message
- Details

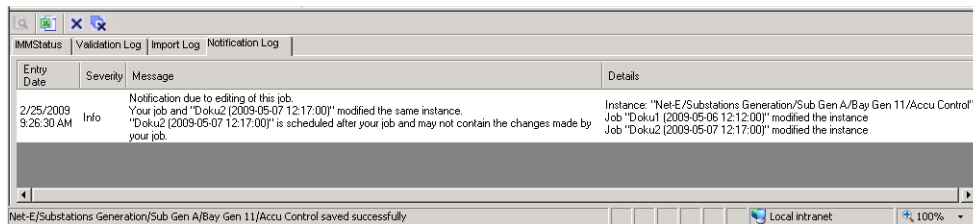
3.2.11.9 Notification Log



Note

This log is only available if the time-based job model is used.

The Notification Log tab contains all messages that occur when the modifications in one job within a time-based job model are in conflict with the modifications made in another job.



[IMMRE039-250209-enUS-01.tif]

Figure 3-52 Notification Log

The Notification log only consists of 1 area, which provides the following information for each notification:

- Time Stamp
- Severity
- Message
- Details

3.3 Instance Editors

3.3.1 Single-Instance Editor

3.3.1.1 Introduction

The Single-Instance Editor is used to view a single instance, to modify an instance, or to create an instance (see [3.3.1.7 Instance Creation Dialog](#)).

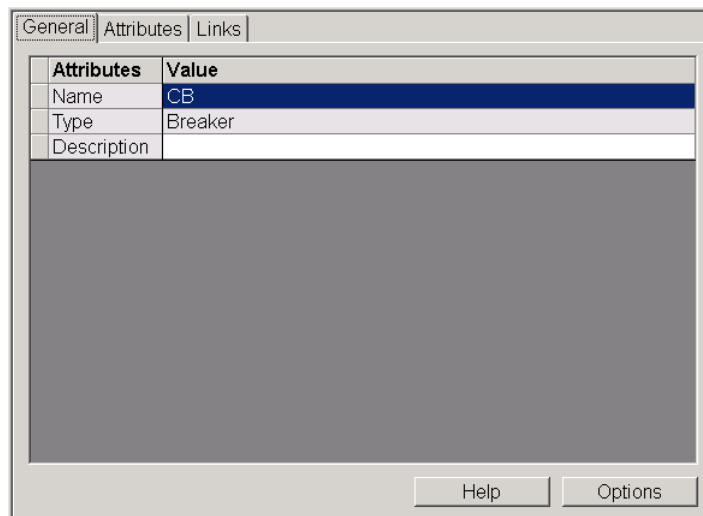
Selecting the Single-Instance Editor

The Single-Instance Editor is opened automatically after:

- An entry on the instance overview tree has been selected or
- An instance creation has been invoked in the Type Selection Display (see [3.3.1.7 Instance Creation Dialog](#)).

The Single-Instance Editor can be also invoked by selecting the menu item **View > Instance Editors > Single View** or by clicking the appropriate toolbar button.

When the Single-Instance Editor opens to display an existing instance, the **Attributes** tab is exposed automatically. When the Single-Instance Editor opens to create an instance, the **General** tab is exposed.



[IMMRF008-051108-enUS-01.tif]

Figure 3-53 Single-Instance Editor

The Single-Instance Editor contains Tabbed Tabs each of which is used to set or modify distinct kinds of instance data.

For additional information on the generic features of tabs, refer to [3.1.5.3 Tabs](#). For additional information on the editor context menu of the Single-Instance Editor, refer to [3.1.5.4 Table Editor](#).

The following Tabbed Tabs are provided on the body of the Single-Instance Editor:

- General tab

- Attributes tab
- Links tab
- Protocol tabs (only available for PI/CFE instances)

Options

This button invokes the **Options** dialog (see [3.3.1.6 Options Dialog](#)).

3.3.1.2 General Tab

The **General** tab is used to configure and modify general instance properties.

Attributes	Value
Name	CB
Type	Breaker
Description	

[IMMRF008-051108-enUS-01.tif]

Figure 3-54 Single-Instance Editor: General Tab

Name

This field contains the name of the respective instance.

Valid values: a character string of up to 100 characters. The following characters are not allowed: " / "and "\"



Note

When the Single-Instance Editor has been opened for the configuration of a new instance, the text box **Name** is empty. In this case, an instance name must be entered before another operation can be performed. Once a new instance has been saved, the instance name cannot be changed and the text box **Name** is disabled.



Note

Some applications require specific names (for example, WinCC requires DigitalMeasurement instances that represent switching devices to be named Status or other name as defined in the presentation setup).

Type

This read-only field displays the label of the type on which the instance is based.

For additional information on this property, refer to the description of the type property **Label** in [3.5.1.4 Attributes Tab](#).

Description

This field allows you to enter a description for the selected instance.

For additional information on this property, refer to the description of the type property **Help Message** in [3.5.1.3 General Tab](#).

3.3.1.3 Attributes Tab

The **Attributes** tab is used to enter attribute values for the selected instance.

The elements provided on the Attributes tab, such as text boxes, form a table whose rows represent instance attributes and whose columns represent properties. Instance attributes as well as properties (except the **Value** property) are automatically retrieved from the type of the instance and are read only. Hence, they cannot be modified or deleted and the related elements are disabled.

Table columns can be unavailable (temporarily removed from the Attributes tab) or made visible using the editor context menu. When the number of shown instance attributes or attribute properties exceeds the Attributes tab, scroll bars are automatically attached to the bottom edge and/or the right edge of the Attributes tab to allow scrolling of the page content.



Note

Despite the fact that an instance gets all attributes and attribute properties from a certain type, some of the attributes may not be available on the Attributes tab due to the setting of the type property **Internal**. For additional information on this topic, refer to the description of the type property **Internal** in [3.5.1.4 Attributes Tab](#).

Attributes	Value
Adaptive Forecast Flag	False
Alias Name	
Area Of Responsibility Id	Area A
Breaker Type	Non-specific
Continuous Rating	0
Control Enable	True
Current Flow	
DMS Flag	False
Disconnects two Voltage Levels	False
Display Order	
Fault Level Rating	0
Flow Breaker Flag	False
In Transit Time	0
Local Name	
Message Configuration	Default

[IMMRF009-051108-enUS-01.tif]

Figure 3-55 Attributes Tab

The leftmost table column consists of static text boxes showing the labels of the respective instance attributes. The top row of the table consists of static text boxes containing character strings that denote the purpose of the respective properties. Both the top row and the leftmost column are read only. They are visible at all times.

The number of attributes is not static and can change depending on the type. Normally, at least 1 attribute (for example, **Description**) is shown. The number of properties is fixed.

Value

The only property that can be modified is the **Value** property. Value contains the actual value of an instance attribute.

Property settings, such as the settings of the properties **Data Type**, **Required**, and so on, can require a value for the **Value** property and can apply restrictions regarding type, length and format of the entered value.

When no value was entered for the **Value** property, and a default value (property **Default Value**) is available, the default value is used.

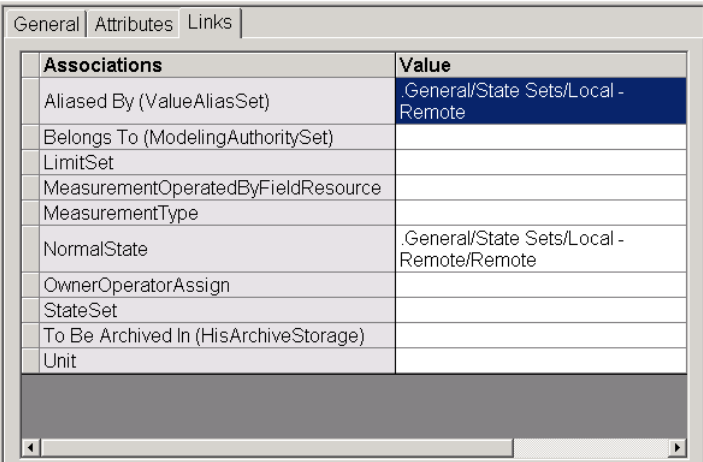
Valid values: depend on the setting of the property **Data Type**.

3.3.1.4 Links Tab

The **Links** tab allows you to establish links between instances. A link implies an association between the currently edited instance and another instance. These associations are used to assign an instance to another without forcing a hierarchical relationship between them. For example, a switch has a message set.

The reference associations of the underlying type determine the number and the characteristics of the links.

Each link is based on a reference association provided by the type of the respective instance and can refer to one or more instances. The currently edited instance represents 1 side of the link (the referring side), the other sides (the referenced sides) by one or more instances that can be selected from the instance overview tree.



Associations	Value
Aliased By (ValueAliasSet)	.General/State Sets/Local - Remote
Belongs To (ModelingAuthoritySet)	
LimitSet	
MeasurementOperatedByFieldResource	
MeasurementType	
NormalState	.General/State Sets/Local - Remote/Remote
OwnerOperatorAssign	
StateSet	
To Be Archived In (HisArchiveStorage)	
Unit	

[IMMRF010-051108-enUS-01.tif]

Figure 3-56 Links Tab

Association

This column shows the labels of all available hierarchy and reference associations of the respective instance in alphabetical order.

The type on which the respective instance is based determines the number of available associations and their properties.

Value

The entries of this column represent the referenced sides of the respective link (the entire path name of an existing instance). You can edit these entries.

The type underlying the referenced instance must match the type defined by the reference association. If not, the selection is ignored and a message box opens that indicates the permitted type.

3.3.1.5 Protocol Tabs

The Protocol Tabs are only available for PI/CFE instances and allow you to enter protocol-specific attribute values for the selected instance.

Examples for provided Protocol Tabs are:

- CDC2
- DNP30 ASE
- IEC 101
- IEC 104
- SINAUT 8FW
- TG 800
- TG 8979

**Note**

PI/CFE protocol pages are based on IMM views. IMM views are special instances of the Application Configuration (AC) dataset used to model the appearance of a certain tab of the Single-Instance Editor. The number, scope, and appearance of IMM views, and thus, the PI/CFE protocol pages, is pre-defined. However, you can customize a PI/CFE protocol page by modifying the related IMM view when you have appropriate data modification rights for the AC dataset.

The Protocol tab is generated dynamically. The protocol is determined based on the data point and the name of the RTU and the corresponding Protocol tab is displayed. For additional information on the provided elements of the Protocol Tabs, refer to description of the Attributes tab in [3.3.1.3 Attributes Tab](#). For additional information on protocol attributes, refer to the following documents: [Reference Guide Communication Front End \(CFE\)](#) and [Reference Guide Telecontrol Interface \(TCI\)](#).

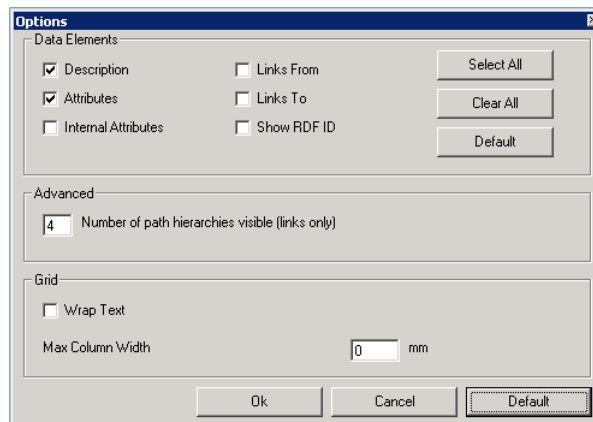
Attributes	Value
Acknowledge Time Out	150
Acknowledge Type	Single character
Alias Name	
Area Of Responsibility Id	System
Baud Rate	19200 Baud: TG800, TG8979, CDC 2, IEC 101
Constant Carrier	True
Data Length	255
Description	
Listen Mode	False
Message Configuration	Default
Polling Time	0
Postconditioning Delay	0
Preconditioning Delay	0
Serial Port	3

[IMMRF011-051108-enUS-01.tif]

Figure 3-57 Example of a Protocol Tab (IEC Protocol Tab)

3.3.1.6 Options Dialog

The **Options** dialog allows you to specify what data types shall be shown in the Single-Instance Editor.



[IMMRF012-051108-enUS-01.tif]

Figure 3-58 Options Dialog

3.3.1.7 Instance Creation Dialog

You create an instance in the following steps:

- Type Selection
You select the type of the new instance.
- Instance Configuration
You configure the new instance in the **General** tab of the Single-Instance Editor.

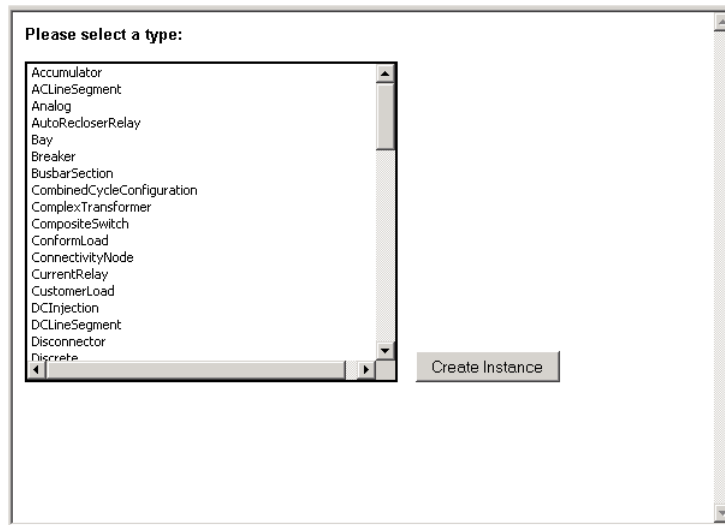
You can invoke the display for type selection in the following ways:

- Selecting an instance in the instance overview tree and clicking **New Instance** on the **Insert** menu
- Right-clicking an instance in the instance overview tree and clicking the menu item **New** in the context menu of the tree



Note

The selected instance represents the parent of the new instance. If no instance has been selected from the instance overview tree before instance creation is invoked then the most recently selected type (still highlighted in the instance overview tree) is used as parent. This can be the root instance.



[IMMRF013-051108-enUS-01.tif]

Figure 3-59 Type Selection Dialog

Type List

In this list all types that can be used to create an instance are displayed.

The hierarchy associations of the type underlying the selected instance determine the list content. Only types with a hierarchy association that refer to the type of the selected instance are considered. For additional information on hierarchy associations, refer to [3.5.1.6 Hierarchy Tab](#).

Entries of the Type List can be double-clicked to invoke immediate instance creation (refer to the description of the button **Create Instance** for further details on this procedure). Single-clicking an entry only selects the respective type. Then, instance creation can be invoked by clicking the button **Create Instance**.

Create Instance

By this button you select the Single-Instance Editor for the configuration of the new instance.

Configuration of a New Instance

When you have clicked the button **Create Instance**, the Type Selection is closed and the Single-Instance Editor is opened allowing you to configure the new instance.



Note

When the Type Selection was invoked by selecting a tree entry in the instance overview tree and clicking **New Type** on the context menu of the tree, the name of the selected instance is passed on to the Single-Instance Editor and is shown as additional information in the Header of the Single-Instance Editor.

You configure a new instance in the **General** tab of the Single-Instance Editor (see [3.3.1.2 General Tab](#)).

3.3.2 Table Editor

3.3.2.1 Introduction

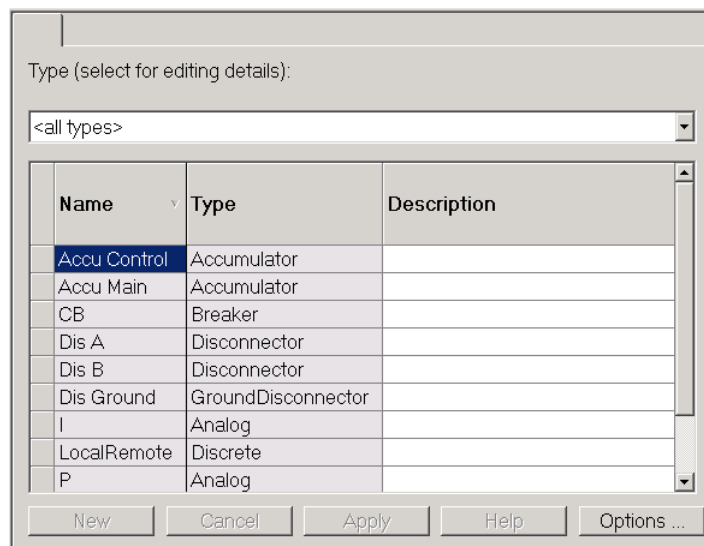
The Table Editor is used to view all subordinated instances and their associated attributes of a selected instance at a glance. These data are presented as a table. Attribute modifications can be performed immediately by changing the table cell contents.

The Table Editor is used to:

- Display and modify child instances of the selected instance
- Create instances

The Table Editor provides the following views to display instance information in a tabular form:

- Table view
- Communication view



[IMMRF014-051108-enUS-01.tif]

Figure 3-60 Table Editor

The Table Editor displays the selected instance information in a tabular form. Depending on the selected view different information and different kinds of tables are displayed. In the following sections the different views and features of the Table Editor are described, respectively.

3.3.2.2 Table View

The Table view of the Table Editor provides 2 kinds of views:

- Summary view
- Detailed view

Summary View

In the Summary view, all subordinated instances (child instances) of the instance selected on the tree view are displayed. In the Summary view, there can be instances of different types. This means only the instance name

(path), the type and the description can be shown. For more details, it is necessary to select a type and open the Detailed view.

Detailed View

In the Detailed view, all attributes and links for a selected type for all instances of this type underneath the selected instance are displayed. In this view, it is possible to enter a new instance of a specific type or to modify attributes and links. The type can only be selected from a list of types, which are allowed under the selected node. The table shows attributes and links only if the corresponding options are set (see [3.3.1.6 Options Dialog](#)).



Note

The list of instances can show the complete subtree of the selected node (optionally). A new instance can only be created under the selected node. After the node is specified only the types that are allowed under the node can be selected. The information of what types can be selected is part of the hierarchical associations defined for types.

The Table view is invoked by selecting the menu item **View > Instance Editors > Table View** or clicking the appropriate toolbar button. The Summary View of the Table Editor opens. The Detailed view of the Table Editor can be invoked by double-clicking any Type row.

Name	Alias Name	Area Of Responsibility Id	Breaker Conf
Bay Gen 11		Area A	
Bay Gen 12		Area A	
Bay Gen 13		Area A	
Bay Line A1		Area A	
Bay Line A2		Area A	
*			

[IMMRF015-051108-enUS-01.tif]

Figure 3-61 Table Editor: Detailed View

Type (Select for Editing Details)

A list box that shows all allowed types.

The **Type** list can be used to filter the content of the table grid. When a particular type is selected from the **Type** list, the content of the Table Grid is updated to show only those instances that are based on the selected type.



Note

A similar filtering can be achieved by double-clicking an entry in the Type column of the table grid.

Table Grid

The table grid shows subordinated instance information of the selected instance in a tabular form. The columns represent the attributes. The rows represent the instances or types. Each row displays attribute values of one single instance. By default, the table grid shows the attributes that are contained in all subordinated instances. By default, the shown attributes are **Name**, **Type** and **Description**. These attributes are contained in all instances.

The settings specified in the **Advanced** block of the **Options** dialog determine the number of shown instances. For additional information on these settings, refer to [3.3.2.5 Options Dialog](#).



Note

According to the settings made on the Options dialog, attributes can be displayed or can become invisible.

New

This button creates an instance from the attribute values entered into the grid cells of the row that shows the row label *.

Cancel

This button discards all changes made in the currently selected row of the table grid.

Apply

This button commits all changes made in the table grid. A new (empty) row is appended to the table grid.

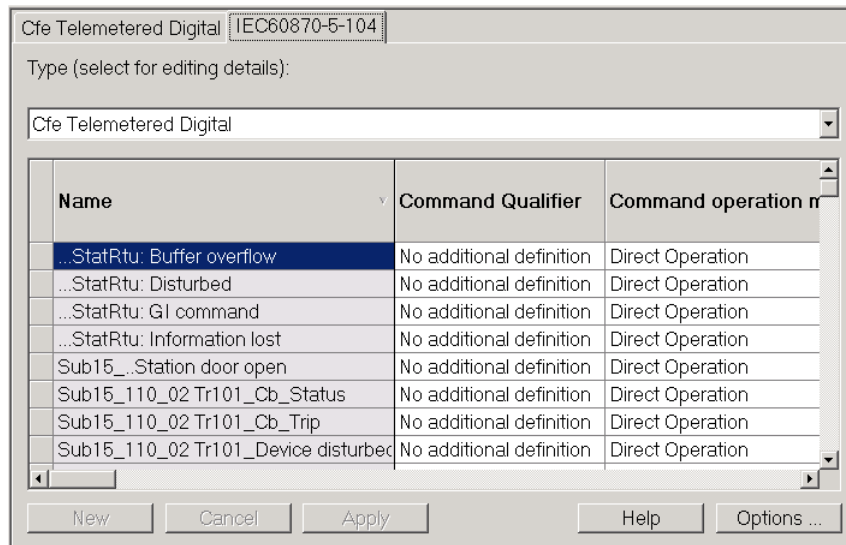
Options ...

This button opens the **Options** dialog. See [3.3.1.6 Options Dialog](#).

3.3.2.3 Protocol Tab

For PI/CFE instances the Table Editor provides an additional tab, the so-called Protocol tab (see the following figure). The Protocol tab is generated dynamically. The protocol is determined based on the data point and the name of the RTU and the corresponding Protocol tab is displayed. The Protocol tab allows you to enter protocol-specific attribute values for the selected instance.

For additional information on protocol attributes, refer to the following documents: Reference Guide Communication Front End (CFE) and Reference Guide Telecontrol Interface Reference (TCI).



[IMMRF016-051108-enUS-01.tif]

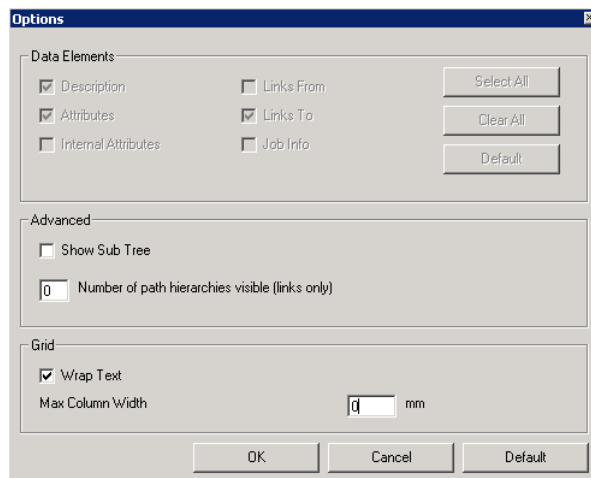
Figure 3-62 Protocol Tab (IEC Protocol)

3.3.2.4 Communication View

- CFE
- ICCP

3.3.2.5 Options Dialog

The **Options** dialog allows you to customize the content of the Table Grid and to specify what data types shall be displayed in the Table Editor.



[IMMED033-180108-enUS-01.tif]

Figure 3-63 Options Dialog

OK

Applies the configured settings.

The appearance of the table grid and the number of shown instances change accordingly. The Options dialog is closed.

Cancel

Discards the option changes. The Options dialog is closed.

Default

Resets all option settings and applies default settings to the options of the Options dialog.

The following default options are set:

The check box **Attributes** is checked, the check box **Wrap Text** is unchecked.

The maximum column width is set to zero.

Description

View or make the attribute **Description** unavailable on the table grid.

The attribute **Description** is shown on the table grid when checked.

Attributes

Show all attributes of the selected instances.

When this check box is checked, all attributes of the selected instances are shown on the table grid. Otherwise, only the attributes **Name**, **Type** and **Description** are shown on the table grid.

Internal Attributes

Show all internal attributes of the selected instances.

When this check box is checked, the table grid shows the attributes **Name**, **Type** and **Description** and all internal attributes of the selected instances. Otherwise, the internal attributes are not shown on the table Grid.

Links From

Shows all links of the selected network instances that point to these instances.

When this check box is checked, the table grid shows all links that point way from the selected instances. Otherwise, the links are not shown on the table grid.

Links To

Shows all links of the selected network instances that point to other instances.

When this check box is checked, the table grid shows all links that point to the selected instances. Otherwise, the links are not shown on the table grid.

Select All

Checks all check boxes of the **Data Elements** group.

Clear All

Unchecks all check boxes of the **Data Elements** group.

Default

Checks the **Attributes** check box of the **Data Elements** group.

Show Subtree

Shows further subordinated instances.

When this check box is checked, the table grid shows all descendant instances of the selected instance. Otherwise, only the first hierarchical level of subordinated instances is displayed on the table grid.

Number of Path Hierarchies Visible (Links Only)

Specifies the number of hierarchy levels shown in a link path on the table grid. For example, when **Number of path hierarchies visible (links only)** is set to 3, the lowest 3 hierarchy levels of the path name are displayed.

Valid values: integer value (practically unlimited)



Note

When this text box is empty or set to zero, all hierarchy levels are shown.

Wrap Text

Specifies whether text shown in a cell of the table grid may wrap.

When this check box is checked, the character string shown in a cell of the table grid is wrapped when it exceeds the maximum column width. The setting of the text box Max Column Width determines the maximum column width.

Max Column Width

Specifies the maximum width of a table cell in Millimeters.

Valid values: real value



Note

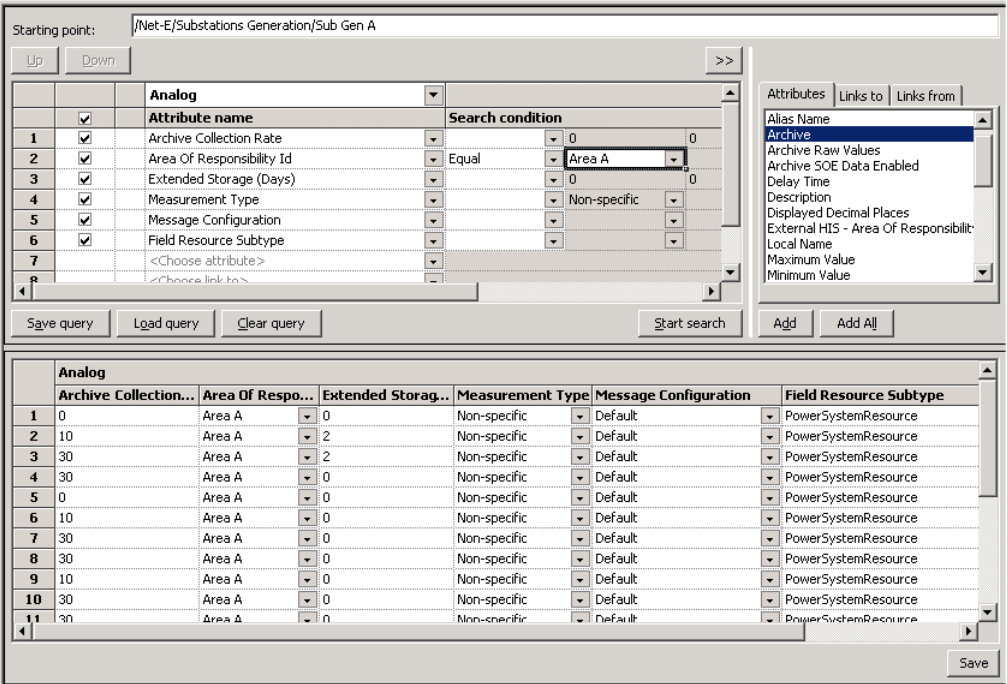
When this text box is empty or set to zero, a column of the table grid is as wide as its widest cell (the cell of that column that contains the longest text).

3.3.3 Bulk-Instance Editor

3.3.3.1 Introduction

Instead of editing single instances in the Single-Instance Editor, the Bulk-Instance Editor allows you to modify multiple instances in a table view.

The Bulk-Instance Editor allows you to define queries. The results of the query can then be modified in a tabular list.

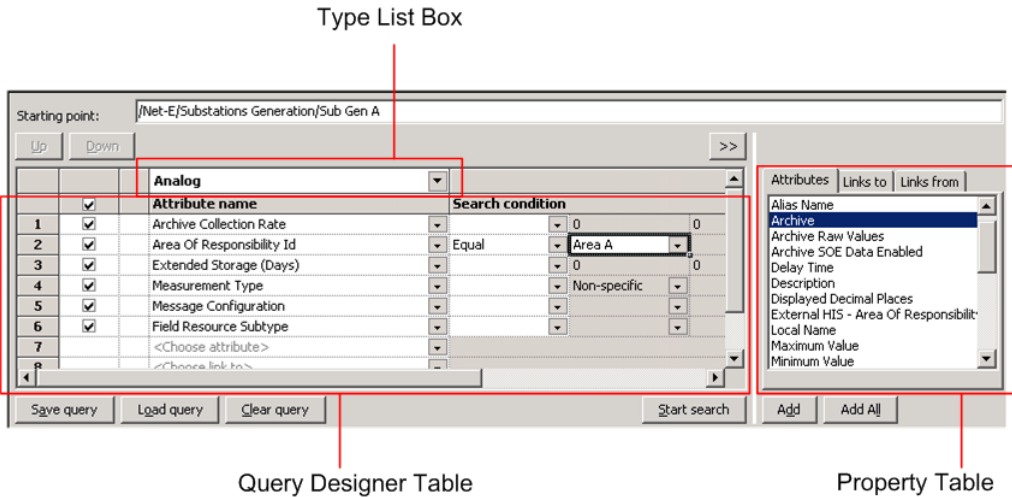


[IMMRF003-051108-enUS-01.tif]

Figure 3-64 Bulk-Instance Editor

3.3.3.2 Query Designer

The Query Designer section of the table editor allows you to search and filter for the desired instances.



[IMMRF051-110411-enUS-01.tif]

Figure 3-65 Query Designer

Starting Point

This text box shows the starting path of the query. When the table editor is started, this field is empty. Input into this field is possible via drag and drop from the tree view.

Up

After selecting a row in the Query Designer Table, clicking this button moves up the selected row 1 step.

Down

After selecting a row in the Query Designer Table, clicking this button moves down the selected row 1 step.

Type List Box

This list box located above the Query Designer table shows all types that belong to the selected **Starting point** and its subordinate instances.

You can select a type from this list box. All displayed types belong to the selected **Starting Point** and its subordinate instances. In the previous figure, the selected type is **Analog**. After selecting the type, you can define a query by selecting an attribute in the **Attributes** tab of the **Property** table. An assigned query attribute can be deleted by selecting the attribute and selecting the corresponding item of the context menu.

Arrow Button

The arrow button on top of the Query Designer table pointing to the right side makes the Query Designer table unavailable to enlarge the area for the Property table. When the Query Designer table is unavailable, this button points to the left side. Clicking it displays the Query Designer table again.

Query Designer Table

The Query Designer table is used to define a query. It consists of 2 parts: the upper part containing the selected attributes and the lower part containing the selected links. Attributes and links can be added by selecting the desired attributes/links from the Property table.

Each line represents 1 search criterion and consists of

- Check box that is used to filter whether the respective attribute/link shall be displayed in the result table
- Name of the attribute/link
- Text box containing the search criterion
- Text box for entering a value for the search criterion

For each attribute, you can define a search criterion. A search criterion defines a rule how an object instance must be identified during the search. Depending on the attribute type, you can set the search criterion. For example, if the attribute is a number, the criterion is a number or a range of numbers. If the attribute type is a string, you can enter a character string.

By setting a hook on the left side you can define for each attribute, whether it shall be added to the filter. After adding the attribute to the Query Designer table, the checkbox is set by default. That means, the attribute is displayed in the **Result Table** as a column. If you deselect the checkbox, the attribute is only used for the search process, but it is not visible in the **Result Table**.

To define a query for an attribute:

- Select an attribute from the **Property** table
- Select a search criterion from the list box
- Enter a value for the search criterion

Property Table

The property table shows a list of all attributes/links of the type selected in the Query Designer table header. It consists of 3 tabs:

- **Attributes**

This tab shows a list of all not assigned attributes of the selected type. When you select an attribute for a query, this attribute is removed from the **Property** table and added to the **Query Designer** table.

- **Links To**

This tab shows a list of all not assigned associations that link to the selected type. When you select a link for a query, this attribute is removed from the **Property** table and added to the **Query Designer** table.

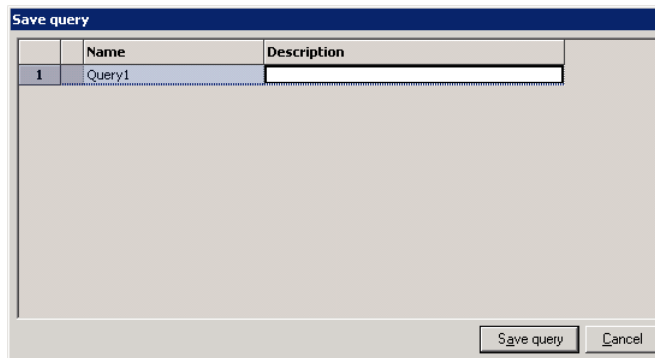
- **Links From**

This tab shows a list of all not assigned associations that link from the selected type. When you select a link for a query, this attribute is removed from the **Property** table and added to the **Query Designer** table.

Selecting an attribute/link and clicking the button **Add** removes this attribute/link from the **Property** table and adds it to the **Query Designer** table. Clicking the button **Add all** adds all attributes/links to the Query Designer table.

Save Query

Opens a display for saving the currently defined query.



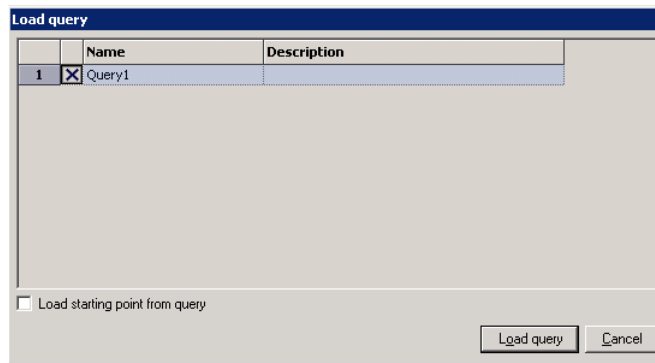
[IMMRF006-051108-enUS-01.tif]

Figure 3-66 Save Query

When the query has been saved, it is available to all users on all clients.

Load Query

Opens a display for selecting a saved query that shall be loaded into the Query Designer.



[IMMRF007-051108-enUS-01.tif]

Figure 3-67 Load Query

Clear Query

Clicking this button clears the Query Designer Table for entering a new query.

Start Search

Clicking this button starts the search according to the defined query. The results are displayed in the Result Table.

3.3.3.3 Result Table

Analog						
	Archive Collection...	Area Of Respo...	Extended Stora...	Measurement Type	Message Configuration	Field Resource Subtype
1	0	Area A	0	Non-specific	Default	PowerSystemResource
2	10	Area A	2	Non-specific	Default	PowerSystemResource
3	30	Area A	2	Non-specific	Default	PowerSystemResource
4	30	Area A	0	Non-specific	Default	PowerSystemResource
5	0	Area A	0	Non-specific	Default	PowerSystemResource
6	10	Area A	0	Non-specific	Default	PowerSystemResource
7	30	Area A	0	Non-specific	Default	PowerSystemResource
8	30	Area A	0	Non-specific	Default	PowerSystemResource
9	10	Area A	0	Non-specific	Default	PowerSystemResource
10	30	Area A	0	Non-specific	Default	PowerSystemResource
11	30	Area A	0	Non-specific	Default	PowerSystemResource

[IMMRF005-051108-enUS-01.tif]

Figure 3-68 Result Table

The Result Table provides a list of all instances conforming to the query criteria defined in the Query Designer. Each row of this table represents 1 instance. All attributes and associations of an instance defined in the query are displayed in this row. The data in this table can be modified by clicking the desired cell and entering a new value, selecting a value from a list box, or by setting a check box.

When a cell was selected, a surrounding border appears, containing a square in the lower right corner allowing further mouse operations. When moving the mouse pointer over the square, the mouse pointer offers the possibility to change cells based on the content of the currently selected cells. It can be dragged in any direction and with any amount of selected cells, for example, one cell can be dragged to the right, left, top, or bottom.

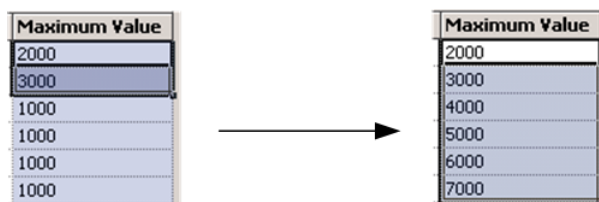
The result varies depending on the selected cells. Values can be copied from one cell to several others.



[IMMRF052-110411-enUS-01.tif]

Figure 3-69 Values are copied from one cell to several other cells

If 2 or more cells are selected and offer a pattern, this pattern is applied to the other cells, to which the mouse pointer is dragged. The following figure shows an example, where 2 cells are selected (the second one is the value of the first cell plus 1000). This pattern is applied to the 4 cells below, to which the lower right corner has been dragged.



[IMMRF053-110411-enUS-01.tif]

Figure 3-70 Values are copied from 2 cells (values ascending in steps of 1000) to other cells

Various combinations of cell, row, or column selections may also be copied to the clipboard using the context menu and pasted to another selection.

Save

Saves the changes made to the instances in the Result Table.

3.3.4 Curve Editor

3.3.4.1 Introduction

The Curve Editor is a particular kind of the Single-Instance Editor and is used to:

- View the curve plot of a selected (curve) instance
- Modify the curve plot of a selected (curve) instance
- Create a curve instance

Curve Instance

A curve instance is an instance that is displayed as curve plot. The interpolation points of a curve are subordinate instances of the curve instance. The number of curves belonging to a curve instance depends on the type of the curve instance.

Curve instances are for example:

- **PII .CFE Characteristics/ Characteristic**
- **Generation/ Hydro Plants/ BigTree/ Unit1**

The following Curve Editor is provided:

- Curve Definition Editor

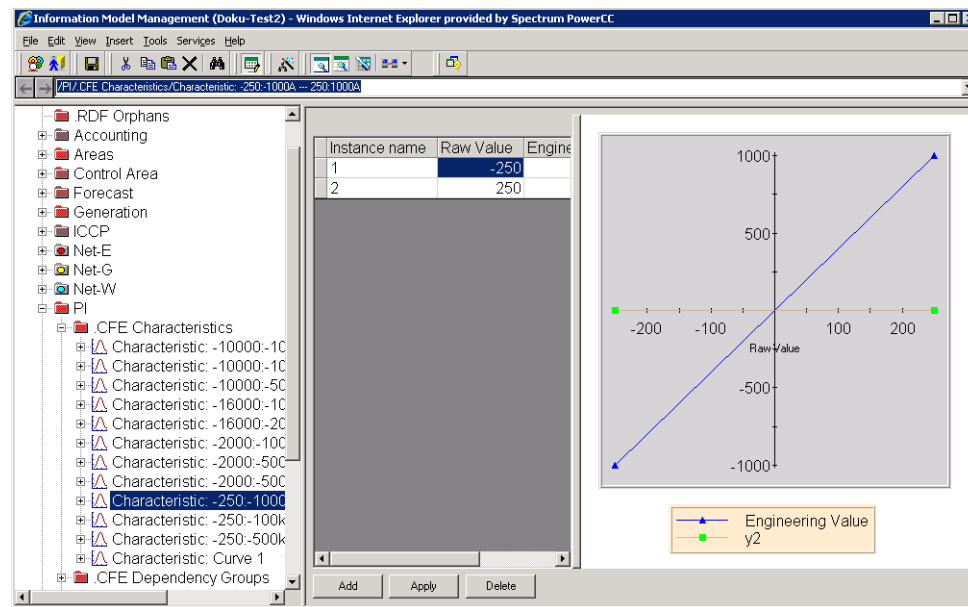
3.3.4.2 Curve Definition Editor

The Curve Definition Editor displays all interpolation points of a curve instance numerically in a table and graphically in a curve plot (The number of curves depends on the type of the curve instance). This editor is used to

- View and modify the interpolation point values
- Add interpolation points

Selecting the Curve Definition Editor

The Curve Definition Editor is invoked automatically by clicking the desired instance on the instance overview tree if the type of the instance is derived from the type Curve (**Domain/IdentifiedObject/Curve**).



[IMMRF049-210909-enUS-01.tif]

Figure 3-71 Curve Definition Editor

Body

The body consists of:

- Table of interpolation points
- Curve plot

The table of interpolation points displays the interpolation points (subordinate instances) of the selected curve instance. Depending on the type of the curve instance, 2 (single curve) or more coordinates (multiple curves) are associated with a curve instance. For each interpolation point the following information is displayed:

- Instance name
- Coordinate values

Instance Name

This column displays the names of the subordinate instances (interpolation points).

You can modify these names. The modified name of a subordinate instance is displayed in the refreshed instance overview tree.

Coordinate Values

The number and the names of the columns correspond to the number and the names of the coordinates of a curve instance.

The number and the names of the coordinates depend on the type of the curve instance. Each interpolation point (subordinate instance) of the selected curve instance consists of the following coordinates:

- X-axis (for example, raw value in [Figure 3-71](#))
- Y2-axis (for example, scaled value in [Figure 3-71](#))
- ...
- .(the existence depends on the type of the curve instance)
- ...
- Yn-axis (The number of y-axes depends on the type of curve)

You can modify the coordinate values.

Additionally, the following buttons are provided below the table:

Add

This button adds a new line for an additional interpolation point.

Apply

This button saves the modifications in the table.

Delete

This button deletes the selected lines of an interpolation point.

Curve Plot

The curve plot displays the graphic representation of the curve values. Depending on the type of the curve instance, multiple curves are displayed. Different markers indicate the curves. In this graphic diagram, the values cannot be modified.

3.3.5 Color Picker

3.3.5.1 Introduction

The Color Picker is a particular the Single-Instance Editor and is used to view, modify, and define the color instances.

Color Instance

The color instances define the colors in the summaries and displays. The following types of color instances can be distinguished:

- Color
- ColorGroup
- NetworkColoringGroup

- sysPresColoringGro
- sysPresColorPalettes

Selecting the Color Picker

The Color Picker is automatically invoked, when a color instance is selected in the instance overview tree.

Attribute Label	Data Type	Value
Documentation	String	Defines the background color for e
Color	Hex String	#F3F3F3 ...

[IMMRF020-061108-enUS-01.tif]

Figure 3-72 Color Picker

Body

In the body of the Color Picker, the following information is displayed:

- **Name**
In this field, the name of the selected instance is displayed.
- **Documentation**
In this field, an instance comment is displayed. You can modify the comment.
- **Color**
In this field, the color and the color hex code is displayed. You can modify the color by entering a new hex code or by clicking the Color Palette button right the hex code. Clicking this button opens the Color Palette.

3.3.5.2 Color Palette

The Color Palette is used to define the color with that the selected instance shall be displayed. The Color Palette is an MS Windows Control.

Selecting the Color Palette

The Color Palette is invoked by clicking the Color Palette button.



[IMMRF021-061108-enUS-01.tif]

Figure 3-73 Color Palette

Definition of the Instance Color

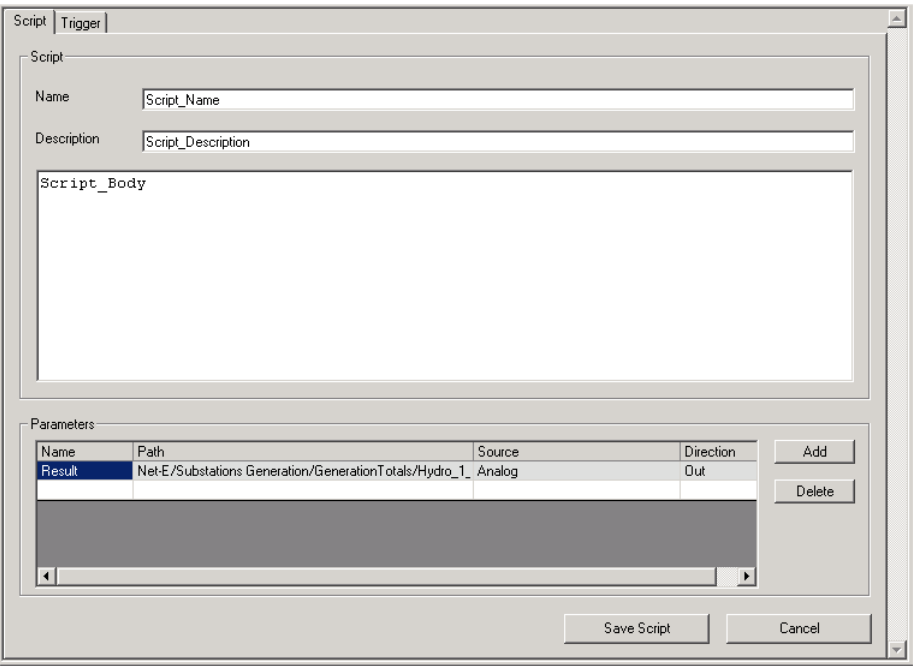
- You can define the color for the instance in the following ways
- Clicking one of the basic colors and by clicking the button **OK**, or
 - Clicking one of the custom colors and by clicking the button **OK**.
- By clicking the button **Define Custom Colors** you open a color scheme. In this scheme, you select the color nuance. Then this color nuance is displayed in one of the fields of the area **Custom Colors** of the Color Palette. In this way, you can define 16 custom colors.

3.3.6 Common Calculation Editor

3.3.6.1 Introduction

The Common Calculation Editor allows you to define a script or calculation and create an instance of this script. For definition of scripts or calculations to be used more than once, refer to [3.6.3 Script Definition Wizard](#), [3.6.4 Calculation Wizard](#), and [3.6.5 Script Wizard](#).

3.3.6.2 Script Tab



[IMMRF001-051108-enUS-01.tif]

Figure 3-74 Script Tab

Script Name/Description

Enter a name (unique among all common calculations) and, optionally, a description.

Script Body

This section is used for entering the script code. Besides comment lines, which start with '`'`, the script may only contain the parameter names defined in the parameter definition section and the following operators:

- `"+" or "-"`
- `Add_If_Positive(a, b), Add_If_Negative(a, b)`
- `ABS(a)`
- `"*" , Multiply_If_Positive(a, b), Multiply_If_Negative(a, b)`
- `":", Divide_If_Positive(a, b), Divide_If_Negative(a, b)`
- `SQR(a)`
- `EXP(a)`
- `>, ≥, =, ≠, <, ≤`
- `AND, OR, XOR`
- `Log (A) Logarithmic function ln (A), A>0`
- `Ceil (A) Rounding off upwards – the minimal integer not under A`
- `Floor (A) Rounding off downwards – the maximal integer not over A`
- `Max (A, B, C, D, E) Maximum from the list of values (the function takes 5 arguments)`
- `Min (A, B, C, D, E) Minimum from the list of values (the function takes 5 arguments)`
- `Hypo (x, y) Euclidian distance or hypotenuse – square root from the sum of squares`
- `MVA (P, Q) Full capacity, where P is active capacity, Q is reactive capacity expressed in respective units (essentially does not differ from hypo function)`
- `Amp (S, U) Value of current calculated as relation of full capacity to voltage, multiplied by the square root from 3, where S is full capacity in MVA, U – voltage in KV`
- `Pf (P, Q) Capacity factor – the relation of active capacity to full capacity, where P is active Capacity, Q – reactive capacity expressed in respective units (both MW or KW and Var)`
- `Round (A, n) Rounding of the real number A up to n digits before or after decimal point (depending on n)`
- `Nested If, Then, Else`
- `Trigonometric functions`

Parameter Definition

The Parameter section always contains 1 parameter with the direction **Out**. This parameter is entered automatically. The column **Path** shows the result path and name that are entered in the **Trigger** tab.

Any number of parameters with the direction **In** can be entered. The path can be entered manually or via drag and drop from the instance tree.

The button **Add** is used to add a new line to the parameter list for entering a new parameter.

The button **Delete** is used to delete the selected line from the parameter list.

3.3.6.3 Trigger Tab

The input in this tab defines, when the calculation defined on the script tab shall be executed.

Result Name	Hydro_1_MW
Result Type	Analog
Population Rule	HIS and SCADA
Trigger Type	On Timer
Periodicity	0
Periodicity Unit	Hour
At Hour	
At Minute	
At Second	
Activate at Runup	

[IMMRF002-051108-enUS-01.tif]

Figure 3-75 Trigger Tab

Result Name and Type

You can enter a name for the result and select a type from the list.

Population Rule

The Population Rule determines, where the script is populated. The following options are provided:

- SCADA
- HIS
- SCADA and HIS
- None

For additional information on HIS calculations, refer to the [Reference Guide Historical Information System \(HIS\)](#).

Trigger Definition

The allowed triggers depend on the population rule. If **SCADA** is selected, the following possibilities are provided:

- On timer
The script is triggered according to a timer.
- On change
The script is triggered whenever a value changes.
- On timer or On change
The script is triggered according to a timer or when a value changes.
- On timer and On change
The script is triggered according to a timer and when a value changes.

If timer is selected, the text boxes are activated.

If **HIS** or **HIS and SCADA** is selected in the Population rule section, only **On Timer** can be selected. All other text boxes are disabled.

3.3.6.4 Buttons

Save Script/Run Script

When editing a script, the buttons **Save Script** is used to save the changes in the script. When the script has been saved, the button **Run Script** can be used to execute the script.

Cancel

Clicking the button **Cancel** discards all changes and closes the Common Calculation Editor.

3.4 Special Editors

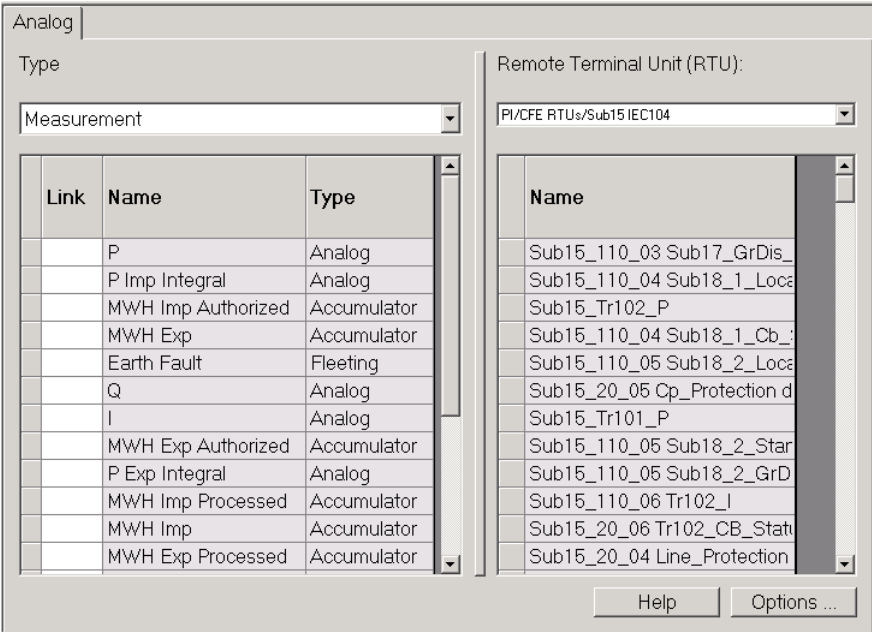
3.4.1 Communication Editor for CFE

3.4.1.1 Introduction

This user interface is a particular Table Editor that simplifies the creation and linkage of CFE data points.

The Communication Editor for CFE is invoked either by clicking the menu item **View > Instance Editors > CFE** or by clicking the toolbar button **Communication Editors** and selecting the menu item **CFE** from the submenu.

The Communication Editor for CFE mainly consists of 2 table grids - the Measurement Table on the left side and the RTU Table on the right side.



[IMMOP023-301008-enUS-01.tif]

Figure 3-76 Communication Editor for CFE



Note

The subsequent description is only a summary of the editor functionality. For additional information on user interface components and interactions, refer to [Reference Guide Communication Front End \(CFE\)](#).

3.4.1.2 Measurement Table

The Measurement Table shows the measurement instances under the subtree of the selected node.

The column **Link** determines whether the measurement is already linked to an RTU data point. It can show **ICCP** or **CFE** when a link is already established, or can be empty when the measurement is not linked yet.

The column **Name** shows the name of the measurement.

The column **Type** indicates what measurement type it is.

Type List

You can use the **Type** list to filter for one specific type: Accumulator, Analog, Digital, Fleeting, or Position. All types of measurements can be displayed by selecting **<all types>**.

Selection of Rows

You can select a single row of the Measurement Table by clicking the corresponding row menu section. Multiple rows can be selected by pressing the SHIFT key and by clicking the corresponding row menu section at the same time. All rows can be selected by clicking the table menu section (see [3.1.5.4 Table Editor](#)).

You can select a measurement and Drag & Drop the path name into the RTU Table. The link between the CFE data point and the measurement is created automatically. You can only drop it into a row where the CFE data point is not linked to a measurement.

Context Menu of the Measurement Table

Clicking the row menu section of the selected rows with the right mouse button opens the context menu of the Measurement Table which contains the menu items **Edit** and **Create**:

- **Edit**
Opens the Single-Instance Editor (see [3.3.1 Single-Instance Editor](#)) showing the measurement instance of the selected row. This menu item is only enabled when a single row is selected.
- **Create**
Creates new CFE data points. The names of the new data points are derived from the selected measurement name. After creation, you can rename these data points in the RTU Table manually. The links are created automatically from the measurements to these CFE data points.

3.4.1.3 RTU Table

Remote Terminal Unit (RTU) List

In the right list (RTU), all instances of the type RTU are listed. You can select a particular entry to display all CFE data points which are defined for the selected RTU in the RTU Table.

The column **Name** lists all CFE data points found underneath the selected RTU.

The column **SendsValueTo** shows whether the CFE data point is already linked to a measurement for the monitoring direction.

The column **ReceivesControlFrom** shows whether the CFE data point is already linked to a measurement for controlling.

Context Menu of the RTU Table

Clicking the row menu section of the selected rows (for row selection refer to description of the Measurement Table) with the right mouse button opens the context menu of the RTU Table which contains the menu items **Delete**, **Edit** and **Unlink**:

- **Delete**
Removes the selected instance from the RTU Table.
- **Edit**
Opens the Single-Instance Editor (see [3.3.1 Single-Instance Editor](#)) showing the instance of the selected row.

- Unlink

Removes the link from the selected CFE data point to the measurement. A message opens where you have the option to delete the CFE data point additionally.

Options ...

Clicking this button opens the **Options** dialog that allows you to customize the table grid of the Communication Editor for CFE. For additional information, refer to [3.3.2.5 Options Dialog](#).

3.4.2 Communication Editor for ICCP

3.4.2.1 Introduction

The ICCP protocol is designed to allow data exchange over a WAN, a Local Area Network (LAN), and so on, between utility control centers. Data exchanged is real-time power system information. An individual control center, in its interactions with other control centers, can act as

- Client
- Server
- Both

For detailed information on ICCP, refer to Reference Guide ICCP Spectrum Power ICCP currently supports Blocks 1, 2, and 5.

These Blocks are:

- Block 1 – Status Information cyclically

This block function is used to exchange the real-time power system data between Spectrum Power and the remote systems.

- Block 2 – Status Information on change

This block function is used to exchange the real-time Exception Data between Spectrum Power and the remote systems.

- Block 5 – Supervisory Control

The Spectrum Power Supervisory Control module issues control commands. This causes device control messages to be sent to a Remote Control Center. Block 5 controls received from the remote node are passed to the Supervisory Control subsystem for additional processing. Control actions initiated by an operator or application that must be sent to the remote node come from Supervisory Control subsystem.

ICCP Variable Name

Each data object that is passed between the local and remote control center via ICCP must be identified by a unique name. This identification is called Variable Name in Spectrum Power. The variable name can have up to 32 characters and can be different between different control centers. Instance names represent variable names of ICCP data objects.

ICCP Data Entry

Spectrum Power IMM provides a particular Table Editor to simplify the data entry and handling of data related to ICCP Block 1,2, and 5. These data are ICCP Server Data Objects and ICCP Transfer Data Objects. The editor provided to configure these data is the Communication Editor for ICCP (refer to the next chapter).

3.4.2.2 Overview

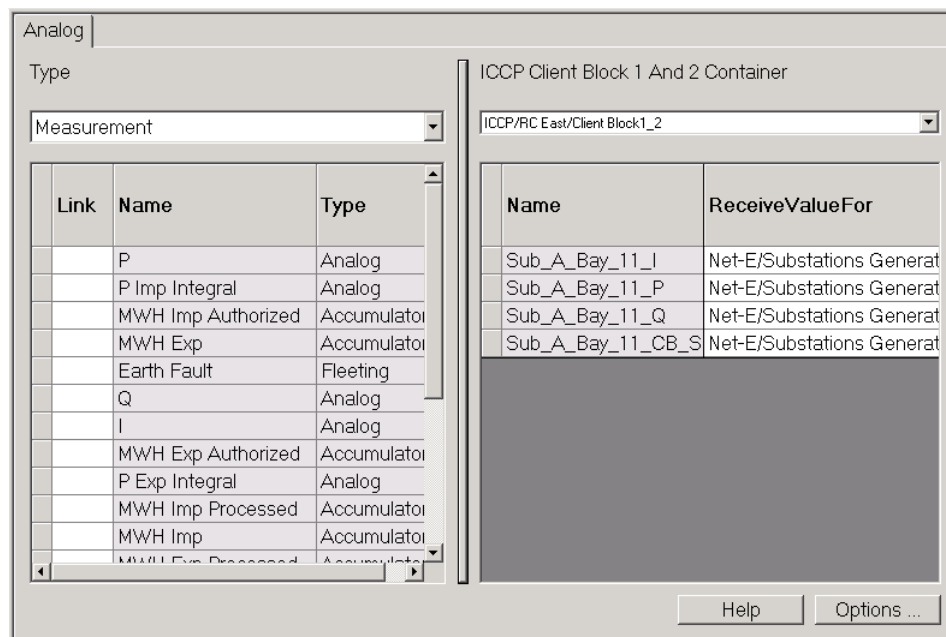
The Communication Editor for ICCP provides the following views for handling ICCP data objects:

- ICCP Client View
 - ICCP Client Block 1 and 2
 - ICCP Client Block 5
- ICCP Server View
 - ICCP Server Block 1 and 2
 - ICCP Server Block 5

A view of the Communication Editor for ICCP is invoked by selecting the corresponding menu item of the menu **View > Instance Editors** or clicking the appropriate item of the toolbar button. When you select an instance in the tree view, the Communication Editor for ICCP is loaded into the working area.

3.4.2.3 Basic Editor Components

The basic layout of the Communication Editor for ICCP is the same for all views, but the meaning of the displayed components can vary depending on the selected ICCP instance and on the ICCP-selected view.



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Figure 3-77 Communication Editor for ICCP

The Communication Editor for ICCP mainly consists of 2 list boxes and 2 table grids.

The button **Options ...** (located in the lower right corner of the body) opens the **Options** dialog. The Options dialog allows you to customize the content of the table grids and to specify what data types to be shown in the Communication Editor for ICCP. For additional information on the Options dialog, refer to [3.3.2.5 Options Dialog](#).

**Note**

For additional information on the user interface of the Communication Editor for ICCP and possible interactions, refer to [Reference Guide ICCP](#).

3.4.2.4 ICCP Client View

A client block object can be either linked to a control or a value. It is not possible that it is linked to both. For instances under the Client Block 1 and 2 Container the measured value is used (menu **View > Instance Editors > ICCP Client Block 1 and 2**), for instances under the Client Block 5 Container the measurement control is used (menu **View > Instance Editors > ICCP Client Block 5**).

When you select an instance in the tree view, all instances of type measurement underneath that node are displayed in the Measurement Table (left part of the ICCP client view). This includes the complete subtree of the selected node.

Type

This list box can be used to filter for one specific type: Accumulator, Analog, Digital, or Position. You can also display all types of measurements at once by selecting the parent type **Measurements**.

**Note**

Fleeting data points are not supported by ICCP, they are not displayed.

Link Column

The **Link** column shows if the measurement is already linked to a CFE data point or an ICCP Client Block 1 and 2 object.

**Note**

A measurement can only be linked once. Therefore the rows for measurements, which are already linked, are read-only rows.

Calculated measurements are marked with an underscore () to indicate that they cannot be linked.

Name Column

The **Name** column shows the name and the path of the measurement.

Type Column

The **Type** column shows the measurement type of the instance.

Selection of Rows

You can select a single row of the Measurement Table by clicking the corresponding row menu section. Multiple rows can be selected by pressing the SHIFT key and by clicking the corresponding row menu section at the same time. All rows can be selected by clicking the table menu section (see [3.1.5.4 Table Editor](#)).

You can select a measurement and Drag & Drop the path name into the Client Object Table (right part of the ICCP client view). The link between the client block object and the measurement is created automatically.

Context Menu of the Measurement Table

Clicking the row menu section of the selected rows with the right mouse button opens the context menu of the Measurement Table which contains the menu items **Edit** and **Create**:

- **Edit**
Opens the Single-Instance Editor showing the measurement instance of the selected row. This menu item is only enabled when a single row is selected.
- **Create**
Creates new ICCP client block objects. After creation, you can rename these objects in the Client Object Table manually. The links are created automatically from the measurements to the ICCP client block objects.

ICCP Client Block 1 and 2 Container/ICCP Client Block 5 Container

This drop-down shows container of the type ICCP Client Block 1 and 2 container instances (Client Monitoring) and ICCP Client Block 5 container instances (Client Control), respectively. You can select a container entry from this list box. After a client container has been selected all client objects of the selected container are listed in the Client Object Table.

Name

The Name column shows all Client Block 1 and 2 or Client Block 5 instances underneath the selected client container instance.

ReceiveValueFor

This column shows the path of the instance for which the Client Block 1 and 2 instance receives data.

SendsValueFrom

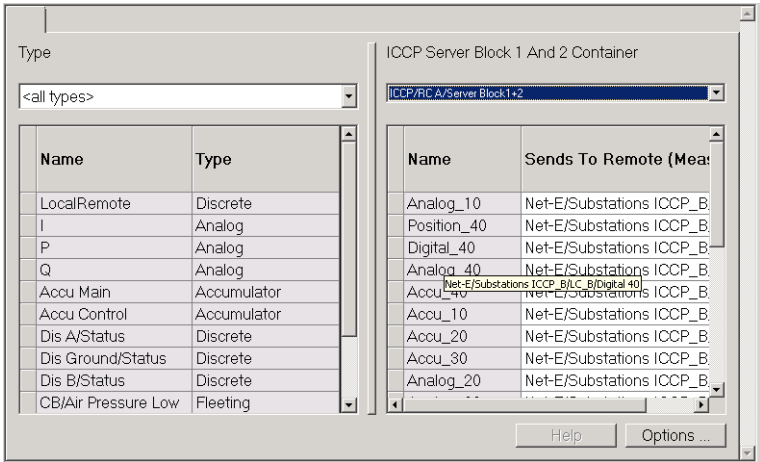
This column shows the path of the instance which sends data (Client Block 5 container instances).

Context Menu of the Client Object Table

Clicking the row menu section of the selected rows (for row selection refer to description of the Measurement Table) with the right mouse button opens the context menu of the Client Object Table which contains the menu items **Delete**, **Edit** and **Unlink**:

- **Delete**
Removes the selected instance from the Client Object Table.
- **Edit**
Opens the Single-Instance Editor showing the instance of the selected row.
- **Unlink**
Removes the link of the selected client block object to the measurement. A message opens where you have the option to delete the client block object additionally.

3.4.2.5 ICCP Server View



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Figure 3-78 ICCP Server View

A measurement can be linked to multiple server block objects. The indication that a server block object is linked to a measurement is in the right table. The left table does not show this information. As described in [3.4.2.4 ICCP Client View](#), you can select 2 modes to view server block objects: **ICCP Server Block 1 and 2** and **ICCP Server Block 5**.

When you select an instance in the tree view, all instances of type measurement underneath that node are displayed in the measurement table. This includes the complete subtree of the selected node.

Type

This list box can be used to filter for one specific type: Accumulator, Analog, Digital, or Position. You can also display all types of measurements at once by selecting the parent type **Measurements**.



Note

ICCP does not support fleeting data points, they are not displayed.

Name Column

The **Name** column shows the name and the path of the measurement.

Type Column

The **Type** column shows the measurement type of the instance.

Selection of Rows

See [3.4.2.4 ICCP Client View](#).

You can select a measurement and Drag & Drop the path name into the Server Object Table (right part of the ICCP server view). The link between the server block object and the measurement is created automatically.

Context Menu of the Measurement Table

Clicking the row menu section of the selected rows with the right mouse button opens the context menu of the Measurement Table which contains the menu items **Edit** and **Create**:

- Edit
See [3.4.2.4 ICCP Client View](#).
- Create
Creates new ICCP server block objects. After creation, you can rename these objects in the Server Object Table manually). The links are created automatically from the measurements to the ICCP server block objects.

ICCP Server Block 1 and 2 Container/ICCP Server Block 5 Container

In this list box all instances of the type ICCP Server Block 1 and 2 container instances (Server Monitoring) or ICCP Server Block 5 container instances (Server Control) respectively are listed. You can select an instance from this list box. After a server instance has been selected all server objects are listed for the selected instance in the Server Object Table.

Name

The **Name** column shows all Server Block 1 and 2 or Server Block 5 objects underneath the selected server container instance.

SendsValueFrom Measurement/ReceivesValueTo

This column shows the corresponding values of the selected Server Block 1 and 2/Server Block 5 container instance.

Context Menu of the Server Object Table

Clicking the row menu section of the selected rows (see [3.4.2.4 ICCP Client View](#)) with the right mouse button opens the context menu of the Server Object Table which contains the menu items **Delete**, **Edit**, and **Unlink**:

- Delete
Removes the selected instance from the Server Object Table.
- Edit
Opens the Single-Instance Editor showing the instance of the selected row.
- Unlink
Removes the link of the selected server block object to the measurement. A message opens where you have the option to delete the server block object additionally.

3.5 General Editors

3.5.1 Type Editor

3.5.1.1 Introduction

Selection of the Type Editor

The Type Editor is used to:

- View or edit types by clicking a type in the type tree, or
- Create a type
 - By selecting a type in the type tree and selecting the menu item **New Type** on the **Insert** menu, or
 - By selecting a type in the type tree and clicking the toolbar button **New Type**, or
 - By right-clicking a type in the type tree and selecting the menu item **New Type** from the context menu of the tree

When a new type is created, the selected type is the parent of the new type. When no type was selected from the type tree before type creation is invoked, the most recently selected type (still highlighted in the type tree) is used as parent. This can be the root type.



Note

Types can be only modified/created in the real-time job. Additionally, you must have access rights for type modification. For additional information on access rights, refer to [Reference Guide System Software \(SYSW\)](#).

Modifications of attributes or associations of existing types are not recommended.

General Attributes Enumerators Hierarchy References	
Name	PowerSystemResource
Label	PowerSystemResource
Help Message	A power system resource can be an item of equipment such as a Switch, an EquipmentContainer containing many individual items of equipment such as a Substation, or an organisational entity such as Company or SubControlArea. This provides for the nesting of collections of PowerSystemResources within other PowerSystemResources. For example, a Switch could be a member of a Substation and a Substation could be a member of a division of a Company.
Rdf Name	PowerSystemResource
Namespace	cim
Scope	ALL
Graphic Sheet	
Abstract Type	Yes
Rdf Default Name	
CIM children	No

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Figure 3-79 Type Editor - General Tab

The Type Editor provides several tabs each of which is used to set or modify distinct kinds of type attributes and properties.

For additional information on the generic features of tabs, refer to [3.1.5.3 Tabs](#).

The following tabs are provided in the Type Editor:

- **General** tab
- **Attributes** tab
- **Enumerations** tab
- **Hierarchy** tab
- **References** tab

3.5.1.2 Editor Context Menu

The editor context menu of the Type Editor is opened by clicking the right mouse button on a tab of the Type Editor. The content of the editor context menu (**Delete Attribute** or **Delete Association**) depends on the tab from which the context menu was invoked.

Delete Attribute

Deletes the selected type attribute from the respective type and, thus, from all instances based on that particular type.

The related attribute column is removed from the **Attributes** tab of the Type Editor.



Note

The concerned type must be saved after the deletion to make the deletion effective.

Delete Association

Deletes the selected hierarchy association or reference association from the respective type and, thus, from all instances based on that particular type.

The related association column is removed from the **Hierarchy** tab or the **References** tab of the Type Editor.



Note

The concerned type must be saved after the deletion to make the deletion effective.

3.5.1.3 General Tab

The **General** tab provides elements for entering and modifying general type properties.

Name

Contains the name of the respective type.

The type name is a unique character string that identifies a certain type. It is used for internal purposes and normally not visible at runtime except for the type tree. In the type tree, the type names are displayed as tree entries.

Valid values: a character string of up to 100 characters.

Type names must not contain the following special characters:

Table 3-7 Special Characters Which Must Not Be Used in Type Names

/	Slash
\	Backslash
	Space
&	Ampersand
#	Pound sign
+	Plus sign



Note

Once a new type is saved, the type name cannot be changed. Thus, the text box containing the type name is disabled except when defining a new type. In order to change the name of an already existing instance, you can use the item **Rename** from the **Edit** menu.

Label

Contains the type label.

The type label is a language-dependent character string which is displayed at runtime to identify a certain type. Language-dependent means, a type can have different language-specific type labels. Depending on the user interface language setting of the operating system, the appropriate type label is selected automatically when a type is retrieved from the database.

Help Message

A short instruction which outlines the purpose of the type.

Like type labels, help messages are language-dependent. A type can have different help messages with the appropriate help message automatically selected depending on the current user interface language setting of the operating system.

Rdf Name

RDF name of the type.

Name Space

The value of this property determines whether the respective type is part of cim, sp5, sp3 or a project-specific extension.

Scope

A comma-separated list of scopes (applications that use this association, for example, AGC, HIS, UC, ...)

Graphic Sheet

Contains the file name of a particular editor that is opened in the working area when the respective type is selected from the type tree.

Table 3-8 Valid Value

rsdectable.asp	The Decision-Table Editor opens in the working area when the type is double-clicked in the tree view.
----------------	---

**Note**

The specified asp-file must exist.

Abstract Type

Specifies if a type is a so-called Abstract Type. Abstract types cannot be used to create instances.

When you create an instance, the type list of the type selection only shows non-abstract types.

Table 3-9 Valid Values

Yes	Type is an abstract type.
No	Type is not an abstract type.

Rdf Default Name

Defines the name of the attribute whose value shall be used as the name of an instance, if the name is not given.

CIM Children

Valid values are Yes and No. Yes means that although this type does not belong to CIM, an instance of this type can have child instances that are relevant to CIM. When exporting in CIM-RDF format, instances of this type are not exported but its CIM children are exported.

3.5.1.4 Attributes Tab

The **Attributes** tab is used to define type attributes and enter properties of type attributes.

The elements provided on the Attributes tab, such as, text boxes, form a table with each column representing a particular type attribute and each row representing a certain property. Hence, a single table cell represents a distinct property of a certain type attribute. Table cells and text boxes can thus be considered synonymous on the Attributes tab.

Some type attributes are inserted automatically. These type attributes are inherited from the parent types and are read only. Inherited type attributes cannot be modified or deleted in a derived (child) type. Text boxes related to such type attributes are disabled on the Attributes tab. For example, the type attribute **AreaOfResponsibility** in the following figure is a type attribute that is inherited from a parent type.

**Note**

Interactive elements of a table column can be disabled or changed depending on property settings made in the same table column. For example: the text box related to the property **String Length** can be disabled depending on the setting of the property **Data Type** or the interactive element related to the property **Default Value** can be a text box or a list box depending on the setting of the property **Data Type**.

Name	pathName	AreaOfResponsibility	MessageConfigurati	DisplayOrder
Label	Path Name	Area Of Responsibili	Message Configurati	Display Order
Help Message	The pathname is a	Identification of an Ar	Relates the FieldRes	This value is used to
Rdf Name	pathName	AreaOfResponsibility	MessageConfigurati	DisplayOrder
Data type	String	EnumAreaOfRespon	EnumMessageConfi	DoubleFloat
String Length	500			
Default Value		AreaA	Default	
Units of Meas			Cost_per_Hour	
Scope	ALL	ALL	ALL	ALL
Namespace	cim	spc	spc	spc
Internal	Yes	No	No	No
Do Not Copy	No	No	No	Yes
Nerc Profile	Yes	No	No	No

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Figure 3-80 Attributes Tab

The leftmost table column consists of static text boxes which represent row headings for the property rows. The label of each static text box denotes the purpose of the respective property. The following properties are included in a type attribute:

Name

Contains the attribute name.

The entered attribute name is automatically displayed as a default value in the text box **Label**, when the text box **Label** is empty. The same procedure applies if the content of the text box **Name** was selected and the mouse pointer has been moved from **Name** to **Label** or another text box on the Attributes tab. When the text box **Label** is not empty, its content remains.

After an attribute name is entered, a new (empty) column is automatically appended to the right side of the table so you can insert another type attribute.



Note

The entered attribute name must be unique among all other attributes of the respective type. After the definition of a new type attribute was saved, the attribute name cannot be changed. Thus, the text box containing the attribute name is disabled except for defining a new attribute.

Valid values: A character string of up to 28 characters.

Label

Contains the attribute label.

Depending on the content of the text box **Label** and operations performed on the text box **Name**, the text box **Label** can automatically be populated with a default value. Refer to the description of the text box **Name** for further information.

The attribute label is language-dependent. A type attribute can therefore have different language-specific type labels. Depending on the user interface language of the operating system, the appropriate type attribute label is selected automatically when a type is retrieved from the database.

Valid values: A character string of up to 255 characters.

RDF Name

RDF attribute name.

Help Message

A short instruction that outlines the purpose of the respective type attribute.

Like type attribute labels, help messages of type attributes are language-dependent. A type attribute can have different help messages with the appropriate help message automatically selected depending on the current user interface language setting of the operating system.

Valid values: A character string of up to 2000 characters.

Data Type

Determines the data type of the respective type attribute.

The **Data Type** property is represented by a list box from which a certain data type can be selected. Besides standard data types, such as, LongInt, String, and so on, the list also contains custom data types defined on the **Enumerations** tab (see [3.5.1.5 Enumerations Tab](#)).

Table 3-10 Available Standard Data Types

Bool	Boolean Value (True, False)
Currency	Currency
Date	Date only
DateTime	Date and Time
DoubleFloat	8-byte floating-point number (real)
Enum ...	All enumeration data types
Float	4-byte floating-point number (real)
LongInt	Signed 4-byteinteger value
LongString	Variable length Unicode string (up to 1 GB)
ShortInt	Signed 2-byte integer value
String	Unicode string
Time	Time

Depending on the selected data type, the type, the accessibility and the content of interactive elements related with the properties **String Length**, **Default Value** and **Data Format** can change or can be subject to certain restrictions:

- The text box **String Length** is enabled only when the selected data type is LongString or String. Otherwise it is disabled. Also, when the attribute is a string attribute, the range of value of the string length changes depending on the selected data type.
- The text box **Data Format** is enabled only when the selected data type is Currency, DoubleFloat, Float, LongInt, or ShortInt. Otherwise it is disabled.
- When the selected data type is a custom data type originating from the **Enumerations** tab or a data type that provides a fixed set of values (for example, Bool), a list box is displayed for the property **Default Value** instead of a text box. The selected data type also restricts the content of the property **Default Value**. For example, if the data type has been set to LongInt, only numerical values would be accepted when entering a default value.
- Certain data types, such as Currency is associated with distinct input masks to control data entry for the property **Default Value**. An input mask is a character string composed of literal display characters and can be understood to be a template which forces entered data to show up in a specific notation. For example, the input mask associated with the data type Currency would force the entered value **123456** to show up as **\$ 1,234.56**.

String Length

Determines the maximum length of the character string a string attribute can hold.

This text box is valid only when the data type identified is a string or a binary attribute. Otherwise, the text box is disabled.

Default Value

Specifies a default value for the respective attribute.

Initially, the interactive element related to the **Default Value** property is a text box, but a list box can be used instead depending on the selected data type.

When the selected data type specifies the type attribute to be a date/time attribute, a special spin box replaces the text box. This element provides extended selection tools for selecting date and time values.

General	Attributes	Enumerators	Hierarchy	References
Name	MessageConfigurati	DisplayOrder	adfadf	
Label	Message Configurati	Display Order	adfadf	
Help Message	Relates the FieldRes	This value is used to		
Rdf Name	MessageConfigurati	DisplayOrder		
Data type	EnumMessageConfi	DoubleFloat	DateTime	
String Length				
Default Value	Default		06:00 06 Nov 2008	
Units of Meas	Cost_per_Hour			
Scope	ALL	ALL		
Namespace	spc	spc		
Internal	No	No		
Do Not Copy	No	Yes		
Nerc Profile	No	No		

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Figure 3-81 Selection Tools for Date/Time Selection

The spin box consists of a text box, up/down arrows, and an arrow button. The text box shows the selected date/time value in the format hh:mm DD MMM YYYY. Each single component of the date/time value can be selected by clicking and can then be increased or decreased individually by clicking the up/down arrows on the spin box. When the up/down arrows are clicked before a specific component of the date/time value is selected, the hour component of the date/time value is automatically selected and modified.

To select a specific date, the arrow button can be clicked to open a calendar. The calendar provides tools to page by month or by year. When the calendar is opened, it is automatically adjusted to show the days of the current month. A surrounding square indicates the current day. Days of the current month are displayed in black color, whereas days of the previous and next month, which can also show up, appear dimmed.

Every date shown in the calendar can be clicked to be used for the **Default Value** property. After a date has been selected, it is displayed in the text box of the spin box and the calendar is closed.

Valid values: Depends on the setting of the property **Data Type**.

Units of Meas

Determines the unit of measure label of the respective attribute. For example: Percent, Degrees Fahrenheit, and so on.

Scope

A comma-separated list of scopes (applications that use this attribute, for example, AGC, UC, ...)

Name Space

Name space the attribute belongs to. Indicates whether it is part of cim, sp5, sp3 or a project-specific extension.

Internal

Determines whether the respective attribute can become visible on the **Attributes** tab of the Single-Instance Editor when it is an instance of the respective type.

Table 3-11 Valid Values

Yes	Attribute is not visible.
No	Attribute is visible on the Attributes tab of the Single-Instance Editor.

Do Not Copy

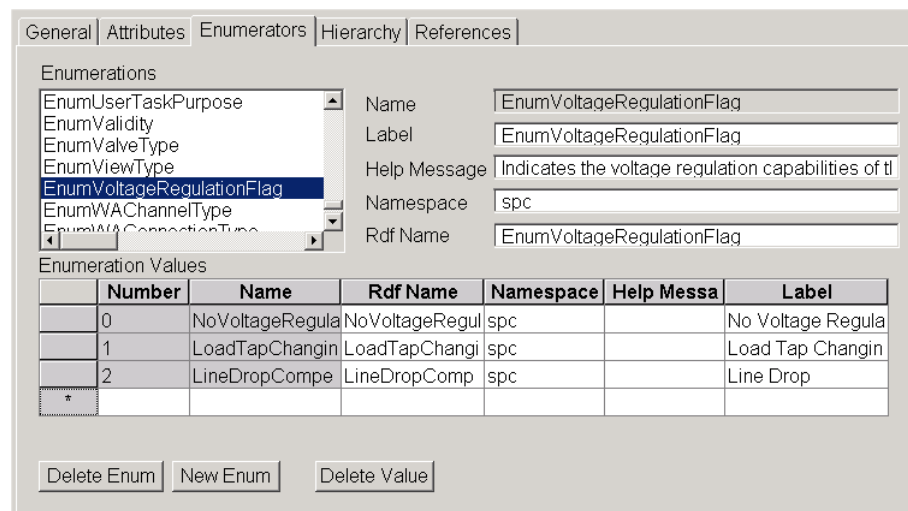
Defines whether the value of an attribute is copied during a Copy/Paste request.

3.5.1.5 Enumerations Tab

The **Enumerations** tab is used for defining and managing enumerations.

Enumerations are predefined data types that can be applied for the property **Data Type** in the **Attributes tab**.

After defining an enumeration, its name shows up in the list box **Data Type** in the **Attributes tab**.



[IMMRF025-061108-enUS-01.tif]

Figure 3-82 Enumerations Tab

The Enumerations tab consists of 2 main areas:

- Enumeration Area
- List of Enumeration Values

The Enumeration Area provides tools for adding and deleting enumerations.

Enumerations

The list shows the names of all enumerations in alphabetical order. Enumerations are inherited like attributes and associations. Each list entry can be clicked to see the definition of the enumeration.

Name

Name of the enumeration. It is recommended that the enumeration name begins with the string Enum.

Label

Defines the label that is displayed on the IMM user interface.

Help Message

Defines the help message that shall be displayed on the IMM user interface.

Name Space

Name space the enumeration belongs to. Indicates whether it is part of cim, sp5, sp3 or a project-specific extension.

Rdf Name

RDF name of the enumeration.

Buttons

- **New Enum**

Appends a new custom enumeration.

After clicking the button **New Enum**, a Name Entry dialog pops up which is used to enter a name for the new enumeration. The entered enumeration name is inserted in the list of values.



Note

The name of the new enumeration must be unique.

- **Delete Enum**

Removes the selected custom enumeration and its associated enumeration values.

- **Delete Value**

Removes the selected enumeration value from the currently selected enumeration.



Note

When an entire table column was selected, all table rows except 1 row (all enumeration values) are deleted. The remaining row is cleared and a key value is automatically inserted in the **Number** column. The displayed key value can be manually replaced. The deletion must be saved to make it effective.

A row of the Entries table can also be removed by placing the mouse pointer in the respective row and pressing the ENTER key on the keyboard.

The Enumeration Area provides interactive elements for adding and deleting enumeration value properties.

Entries Table

Table containing the enumeration values of the selected enumeration. Each table row represents a single enumeration value. The table columns represent enumeration value properties and provide selection tools for entering and modifying those properties.

The heading row of the table consists of buttons each of which can be clicked to select an entire table column. The label of each button denotes the purpose of the respective property. The left-hand button of that row selects the entire table when clicked.

The left-hand table column consists of buttons which can be used to select an entire table row (a certain enumeration value). A range of table rows can be selected by dragging the mouse pointer over the buttons of the left-hand column while keeping the left mouse button pressed.

- **Number**
This column represents a key value used to identify the enumeration value.
Valid values: Integer value.
- **Name**
Name of the enumeration value.
- **Rdf Name**
RDF name of the enumeration value.
- **Help Message**
Defines the help message that shall be displayed on the IMM user interface.
- **Name Space**
Name space the enumeration value belongs to. Indicates whether it is part of cim, sp5, sp3 or a project-specific extension.
- **Label**
Defines the label that is displayed on the IMM user interface.

3.5.1.6 Hierarchy Tab

The **Hierarchy** tab allows you to establish hierarchy associations between types. A hierarchy association implies a parent-child relationship between instances of the currently edited type and instances of another type. For example: a substation has a bay. This relationship is of particular interest for instance creation since it forces instances to arrange in a certain hierarchical order when defined.

General Attributes Enumerators Hierarchy References					
Association Name	DomainMemberOfsy	PowerSystemResour	EquipmentMemberO	EquipmentMemberC	
Label	sysOrphan	Company	MemberOf	EquipmentContaine	
Help Message		A power system reso		The association is u	
Child Cardinality	0..n	0..n	0..n	0..n	
Parent Type	sysOrphan	Company	VoltageLevel	EquipmentContaine	
Parent Cardinality	0..1	0..1	0..1	0..1	
Rdf Name Role A	Domain_MemberOf	Operates_PSRs	Equipment_Member	Contains_Equipmer	
Rdf Name Role B	MemberOf_sysOrph	OperatedBy_Compa	MemberOf_VoltageL	MemberOf_Equipm	
Namespace	spc	cim	spc	cim	
Scope	ALL	ALL	ALL	ALL	

[IMMRF047-140909-enUS-01.tif]

Figure 3-83 Hierarchy Tab

The elements provided on the Hierarchy tab form a table where each column represents a single association and each row represents a certain property. A table cell therefore represents a distinct property of a hierarchy association.

Some associations are inserted automatically when a new type is created. These associations are inherited from the parent types and are read only. Inherited associations cannot be modified or deleted in a subordinate type. Text boxes related to such associations are disabled.

Association Name

Contains the name of the respective hierarchy association.

Valid values: a character string of up to 255 characters.

Label

A short character string that denotes the identification of the hierarchy association for documentation purposes.

The label of a hierarchy association is a language-dependent. Depending on the user interface language setting of the operating system, the appropriate label is selected automatically when a type is retrieved from the database.

Valid values: A character string of up to 64 characters.

Help Message

A comment that outlines the purpose of the respective hierarchy association.

Like the association label, the help message is language-dependent.

A hierarchical association can have different help messages with the appropriate help message automatically selected depending on the current user interface language setting of the operating system.

Child Cardinality

Defines the cardinality on the child side of the hierarchy association (the current object side of the association).

Table 3-12 Valid Values

0..1	Zero to One
1..1	One to One
0..n	Zero to Many
1..n	One to Many

Parent Type

Specifies the type on the parent side of the hierarchy association (the current type represents the child side of the hierarchy association).

The property **Parent Type** is represented by a list box from which you can select a type name.

Valid values: The name of an existing type.

Parent Cardinality

Defines the cardinality of the association on the parent side of the hierarchy association (the opposite side of the association).

The property **Parent Cardinality** is represented by a list box from which you can select a cardinality range.

Table 3-13 Valid Values

0..1	Zero to One
1..1	One to One

Name Space

Name space the association belongs to. Indicates whether it is part of cim, sp5, sp3 or a project-specific extension.

Scope

A comma-separated list of scopes (applications that use this association, for example, AGC, UC, ...)

Rdf Name Role A

Defines the RDF Role A name of the association used by CIM-RDF import/export. The RDF Role A name must be unique within all RDF Role A and RDF Role B names within a type.

Rdf Name Role B

Defines the RDF Role B name of the association used by CIM-RDF import/export. The RDF Role B name must be unique within all RDF Role A and RDF Role B names within a type.

3.5.1.7 References Tab

General Attributes Enumerators Hierarchy References				
Association Name	DomainHasAssigne	IdentifiedObjectBelo	FieldResourceBelon	Po
Label	OwnerOperatorAssig	Belongs To (Modelin	Belongs To (Multisite Bu	
Help Message		An IdentifiedObject b	Link to a MultisiteGro	
Cardinality On Current Side	0..n	0..n	0..n	0..1
Referenced Object's Type	Percentage	ModelingAuthoritySe	MultisiteGroup	Bu
Cardinality On Referenced Object's Side	0..n	0..1	0..1	0..
Rdf Name Role A	Domain	IdentifiedObjects	FieldResource_Belo	Po
Rdf Name Role B	Percentage	ModelingAuthoritySe	BelongsTo_Multisite	Op
Namespace	spc	cim	spc	sp
Scope	ALL	ALL	ALL	AL
Reference type	Reference	Reference	Reference	Re

[IMMRF048-140909-enUS-01.tif]

Figure 3-84 References Tab

The **References** tab allows you to establish reference associations between types. Reference associations define general-purpose relationships which can later be used to define links between instances.

A reference is an association between 2 types. 2 types of associations can be distinguished: the attributive and the grouping type.

To understand the difference, consider instances of the types. In a normal reference, 2 instances are linked together. If the referring instance is copied the link gets lost.

If it is an attributive link, the link is retained if the instance is copied. An example for an attributive link is the relationship between a DigitalMeasurement and its StateSet. The StateSet defines the values for the DigitalMeasurement, for example, Open and Close. The assumption is that a DigitalMeasurement used for a specific purpose for example, the status of a circuit breaker, should keep its properties when it is copied. One of the properties of the status measurement is a specific state set.

The elements provided on the References tab form a table where each column represents a single reference and each row represents a certain property. A table cell therefore represents a distinct property of a reference association.

Some reference associations are automatically inserted when a new type is created. These associations are inherited from the parent types and are read only. Inherited reference associations cannot be modified or deleted in a derived type. Elements, like text boxes, related to such reference associations are disabled.

Association Name

Contains the name of the respective reference association.

Valid values: A character string of up to 255 characters.

Label

Contains the label of the respective reference association.

Valid values: A character string of up to 64 characters.

Help Message

A short description of the purpose of the association.

Valid values: a character string of up to 2000 characters.

Cardinality on Current Side

Defines the cardinality of the reference association on the side of the current type.

The property **Cardinality On Current Side** is represented by a list box from which you can select a cardinality range.

Table 3-14 Valid Values

0..1	Zero to One
1..1	One to One
0..n	Zero to Many
1..n	One to Many

Referenced Object's Type

Determines which type is allowed for referenced types when a link is established.

The property **Referenced Object's Type** is represented by a list box from which you can select the desired type.

Valid value: A name of an existing type.

Cardinality on Referenced Object's Side

Defines the cardinality of the reference on side of the referenced type (the opposite side of the association).

The property **Cardinality On Referenced Object's Side** is represented by a list box from which you can select a cardinality range.

Table 3-15 Valid Values

0..1	Zero to One
1..1	One to One
0..n	Zero to Many
1..n	One to Many

Reference Type

Defines the reference type of the association.

Table 3-16 Valid Values

Reference	A 'normal' association between types
Attributive	A reference type for an attributive link between 2 instances. In this case the link is retained if 1 instance of types is copied.

Name Space

Name space the association belongs to. Indicates whether it is part of cim, sp5, sp3 or a project-specific extension.

Scope

A comma-separated list of scopes (applications that use this association, for example, AGC, UC, ...)

Rdf Name Role A

Defines the RDF Role A name of the association used by CIM-RDF import/export. The RDF Role A name must be unique within all RDF Role A and RDF Role B names within a type.

Rdf Name Role B

Defines the RDF Role B name of the association used by CIM-RDF import/export. The RDF Role B name must be unique within all RDF Role A and RDF Role B names within a type.

3.5.2 Decision Table Editor

3.5.2.1 Introduction

The Decision-Table Editor is a special purpose editor used to view and edit properties of presentation logic instances as well as to create new presentation logic instances.

A presentation logic instance defines the graphical representation of an instance (for example, a circuit-breaker) for the runtime user interface of Spectrum Power considering different operational situations or purposes. For example: different symbols represent a circuit-breaker depending on its status.

Rules and Conditions

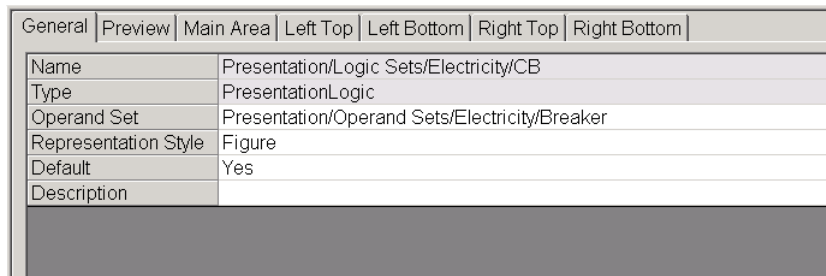
Rules and conditions control the presentation of field resources. Rules determine the presentation of a power system component (graphical symbol, symbol shape, color, flashing, and so on) by evaluating conditions (combinations of component attribute values).

Each operational situation or purpose is related to some conditions and rules which eventually determine the appearance of a field resource for the particular situation or purpose. The conditions and rules form a matrix of logical combinations called decision table. Each decision table covers one particular operational situation or purpose.

Selecting the Decision-Table Editor

The Decision-Table Editor is automatically opened after you have selected a presentation logic instance on the instance tree or after creation for a presentation logic instance was invoked in the type selection. For additional information on how an instance can be forced to open a specific editor, refer to the description of the type property **Graphic Sheet** in [3.5.1.3 General Tab](#).

When the Decision-Table Editor opens, the **General** tab is exposed automatically.



[IMMRF028-071108-enUS-01.tif]

Figure 3-85 Decision-Table Editor

Header

The header of the Decision-Table Editor shows the name of the presentation logic instance that has been selected from the instance overview tree.



Note

When the Decision-Table Editor was invoked by the type selection to create a presentation logic instance, the new presentation logic instance becomes a subordinate instance of the instance shown in the header.

Body

The body of the Decision-Table Editor provides different tabs. Each is used to set or modify a distinct decision table of the respective presentation logic instance.

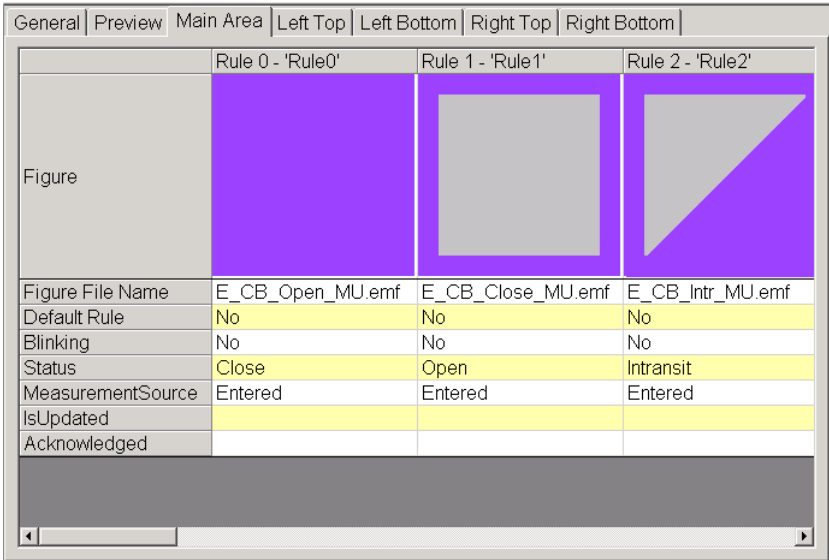
For additional information on the generic features of tabs, refer to [3.1.5.3 Tabs](#).

The following tabs are provided on the body of the Decision-Table Editor:

- **General** tab
- **Preview** tab
- **Left Top** tab
- **Right Top** tab
- **Main Area** tab
- **Left Bottom** tab
- **Right Bottom** tab

3.5.2.2 Editor Context Menu

The editor context menu of the Decision-Table Editor is opened by clicking the tabs of the Decision-Table Editor with the right mouse button. Depending on the clicked area a different menu opens.



[IMMOP030-311008-enUS-01.tif]

Figure 3-86 Areas for Different Context Menus

Edit Zones

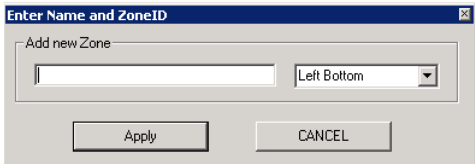
The context menu of the tab header of the Decision-Table Editor contains the following menu items:

- **Add new Zone**
- **Remove Zone**
- **Rename**

Detailed descriptions of the menu items are provided in the subsequent paragraphs.

Add New Zone

Opens the New zone dialog for adding a new zone (presentation area):



[IMMRF029-071108-enUS-01.tif]

Figure 3-87 New Zone Dialog

In this dialog, you can select the corresponding zone and enter the desired zone name. After clicking the button **Apply**, the new zone is displayed as new tab.



Note

In total, there can be only 5 zones.

Remove Zone

Removes the selected zone (tab).

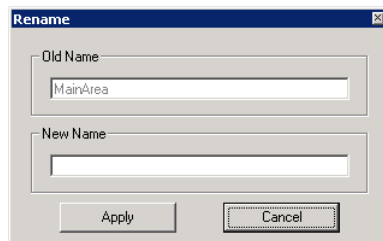


Note

The **General** tab and the **Preview** tab cannot be removed.

Rename

Opens the tab rename dialog for renaming a zone:



[IMMRF030-071108-enUS-01.tif]

Figure 3-88 Tab Rename Dialog

In this dialog, you can enter a new zone name. After clicking the button **Apply**, the new zone name is displayed on the tab.

Edit Rules

The context menu of first column of the rule table of the Decision-Table Editor contains the following menu items:

- Add new rule
- Insert new rule
- Remove Rule
- Rename

Detailed descriptions of the menu items are provided in the subsequent paragraphs.



Note

This context menu cannot be invoked on the General and the Preview tab.

Add New Rule

Adds a new rule column to the decision table.

Insert New Rule

Inserts a new rule column left of the selected rule column.

Remove Rule

Deletes the selected rule of the decision table. All interactive elements related to the concerned rule are removed from the respective tab of the Decision-Table Editor.

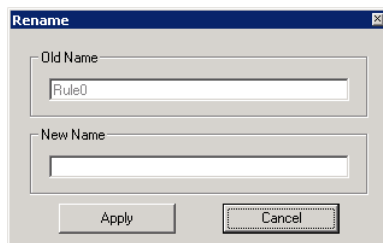


Note

When a rule is deleted, the changes must be saved to make the deletion effective.

Rename

Opens the Rule rename dialog for renaming a rule:



[IMMRF031-071108-enUS-01.tif]

Figure 3-89 Rule Rename Dialog

In this dialog, you can enter a new rule name. After clicking the button **Apply**, the new rule name is displayed on the header of the decision table.

Edit Conditions

The context menu of the condition rows of the Decision-Table Editor contains the following menu items:

- **Add New Condition**
- **Insert New Condition**
- **Remove Condition**
- **Change All Value Formats**

Detailed descriptions of the menu items are provided in the subsequent paragraphs.



Note

The row menu cannot be invoked on the **General** and the **Preview** tab.

Add New Condition

Opens a submenu right of the context menu. This submenu contains the appropriate conditions of the presentation area. The new condition is added below the existing rows of the decision table by clicking the corresponding menu item of the submenu.



Note

Conditions that are already displayed in the decision table are disabled on the submenu.

Insert New Condition

Opens the condition submenu. The new condition is added above the selected condition of the decision table by clicking the corresponding menu item of the submenu.

Remove Condition

Opens the condition submenu. Clicking the corresponding menu item of the submenu deletes the selected condition.



Note

When a condition is deleted, the changes must be saved to make the deletion effective.

All interactive elements related to the concerned condition are removed from the respective tab of the Decision-Table Editor.

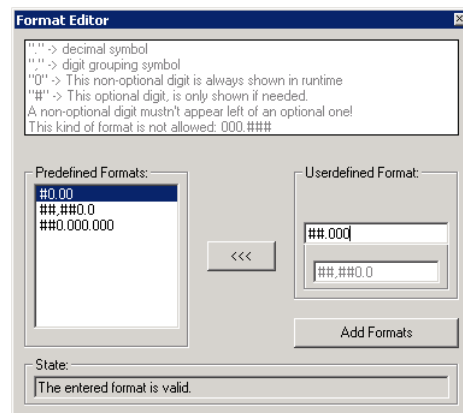
Change All Value Formats

Clicking this menu item opens the Format Editor (see the following figure).



Note

This menu item is only enabled for analog value instances, which contain the condition 'Value'.



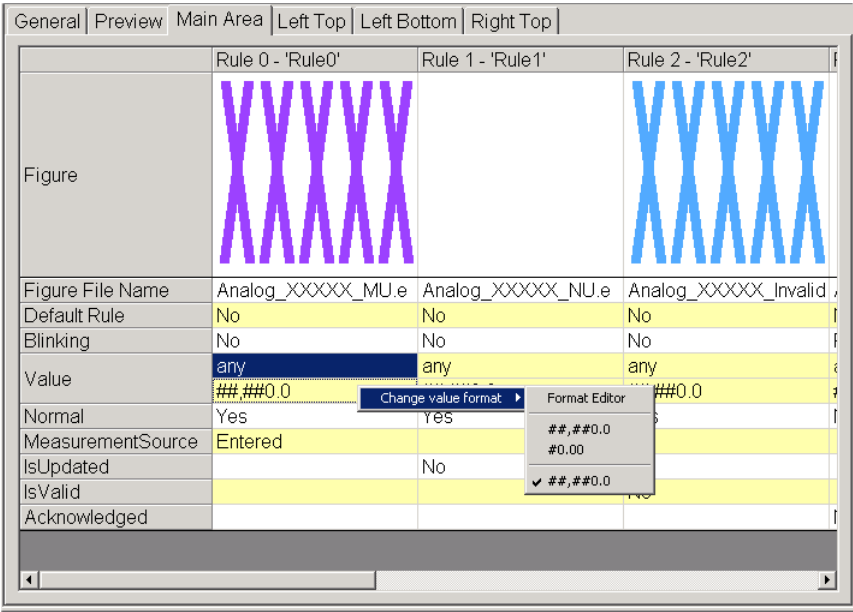
[IMMRF032-071108-enUS-01.tif]

Figure 3-90 Format Editor

The Format Editor provides the definition of user-defined value formats. You can enter the new file format into the field **Userdefined Format**. A description of the value format characters is displayed above the format fields.

Clicking the button <<< moves the specified value format to the field **Predefined Formats**. Clicking the button **Add Formats** adds the user-defined formats (listed in the field Predefined Formats) to the predefined formats, which can be accessed in the text boxes of the condition row **Value** of the decision table.

Value formats of the decision table can be changed by clicking the corresponding field with the right mouse button. A context menu with the predefined value formats opens. You can select a predefined value format from the context menu. After selecting the corresponding menu item, the selected value format is displayed in the appropriate field of the decision table.

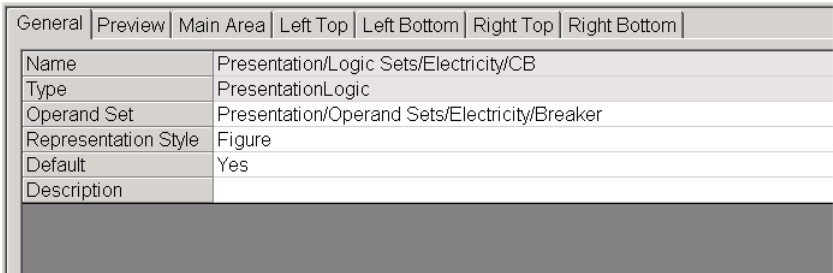


[IMMRF033-071108-enUS-01.tif]

Figure 3-91 Change Value Formats

3.5.2.3 General Tab

The **General** tab provides tools for viewing and editing general properties of a presentation logic instance.



[IMMRF028-071108-enUS-01.tif]

Figure 3-92 Decision-Table Editor: General Tab

Name

Contains the name of the respective presentation logic instance.

Valid values: A character string of up to 255 characters.

Like other instances, some restrictions apply when entering an instance name of a presentation logic instance. For additional information on this topic, refer to the description of the instance property **Name** in [3.3.1.2 General Tab](#).



Note

When a new presentation logic instance is created, the property Name must be set before another tab can be selected. Otherwise, the attempt to select another tab is ignored and a message box pops up that displays an appropriate system message. Once a new presentation logic instance was saved, its name cannot be changed. Therefore, the text box **Name** is disabled except for defining a new presentation logic instance.

Type

Displays the name of the type on which the respective presentation logic instance is based. Read only.

For additional information on this property, refer to description of the type property **Label** in [3.5.1.3 General Tab](#).



Note

Once a new presentation logic instance was saved, its underlying type cannot be changed. Therefore, the text box **Type** is disabled except for defining a new presentation logic instance.

Operand Set

Specifies the operand set of the respective presentation logic instance.

The property **Operand Set** is represented by a list box from which you can select an operand set.

Valid values: The name of an existing instance that is configured as an operand set.

An operand set is a collection of special instances that describe distinct operands. Operands are used in conditions of decision tables. For example, a condition to determine if a circuit breaker is open can look like Status == Open, where Status would be the operand.

Each operand set is composed of a superordinate instance and some subordinate instances that represent the operands of the operand set. The superordinate instance specifies the field resource for which the operand set can be applied. Each subordinate instance is an operand that represents a certain characteristic or status of the field resource specified by the superordinate instance. Different operand sets are provided for different field resources. For example: operand set for circuit breakers, operand set for analog values, and so on.

The setting of the property **Operand Set** determines which operands can be used in conditions of the decision tables on the **Main Area** tab, the **Left Top** tab, the **Left Bottom** tab, the **Right Top** tab, and the **Right Bottom** tab. For additional information on conditions in a decision table, refer to [3.5.2.5 Main Area Tab](#).



Note

Once a new presentation logic instance was saved, its operand set configuration cannot be changed. Therefore, the text box **Operand Set** is disabled except for defining a new presentation logic instance.

Representation Style

Determines whether the state of the instance is displayed by a graphical figure (for example, a symbol for a circuit breaker) or a formatted value (for example, a formatted voltage value).

The setting of the property **Representation Style** affects the accessibility of the representation property **Condition Rows** on the other tabs of the Decision-Table Editor. For additional information on this property, refer to its descriptions in [3.5.2.5 Main Area Tab](#). Once a name was entered for a new presentation logic instance, its basic presentation style cannot be changed. Therefore, the text box Representation Style is disabled except for defining a new presentation logic instance.

Table 3-17 Valid Values

Figure	A graphical symbol represents the related data point.
Formatted Value	A formatted numerical value represents the related data point.

Default

The setting of the property **Default** defines the respective presentation logic instance as the default presentation logic instance of all presentation logic instances using the same operand set.

This setting is evaluated when creating a control for the runtime user interface with the Graphics Editor. After a generic control was created with the Graphics Editor, a certain field resource, represented by its instance, is assigned to the generic control. The type underlying the assigned instance is evaluated to find presentation logic instances with matching operand sets. Among these, the default presentation logic instance determines the presentation of the control on the runtime user interface unless another of the matching presentation logic instances was selected.

Yes	Currently edited presentation logic instance is used as default.
No	Not default

Description

A short instruction which outlines the purpose of the respective presentation logic instance.

Valid values: A character string. The maximum string length depends on the configuration of the **Description** attribute.

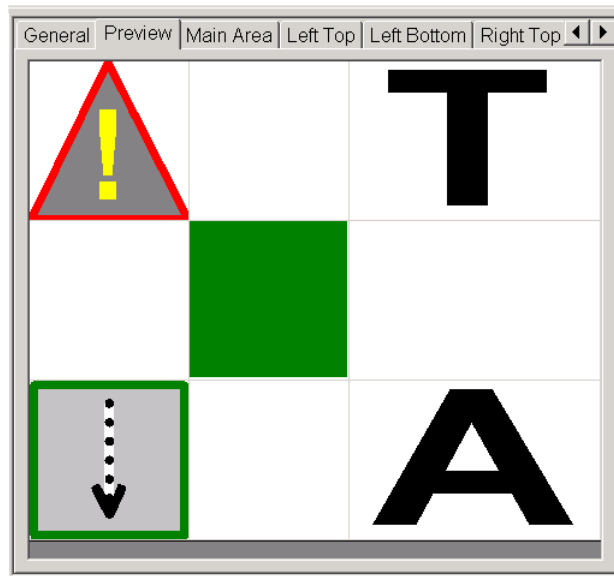
3.5.2.4 Preview Tab

The **Preview** tab shows how the respective field resource would be displayed on the runtime user interface in different operational situations or for different purposes.

Presentation Area

The Preview tab is divided into different presentation areas each of which represents a specific component attribute.

Each presentation area is related to a component attribute and, thus, a decision table determines the appearance (symbol, symbol shape, color, blinking, and so on) of the field resource attribute. 1 rule of a decision table can be configured as default rule. The default rule is used to show a certain appearance of the field resource attribute on the related presentation area.



[IMMOP029-301008-enUS-01.tif]

Figure 3-93 Preview Tab

Individual tabs are provided by the Decision-Table Editor to configure the different decision tables. Each presentation area can be double-clicked to access the related decision table and, therefore, to expose the respective tab.

The following list shows the presentation areas of the Preview tab and tabs to which they are related:

Left Top	Left Top tab
Right Top	Right Top tab
Main	Main Area tab
Left Bottom	Left Bottom tab
Right Bottom	Right Bottom tab



Note

The remaining areas of the Presentation Area, such as Top Middle, Left Middle, and so on, are not used.

3.5.2.5 Main Area Tab

The **Main Area** tab provides interactive elements to define a decision table. The default rule of this decision table determines the content of the presentation area in the center of the **Preview** tab.

The elements provided on the Main Area tab, such as text boxes, form the decision table. The table columns represent the rules of the decision table. The table rows represent presentation properties and conditions of a rule. Therefore, each rule contains conditions and presentation properties. Refer to the following figure to see, which rows represent presentation properties and which rows represent conditions.

To determine the appearance of a field resource, the rules of a decision table are evaluated. 1 rule of a decision table must be fulfilled (be true) to finish decision table processing.

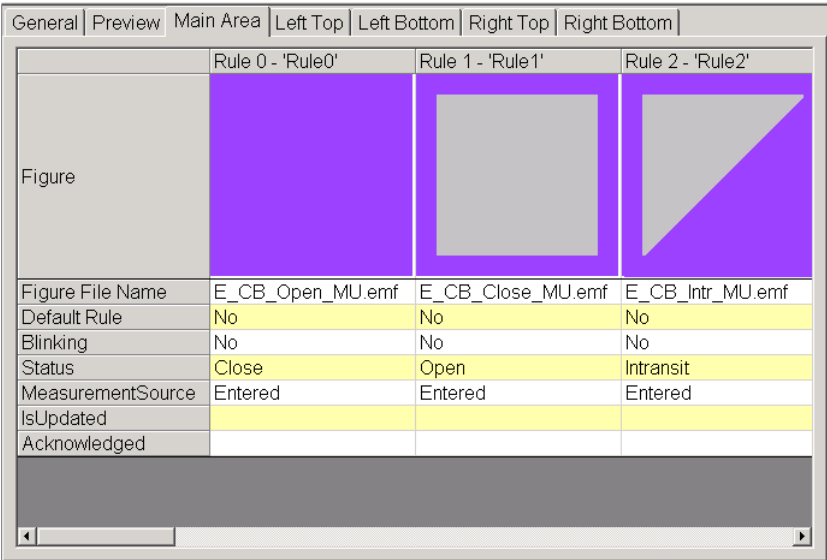
The presentation properties of the fulfilled rule then determine the presentation of the respective resource. When no rule was fulfilled, the default rule determines the presentation of the respective resource.

Rules and conditions can be deleted from a decision table using the editor context menu. When a rule or a condition is deleted, the interactive elements related to that particular column or row are removed from the Main Area tab. When the number of shown rules or conditions exceeds the Main Area tab, scroll bars are automatically attached to the bottom edge and/or the right edge of the Main Area tab to allow scrolling of the page content.



Note

Presentation properties are mandatory and cannot be removed from a decision table using the editor context menu. When rules or conditions are deleted, the deletion must be saved to become effective.



[IMMOP030-311008-enUS-01.tif]

Figure 3-94 Main Area Tab

The following representation properties are provided on the Main Area tab:

Figure

Displays a preview image of the graphic symbol used for representing the respective field resource.

The text box **Figure** can be double-clicked to open the Graphic File Browser dialog. For additional information on the Graphic File Browser dialog, refer to [3.5.2.10 Graphic File Browser Dialog](#).

Figure File Name

An interactive element that contains a text box (**Figure File Name**) and a button (...). The text box **Figure File Name** displays the file name of the EMF graphic file that contains the graphic symbol used for representing the field resource. The text box is read only.

...

A button that opens the Graphic File Browser dialog.

**Note**

This button is only displayed, after you have clicked the **Figure File Name** field.

Default Rule

Determines the respective rule to be the default rule of the decision table.

The default rule determines the content of the **Preview** tab. When the **Preview** tab is displayed, the default rule of each related tab is evaluated and the configured graphical symbol or formatting string is displayed on the corresponding Presentation Area.

Table 3-18 Valid Values

Yes	Respective rule is used as default rule.
No	Not default rule.

Blinking

Determines if the specified graphic symbol blinks when it is displayed on the runtime user interface.

Table 3-19 Valid Values

Yes	Graphic symbol blinks when represented on the runtime user interface.
No	No blinking.

Condition Rows

The decision table rows located beneath the representation property rows represent conditions. Conditions can be added/removed by using the editor context menu of the Decision-Table Editor.

Once a condition was inserted, it is valid for all rules of the decision table. However, whether the condition is considered or not considered when a rule is evaluated, depends on the configuration of the condition in the condition section of the rule.

If an operator and a value are configured in the condition section of a rule, the respective condition is considered when the rule is evaluated. If a condition section in a rule is empty, the respective condition is considered as irrelevant in that rule.

3.5.2.6 Left Top Tab

The **Left Top** tab provides interactive elements to define a decision table. The default rule of this decision table determines the content of the left top presentation area of the **Preview** tab.

The elements provided on the Left Top tab, such as text boxes, form a decision table. For additional information on these elements, refer to [3.5.2.5 Main Area Tab](#).

3.5.2.7 Left Bottom Tab

The **Left Bottom** tab provides interactive elements to define a decision table. The default rule of this decision table determines the content of the left bottom presentation area of the **Preview** tab.

The elements provided on the Left Bottom tab, such as text boxes, form a decision table. For additional information on these elements, refer to [3.5.2.5 Main Area Tab](#).

3.5.2.8 Right Top Tab

The **Right Top** tab provides interactive elements to define a decision table. The default rule of this decision table determines the content of the right top presentation area of the **Preview** tab.

The elements provided on the Right Top tab, such as text boxes, form a decision table. For additional information on these elements, refer to [3.5.2.5 Main Area Tab](#).

3.5.2.9 Right Bottom Tab

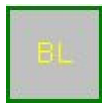
The **Right Bottom** tab provides interactive elements to define a decision table. The default rule of this decision table determines the content of the right bottom presentation area of the **Preview** tab.

The elements provided on the Right Bottom tab, such as text boxes, form a decision table. For additional information on these elements, refer to [3.5.2.5 Main Area Tab](#).

3.5.2.10 Graphic File Browser Dialog

The Graphic File Browser dialog allows you to browse the available Enhanced Metafile Format (EMF) graphic files to select a graphic symbol (figure) or value style for the representation of a certain field resource (for example, a circuit breaker or an analog value). EMF files are created with the Graphics Editor.

Graphic symbols (Figures) determine the geometrical shape, the size, and color of a field resource (for example, a circuit breaker). Value styles determine the font properties (font family, size, angle, weight, and color) of a value resource (see the following figure):



[IMMED019-170108-enUS-01.tif]

Figure 3-95 Example of a Graphic Symbol

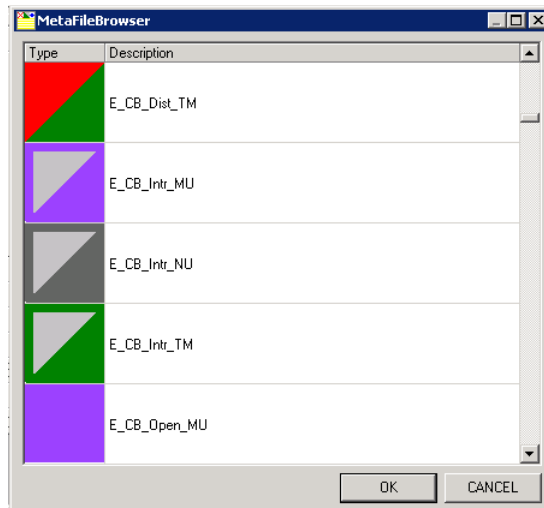


[IMMED020-170108-enUS-01.tif]

Figure 3-96 Example of a Value Style

The dialog is accessed by clicking the button ... or double-clicking the text boxes **Figure** or **Figure File Name** in the **Main Area** tab, the **Left Top** tab, the **Left Bottom** tab, the **Right Top** tab, or the **Right Bottom** tab on the Body of the Decision-Table Editor.

The Graphic File Browser dialog consists of a graphic symbol list that shows preview images and descriptions of the available graphic files. The corresponding image is loaded into the decision table by double-clicking the appropriate graphic symbol.



[IMMOP031-311008-enUS-01.tif]

Figure 3-97 Graphic File Browser

Graphic Symbol List

A single-selection list with 2 columns that lists all available graphic files.

Each list entry represents a single graphic file and can be clicked for selection. The left column (**Type**) shows a preview image of the graphic file content. The right column (**Description**) shows a short comment that describes the purpose of the respective graphic file. When a list entry is selected its background color changes to indicate the selection.



Note

The comment shown in the right column of the graphic symbol list is actually the name of the concerned EMF graphic file without the file extension.

3.6 Wizards

3.6.1 Introduction

Wizards support you when creating new instances to make sure that all required attributes of the generated instances are set according to the corresponding instance requirements.

The Information Model Management provides the following wizards:

- Energy Accounting Wizard
- Script Definition Wizard
- Calculation Wizard
- Script Wizard

3.6.2 Energy Accounting Wizard

3.6.2.1 Introduction

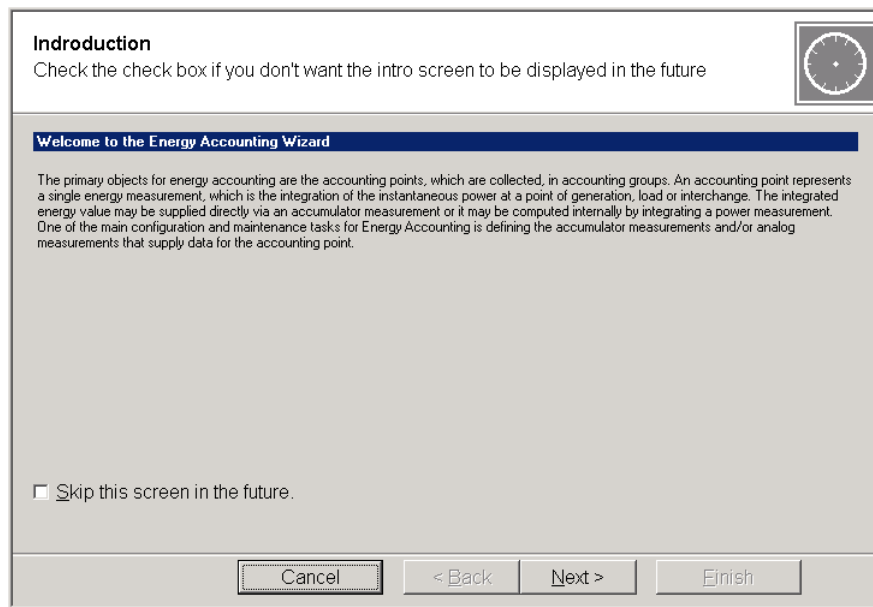
The primary objects for Energy Accounting are the accounting points, which are collected in accounting groups. An accounting point represents a single power measurement, which is the integration of the instantaneous power at a point of generation and load, respectively.

Templates for the accounting points are available to make sure that all required attributes of the auto generated accounting points are set according to the requirements of the project. The same is true for the default path names of the default tariff rule and default control area if applicable.

The wizard guides you through the following 5 steps and supports him/her in creating new accounting points under the primary node:

- Step 1: Definition of the number of import and export accounting points
- Step 2: Configuration of accounting points
- Step 3: Definition of measuring links to accounting points
- Step 4: Definition of tariff rule, control, and forecast area links to accounting points
- Step 5: Saving accounting points

The Energy Accounting Wizard is invoked by selecting the menu item **Accounting Wizard** from the context menu of the tree. It is enabled when a node is selected which can have accounting point instances as children.



[IMMRF034-101108-enUS-01.tif]

Figure 3-98 Energy Accounting Wizard



Note

You can skip the introduction of the Energy Accounting Wizard by clicking the check box in the introduction page. If necessary, it is possible to backtrack all steps. Some data gathered in 1 step can be discarded in the backward step.

3.6.2.2 Definition of the Number of Import and Export Accounting Points

In this step, you can define the accounting points to be created.

Step 1 - Enter number of accounting points
Specify the number of import and export accounting points which should be created!

Already existing Accounting Points

MWH Exp
MWH Imp

Number of export Accounting Points to add:
1

Number of import Accounting Points to add:
1

Note: The primary objects for energy accounting are the accounting points. An accounting point represents a single energy measurement, which is the integration of the instantaneous power at a point of generation, load or interchange.

Cancel < Back Next > Finish

[IMMRF035-101108-enUS-01.tif]

Figure 3-99 Definition: Number of Accounting Points

Number of Export/Import Accounting Points to Add

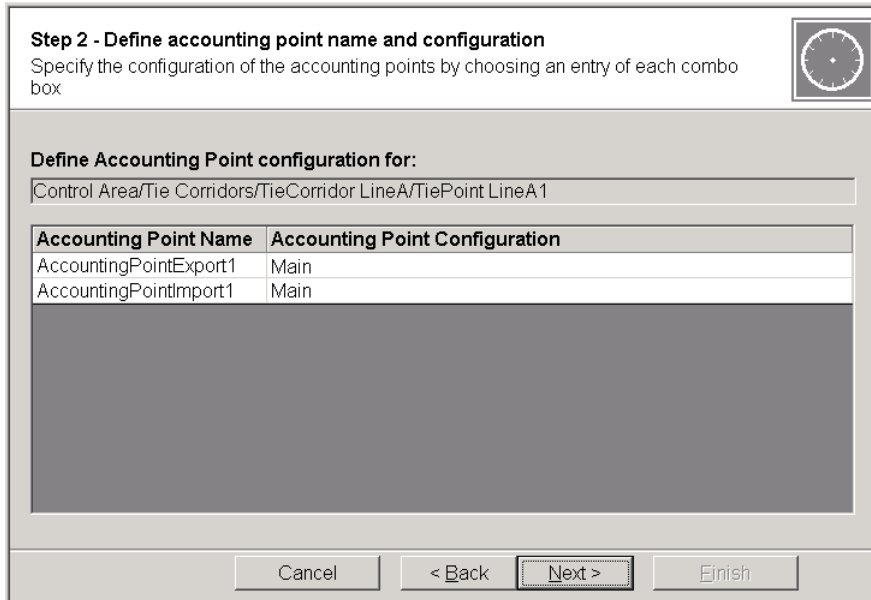
You can define the number of import and export accounting points to be created under primary node. The following name convention is used for the auto-generated instances:

Accounting Point <Import|Export> <number>

Depending on the interchange direction Import or Export is used, <number> is an auto-increment number and starts with the next free number if there are already auto generated accounting points defined for the primary node.

3.6.2.3 Configuration of Accounting Points

In this step, you can configure the accounting points that have been defined in the previous step.



Step 2 - Define accounting point name and configuration
Specify the configuration of the accounting points by choosing an entry of each combo box

Define Accounting Point configuration for:

Control Area/Tie Corridors/TieCorridor LineA/TiePoint LineA1

Accounting Point Name	Accounting Point Configuration
AccountingPointExport1	Main
AccountingPointImport1	Main

Cancel < Back Next > Finish

[IMMRF036-101108-enUS-01.tif]

Figure 3-100 Configuration of Accounting Points

Accounting Point Configuration

You can select the possible configurations from a list box. The following configurations are supported:

- Main
Accounting point has one main accumulator.
- Main, Control
Accounting point has main and control accumulator
- Main, Control, 1 Analogs
Accounting point has main and control accumulator and one integrated analog measurement
- Main, Control, 2 Analogs
Accounting point has main and control accumulator and 2 integrated analog measurements

3.6.2.4 Definition of Measuring Links to Accounting Points

In this step, you can define which measurements shall be linked to the accounting points.

Step 3 - Link accounting points to measurements
Define the links to existing measurements by using drag and drop from the tree in the Engineering Explorer



Link Accounting Point to Measurements for:
Control Area/Tie Corridors/TieCorridor LineA/TiePoint LineA1

Accounting Point Name	Type	Linked to
AccountingPointExport1	Raw Main	<Select Raw Main Accumulator>
	Main	<Select Main Accumulator>
	Authorized	<Select Authorized Accumulator>
AccountingPointImport1	Raw Main	<Select Raw Main Accumulator>
	Main	<Select Main Accumulator>
	Authorized	<Select Authorized Accumulator>

[IMMRF037-101108-enUS-01.tif]

Figure 3-101 Definition of Measuring Links

Linked To

You get the selected configuration for each accounting point and must define the links to existing measurements. Linking can be done through Drag & Drop; from the tree to the **Linked To** column of the Energy Accounting Wizard. The path can also be entered directly.



Note

A configuration change for a single accounting point is possible only by going back 1 step.

3.6.2.5 Definition of Tariff Rule Links, Control, and Forecast Area Links to Accounting Points

In this step, you can assign tariff rules, control areas, and forecast areas to the accounting points.

Step 4 - Link accounting points
Specify links to tariff rule set, control area and forecast area for each Accounting Point

Tariff Rule Set Control Area Forecast Area

Accounting Point Name	Tariff Rule Set
AccountingPointExport1	Accounting/.Tariff Rule Sets/Base Tariff Set
AccountingPointImport1	Accounting/.Tariff Rule Sets/Base Tariff Set

Cancel < Back Next > Finish

[IMMRF038-101108-enUS-01.tif]

Figure 3-102 Definition of Links

Tariff Rule/Control Area/Forecast Area

You can link each accounting point to a Tariff Rule and, optionally, to a control and/or forecast area. Links are defined via Drag & Drop; from the tree to the **Tariff Rule** or **Control Area** or **Forecast Area** tab respectively.

3.6.2.6 Saving Accounting Points

Completing the EA Wizard
The Accounting Points will be created in the configuration shown below. It may take some seconds for the wizard to import the data into the current context.

To complete this Wizard, click finish!

Accounting Point Name	Accounting Point Configuration
AccountingPointExport1	Main
AccountingPointImport1	Main

Cancel < Back Next > Finish

[IMMRF039-101108-enUS-01.tif]

Figure 3-103 Saving Accounting Points

In the last step, you can save the generated instances and links into the job by clicking the button **Finish**. After you have clicked the button **Finish**, a confirmation message is shown. After you have clicked the button **OK**, the wizard restarts with the introduction step and is ready for the next accounting point definition.

3.6.3 Script Definition Wizard

3.6.3.1 Introduction

The Script Definition Wizard creates formal calculation script definitions or generic script definitions under the selected node. Additionally the Script Definition Wizard can also be used for modifying existing script definitions.

The wizard guides you through the following 3 steps and supports him/her when creating the definitions:

- Step 1: Definition of the script and its attributes
- Step 2: Definition of formal script operands
- Step 3: Definition of the script body

The wizard is invoked by selecting the menu item **Script Definition Wizard** from the context menu of the tree.



Note

You can skip the introduction of the Script Definition Wizard by clicking the check box in the introduction page. If necessary, it is possible to backtrack all steps. All data gathered in 1 step must be discarded in the backward step.

3.6.3.2 Definition of the Script and Its Attributes

In this step, you can define the general parameters for the script.

Parent Path of Script Instance:	.General/User Defined Calculation Definitions
Name of Script Instance:	Add Two Operands
Description:	CalculationScriptDef
Script Function Name:	AddTwo
Trigger:	☐ On Timer Or Operand Has Changed ☐ On Timer And Operand Has Changed ☐ On Timer ☒ On Operand Change ☐ Fleeting ☐ GroupAlarmIndicator ☐ Discrete
Allowed Type:	☒ Analog ☐ Accumulator

[IMMRF040-101108-enUS-01.tif]

Figure 3-104 Definition of Script and Attributes

Parent Path of Script Instance

You can define the new script, which is created under the selected node. The text box is read-only and is determined by the node that was selected in the tree, when the wizard was started.

Name of Script Instance

The name of the calculation script definition (CalculationScriptDef) or the generic script definition (GenericScriptDef)

Description

A comment that describes the script. By default, the description field shows the type name of the script instance.

Script Function Name

The name of the function to be defined in step 3 of the wizard.

Trigger

On the level of the script definition the trigger values are only for information purposes and may be used to define a set of admissible triggers for the script or calculation execution instances. This trigger is defined on the level of the script or calculation execution instances. The following types of triggers are possible:

- On Operand Change
The corresponding script is executed after an operand value has changed.
- On Timer
The corresponding script is executed when the specified timer has expired according the specified time interval (see [3.6.4 Calculation Wizard](#) and [3.6.5 Script Wizard](#)).
- On Timer And Operand Has Changed
The corresponding script is executed when the specified timer has expired according the specified time interval and after an operand value has changed.
- On Timer Or Operand Has Changed
The corresponding script is executed when the specified timer has expired according the specified time interval or after an operand value has changed.

Type

A check box list with the allowed types for the result of a calculation. This list is displayed only if an instance of CalculationScriptDef is created.



Note

The **Type** attribute is visible only for calculation scripts. When you define a generic script, the attribute **Type** is not available.

3.6.3.3 Definition of Formal Script Operands

In this step, you can define the operands to be used in the script.

Step 2
Define script formal operands.

Result: .General/User Defined Calculation Definitions/Add Two Operands
Script Function Name: AddTwo

Operands

Operand name:

- a1
- a1
- a2

Add Remove

Operand Label: a1
Order No.: 1
Operand Type: Either

Allowed Type:

- ☐ Fleeting
- ☐ GroupAlarmIndicator
- ☐ Discrete
- ☐ Analog
- ☐ Accumulator
- ☐ Optional
- ☐ Repeatable

Operand Property:

- ☐ Link To State Set

Cancel < Back Next > Finish

[IMMRF041-101108-enUS-01.tif]

Figure 3-105 Definition of Script Operands

Operand Name

You can enter a new name operand name (top part) or select an already existing operand (bottom part).

Add

When you click the **Add** button, a new operand with the name defined in the previously described step is added. The parameters of the operand are added in the right part of the screen. A name which exists in the list is not allowed.

Remove

Clicking this button removes the selected operand from the operand list.

Up Arrow

This button allows you to change the order of the selected operand. Clicking this button moves the selected operand up by one place. The field **Order no.** of the operand automatically reflects this change.

Down Arrow

This button allows you to change the order of the selected operand. Clicking this button moves the selected operand down by 1 place. The field **Order no.** of the operand automatically reflects this change.

Operand Label

The label of the operand. The default is the operand name.

Order No.

This is a read-only field and indicates the order of the operand in the operand list.

Operand Type

You can select from the list box whether the actual operand is of the type **Constant** (constant value), **Dynamic** (dynamic operand - to be linked to a measurement in the instance tree), or **Either** (constant value or dynamic operand).

Allowed Type(s)

You can select the allowed types for the actual operand. These are subtypes of the type Measurement.

Operand Property

- Optional
If the actual operand can be omitted, you can select this item.
- Repeatable
If the actual operand can be repeatable, you can select this item.



Note

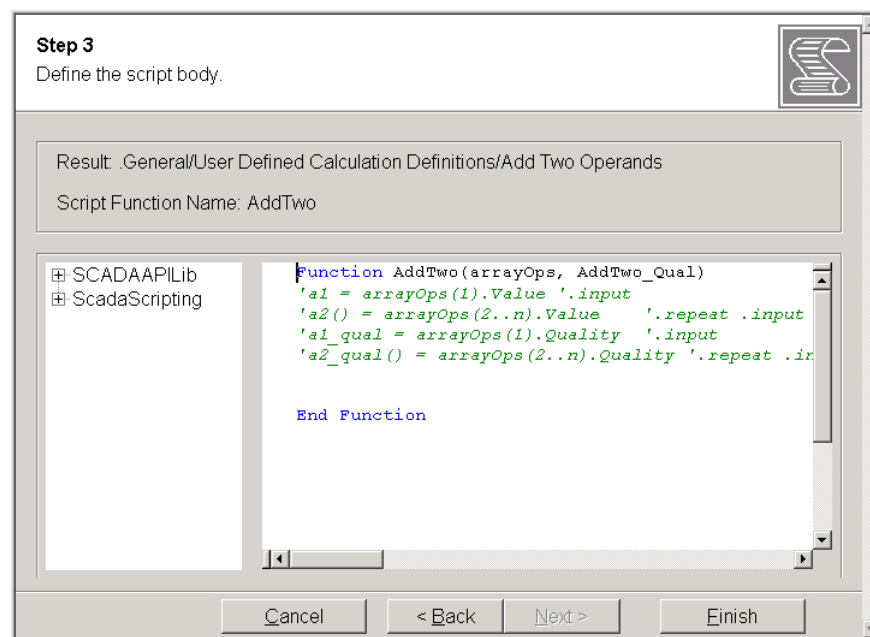
Only the last operand can be optional or repeatable.

Link to State Set

This box is enabled if you have selected **Digital Measurements** as **Result Type** of a calculation. In this case you can select a State Set instance from the list box.

3.6.3.4 Definition of the Script Body

In this step, you can define the script content.



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Figure 3-106 Definition of the Script Body

Parameter Tree

Shows available interfaces and their parameters which can be used in the script. By double-clicking a parameter, the selected parameter is pasted into the Script Editor.

Script Editor

The following script function with the corresponding operands is automatically generated in the Script Editor of the page:

```
Function AddTwo(...)
```

```
End Function
```

You can enter the body of the script between the begin and the end of the function. The operand names are used as parameter names. The last parameter in the parameter list is called **Quality**. This is a required parameter.

You can save the generated instances and links into the job by clicking the button **Finish**. After you have clicked the button **OK** in the message box, the wizard restarts with the first step and is ready for the next calculation or script definition.

The script is based on a subset of VB Script. For additional information, refer to [Programmer's Guide Scripting](#).

3.6.4 Calculation Wizard

3.6.4.1 Introduction

The Calculation Wizard creates calculated measurements based on calculation definitions defined with the script definition wizard..

The wizard guides you through the following 2 steps:

- Step 1: Definition of the measuring result and the calculation method
- Step 2: Definition of operands

The wizard is invoked by selecting the menu item **Calculation Wizard** from the context menu of the tree.



Note

You can skip the introduction of the Calculation Wizard by clicking the check box in the introduction page. If necessary, it is possible to go all steps backward. All data gathered in 1 step must be discarded in the backward step.

3.6.4.2 Definition of the Measuring Result and the Calculation Method

In this step, you can define the measurement for the result and the calculation method to be used.

Step 1
Define the result measurement and select the calculation rule.

Add Result To:	Net-E/Substations Generation/Sub Gen A
Result Name:	Res MW
Calculation:	Pre-defined calculation
Calculation Rule:	Multiplication
Result Type:	Accumulator
Calculation Trigger:	On Operand Change
Periodicity:	
Periodicity Unit:	
At Hour:	
At Minute:	
At Second:	
Activate At System Run Up:	False

Cancel < Back Next > Finish

[IMMRF043-101108-enUS-01.tif]

Figure 3-107 Definition of Measuring Result and Calculation Method

Add Result To

This read-only field displays the selected node, where the measuring result is added.

Result Name

Define the new measuring result, which is created under the selected node.

Calculation

You can select either the predefined or the user-defined calculations rules (specified by the Script Definition Wizard).

Calculation Method

A list box is displayed with the allowed calculation methods (for example, Multiplication, Division, and so on). Depending on the selection in the last step this list box contains either predefined or user-defined calculation methods.

The following predefined calculations are available:

- Aggregation - Average
- Aggregation - Current value
- Aggregation - Integration
- Aggregation - Max
- Aggregation - Min
- Analog Conditional
IF Condition == 1 (Open) THEN result = 0 ELSE result = ReplacementValue
- Capacitor Factor
The relation of active capacity to full capacity: $PF = P / \sqrt{P^2 + Q^2}$

- Combined Switch
Calculation of combined switch from series of switches for network overview displays
- Cosinus
Cosine calculation based on analog values or constants. (Single operand: $\text{Result}=\cos(a)$ or multiple operands $\text{res}=a*\cos(b)*\cos(c)$)
- Current in Amps
Value of the current calculated as relation of full capacity to voltage: $\text{Amps} = (S * 1000.0 / (U * \text{Sqrt}(3.0)))$
- DDC Replay - Bus/Branch to Node
- Digital AND
IF both operands equal 1 (Open) THEN resultant equals 1 (Open)
- Digital OR
IF either one of the operands equals 1 (Open) THEN resultant equals 1 (Open)
- Digital XOR
Logical XOR calculation for digital values. (Result = a OR b; for example, $\text{Open}(01) \text{ XOR } \text{Open}(01) = \text{Close}(10) \mid \text{Open}(01) \text{ XOR } \text{Closed}(10) = \text{Open}(01) \mid \text{Closed}(10) \text{ XOR } \text{Closed}(10) = \text{Closed}(10)$)
- Division
- Extended Rate of Change
Calculates the rate of changes (gradient) based on 2 consecutive filtered values of an analog. It is possible to define additional measurements for abrupt change confirmation.
- Multiplication
- MVA KV
MVA calculation: $\text{MVA} = U [\text{kV}] * \mid * \text{sqrt}(3) / 1000$
- MVA MW
MVA calculation $\text{MVA MW} = \text{sqrt}(P \text{ exp } 2 + Q \text{ exp } 2)$
- Operational Counter
The operational counter calculation counts the time the device is in operation. The unit of the counter can be set to 15 min, 30 min, or 1 h in units of seconds.
- Rate of Change
Calculates the rate of changes (gradient) based on 2 consecutive filtered values of an analog.
- Sinus
- Square Root
- Subtraction
- Summation
- Switch Counter - all changes
- Switch Counter - control response changes
- Switch Counter - entered changes
- Switch Counter - Spontaneous changes
- VB Script Calculation

Result Type

This defines the type of calculated measurement that is created by the wizard. A list box is displayed with the allowed types. This parameter can only be changed when creating a calculation, not when editing an existing calculation.

Calculation Trigger

A list box showing the possible calculation triggers.

- On timer
The script is triggered according to a timer.
- On change
The script is triggered whenever a value changes.
- On Timer Or Operand Has Changed
The script is triggered according to a timer or when a value changes.
- On Timer And Operand Has Changed
The script is triggered according to a timer and when a value changes.



Note

The following fields are enabled for editing, when one of the following trigger values has been selected:

- OnTimer
- OnTimerAndOperandHasChanged
- OnTimerOrOperandHasChanged

Periodicity

You can enter a periodicity value.

The fields Periodicity, Periodicity Unit, At Hour, At Minute, At Second define the configuration of the triggering timer: If for example, Periodicity = 10, Periodicity Unit = Minute and At Minute = 3 and At Second = 15, the script is triggered at the following times every 10 m.: 3:15, 13:15, and so on.

Periodicity Unit

You can select one of the following time units from the list box:

- Day
- Hour
- Minute
- Second

At Hour

You can select the trigger start hour from this list box.

At Minute

You can select the trigger start minute from this list box.

At Second

You can select the trigger start second from this list box.

Activate at System Startup

Possible values are:

- True
Calculation trigger is activated at the system startup.
- False
No activation of the calculation trigger at the system startup.

3.6.4.3 Definition of Operands

Step 2
Define calculation operands.

Result: Net-E/Substations Generation/Sub Gen A

Rule for Multiplication

Op.	Op. Type	Op. Name	Op. Path Or Constant
Multiplier	Constant	Multiplier	

Cancel < Back Next > Finish

[IMMRF044-101108-enUS-01.tif]

Figure 3-108 Definition of Operands

You can define the operands for the selected predefined or user-defined calculation method. The necessary operands are displayed in a grid showing the following columns (the attribute order number of a formal operand defines its position in the grid):

Op.

Label of the formal operand.

Op. Type

The operand type. The list box can show the values **Dynamic** and **Constant**.

Op. Name

The operand name. This name is used for the instance defining the dynamic or constant operand. The default is the label of the formal operand of the selected calculation or script. If multiple operands of the same type are allowed, then the default name is the label of the formal operand and a running number starting with 1. This field is read-only, defined in the calculation definition.

Op. Path or Constant

If **Dynamic** is selected as the operand type then drag and drop is enabled for the row. You can drag a measurement of the allowed type from the instance tree and drop it into the **Op. Path** field. If **Constant** is selected, you can enter a constant value. A check of the value is performed according the allowed type. If a formal operand has Repeat=True then there is always an additional empty row at the end of the table. When you enter a value for the operand type, the field operand is handled as described previously.

Finish

You can save the defined calculation into the session dataset by clicking the button **Finish**. After you have clicked the button **Finish**, a confirmation message box is shown. After you have clicked the button **OK** the wizard restarts with the introduction step and is ready for the next calculation or script definition.

3.6.5 Script Wizard

3.6.5.1 Introduction

The Script Wizard creates new script instances.

The wizard guides you through the following 2 steps:

- Step 1: Definition of the script call/execution
- Step 2: Definition of operands

The wizard is invoked by selecting the menu item **Script Wizard** from the context menu of the tree.



Note

You can skip the introduction of the Script Wizard by clicking the check box in the introduction page. If necessary it is possible to backtrack all steps. All data gathered in one step must be discarded in the backward step.

3.6.5.2 Definition of the Script Call/Execution

In this step, you can define when the script shall be executed.

Step 1

Define the result measurement and select the calculation rule.

Script Execution Will Be Added To:

.General/User Defined Scripts

Name Of Script Execution:

Change Limit

Script:

spcChange Limit

Script Trigger:

On Operand Change

Periodicity:

Periodicity Unit:

At Hour:

At Minute:

At Second:

Activate At System Run Up:

False

Cancel

< Back

Next >

Finish

[IMMRF045-101108-enUS-01.tif]

Figure 3-109 Script Wizard

Script Execution Will Be Added To

This read-only field displays the selected node, where the script execution is added.

Name of Script Execution

Define the new script execution which is created under the selected node.

Script

The script definition as defined in the script definition wizard is selected. All script definitions are provided as list for selection by you.

Script Trigger

The list box shows the possible calculation script trigger values.

The following fields are described in [3.6.4.2 Definition of the Measuring Result and the Calculation Method](#).

■